SEAGIS

Arsitektur Database, Lineage dan Kontrak Data

Bahasa Indonesia

Version 1.0.0
27 October 2026 / Asia Jakarta
Catalog snapshot: 22 October 2026

Audit kode dan snapshot historis. Runtime terkini belum diverifikasi.

Daftar Isi

1 Ringkasan Eksekutif 7

2 Gambaran Arsitektur Sistem dan Data 8

3 Inventaris Database dan Skema 9

4 Kamus Data Lengkap 15

4.1 seagis.dataset . . . 15
4.2 seagis.dataset_parameter . . . 21
4.3 seagis.forecast_run . . . 25
4.4 seagis.forecast_value . . . 30
4.5 seagis.ingestion_run . . . 39
4.6 seagis.observation . . . 43
4.7 seagis.parameter . . . 51
4.8 seagis.plot . . . 53
4.9 seagis.plot_grid_mapping . . . 56
4.10 seagis.quality_event . . . 61
4.11 seagis.sensor . . . 66
4.12 seagis.source . . . 70
4.13 seagis.source_grid . . . 72
4.14 seagis_api.credential . . . 77
4.15 seagis_api.packet_receipt . . . 80
4.16 seagis_api.reader_plot . . . 83
4.17 seagis_api.request_event . . . 84
4.18 seagis_api.temperature_policy . . . 87
4.19 seagis_environment.attempt . . . 91
4.20 seagis_environment.current_speed . . . 94
4.21 seagis_environment.demo_grid_mapping . . . 98
4.22 seagis_environment.policy . . . 103
4.23 seagis_environment.schedule_state . . . 108
4.24 seagis_features.feature_input . . . 112
4.25 seagis_features.feature_snapshot . . . 114
4.26 seagis_quality.alignment_snapshot . . . 120
4.27 seagis_quality.policy . . . 125
4.28 seagis_quality.snapshot_input . . . 127
4.29 seagis_raw.raw_payload . . . 130

5 Hubungan Antarentitas 137

FK1 seagis.source → seagis.dataset . . . 137

FK2 seagis.dataset → seagis.dataset_parameter . . . 137
FK3 seagis.parameter → seagis.dataset_parameter . . . 137
FK4 seagis.dataset → seagis.forecast_run . . . 138
FK5 seagis_raw.raw_payload → seagis.forecast_run . . . 138
FK6 seagis.dataset_parameter → seagis.forecast_value . . . 138
FK7 seagis.forecast_run → seagis.forecast_value . . . 139
FK8 seagis.source_grid → seagis.forecast_value . . . 139
FK9 seagis.dataset → seagis.ingestion_run . . . 139
FK10 seagis.dataset_parameter → seagis.observation . . . 140
FK11 seagis_raw.raw_payload → seagis.observation . . . 140
FK12 seagis.sensor → seagis.observation . . . 140
FK13 seagis.plot → seagis.plot_grid_mapping . . . 141
FK14 seagis.source_grid → seagis.plot_grid_mapping . . . 141
FK15 seagis.forecast_value → seagis.quality_event . . . 141
FK16 seagis.ingestion_run → seagis.quality_event . . . 142
FK17 seagis.observation → seagis.quality_event . . . 142
FK18 seagis.dataset_parameter → seagis.sensor . . . 142
FK19 seagis.dataset → seagis.sensor . . . 143
FK20 seagis.plot → seagis.sensor . . . 143
FK21 seagis.dataset → seagis.source_grid . . . 143
FK22 seagis.sensor → seagis_api.credential . . . 144
FK23 seagis.ingestion_run → seagis_api.packet_receipt . . . 144
FK24 seagis.observation → seagis_api.packet_receipt . . . 144
FK25 seagis.sensor → seagis_api.packet_receipt . . . 145
FK26 seagis_api.credential → seagis_api.reader_plot . . . 145
FK27 seagis.plot → seagis_api.reader_plot . . . 145
FK28 seagis_api.credential → seagis_api.request_event . . . 146
FK29 seagis.sensor → seagis_api.request_event . . . 146
FK30 seagis_environment.policy → seagis_environment.attempt . . . 146
FK31 seagis.forecast_run → seagis_environment.attempt . . . 147
FK32 seagis_environment.policy → seagis_environment.demo_grid_mapping . . . 147
FK33 seagis.plot → seagis_environment.demo_grid_mapping . . . 147
FK34 seagis.source_grid → seagis_environment.demo_grid_mapping . . . 148
FK35 seagis.dataset → seagis_environment.policy . . . 148
FK36 seagis_environment.policy → seagis_environment.schedule_state . . . 148
FK37 seagis.forecast_run → seagis_environment.schedule_state . . . 149
FK38 seagis.forecast_value → seagis_features.feature_input . . . 149
FK39 seagis.observation → seagis_features.feature_input . . . 149
FK40 seagis_features.feature_snapshot → seagis_features.feature_input . . . 150
FK41 seagis.plot → seagis_features.feature_snapshot . . . 150
FK42 seagis.plot → seagis_quality.alignment_snapshot . . . 150

FK43 seagis_quality.policy → seagis_quality.alignment_snapshot . . . 151
FK44 seagis.forecast_value → seagis_quality.snapshot_input . . . 151
FK45 seagis.observation → seagis_quality.snapshot_input . . . 151
FK46 seagis_quality.alignment_snapshot → seagis_quality.snapshot_input . . . 152
FK47 seagis.ingestion_run → seagis_raw.raw_payload . . . 152
seagis . . . 159
seagis_api . . . 163
seagis_environment . . . 165
seagis_features . . . 167
seagis_quality . . . 169
seagis_raw . . . 171

6 Sumber dan Asal-Usul Data 172

7 Ingesti dan Transformasi Data 176

7.1 Pemetaan Sumber–Tujuan . . . 178

8 Kontrak Data 227

C01 A0 source envelope . . . 229
C02 A0 normalized envelope . . . 237
C03 A1 device packet . . . 268
C04 A3 alignment document . . . 271
C05 A1 history/status/replies . . . 289
C06 A2 adapter batch/lease . . . 290
C07 A3 source_bundle . . . 291
C08 CSV experiment-csv-2 . . . 291
C09 iOS Assessment domain . . . 292
C10 Open-Meteo response . . . 292
C11 Copernicus subset . . . 293
Rute HTTP dan respons . . . 293
Payload dan error A1 . . . 294

9 Siklus Hidup Data 296

10 Dependensi Data pada Aplikasi 297

11 Audit Kualitas Data dan Arsitektur 302

F01 Assessment iOS belum terhubung ke backend . . . 302
F02 Mapping immutable setelah 0005 mengubah prosedur penutupan lama . . . 302
F03 Riwayat tumbuh tanpa job retensi yang ditemukan . . . 302
F04 SQL A3 memeriksa lineage tetapi tidak menghitung ulang seluruh QC . . . 302
F05 Metadata-only membatasi reproduksi raw . . . 303

F06 CSV capture bukan backtest historis . . . 303
F07 Dokumen A0 mendahului implementasi A1–A3 . . . 303
F08 Snapshot catalog berumur lima hari . . . 304

12 Glosarium 305

13 Lampiran 306

13.1 Basis bukti dan batas audit . . . 306
13.2 Cakupan terukur . . . 306
13.2.1 Dataset file lokal teramati . . . 307
13.3 Pertanyaan terbuka . . . 309
13.4 SQL read-only ilustratif . . . 309
13.5 Registry fungsi SQL . . . 309
13.6 Register sumber . . . 310

Version: 1.0.0 | Date: 2026-10-27 (Asia/Jakarta) | Catalog snapshot: 2026-10-22T10:42:58.974431+00:00

Status: audit kode dan snapshot historis; runtime terkini UNVERIFIED.

Label	Makna
VERIFIED	Didukung kode/inspeksi; tidak otomatis diuji runtime
INFERRED	Interpretasi masuk akal, tidak eksplisit
UNVERIFIED	Belum dapat dibuktikan dengan akses tersedia
NOT FOUND	Dicari pada lingkup yang disebut, tidak ditemukan
NOT APPLICABLE	Tidak relevan untuk komponen

1 Ringkasan Eksekutif

Audit ini mendokumentasikan implementasi backend dan konsumen lokal yang dapat diperiksa pada 27 Oktober 2026. Satu database PostgreSQL aplikasi ditemukan melalui migrasi dan snapshot catalog 22 Oktober: enam schema, 28 tabel, satu view, 268 kolom termasuk sembilan kolom view, dan 47 FK. Tidak ada koneksi live yang dibuat dalam audit ini; jumlah baris, deployment terkini, backup, role membership dan drift setelah snapshot tetap UNVERIFIED.

A1 menerima paket suhu lewat Edge Function, menetapkan identitas dari credential, menyimpan raw/provenance, observasi dan QC secara atomik, serta mengenali retry semantik. A2 mengambil empat dataset regional, mempertahankan grid dan native cadence, lalu menyimpan vintage immutable. A3 menyusun konteks per jam dengan cutoff ketersediaan dan menyimpan snapshot review. A3 bukan feature builder produksi atau model penyakit.

Aplikasi Seagis dan prototype AppPOC diperiksa sebagai konsumen terpisah. MockAssessmentRepository menghasilkan assessment sintetis; tidak ditemukan jalur produksi dari assessment ke database backend. Status QC, data model regional dan demo tidak boleh diperlakukan sebagai diagnosis atau probabilitas ice-ice. Dokumen ini adalah rujukan audit kode versi tertentu, bukan sertifikasi sistem produksi.

Bukti implementasi utama: Seagis-Model/supabase/migrations/0001_initial_schema.sql:1;
Seagis-Model/supabase/migrations/0002_temperature_api.sql:1;
Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1;
Seagis-Model/supabase/migrations/0004_provider_update_status.sql:1;
Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1.

2 Gambaran Arsitektur Sistem dan Data

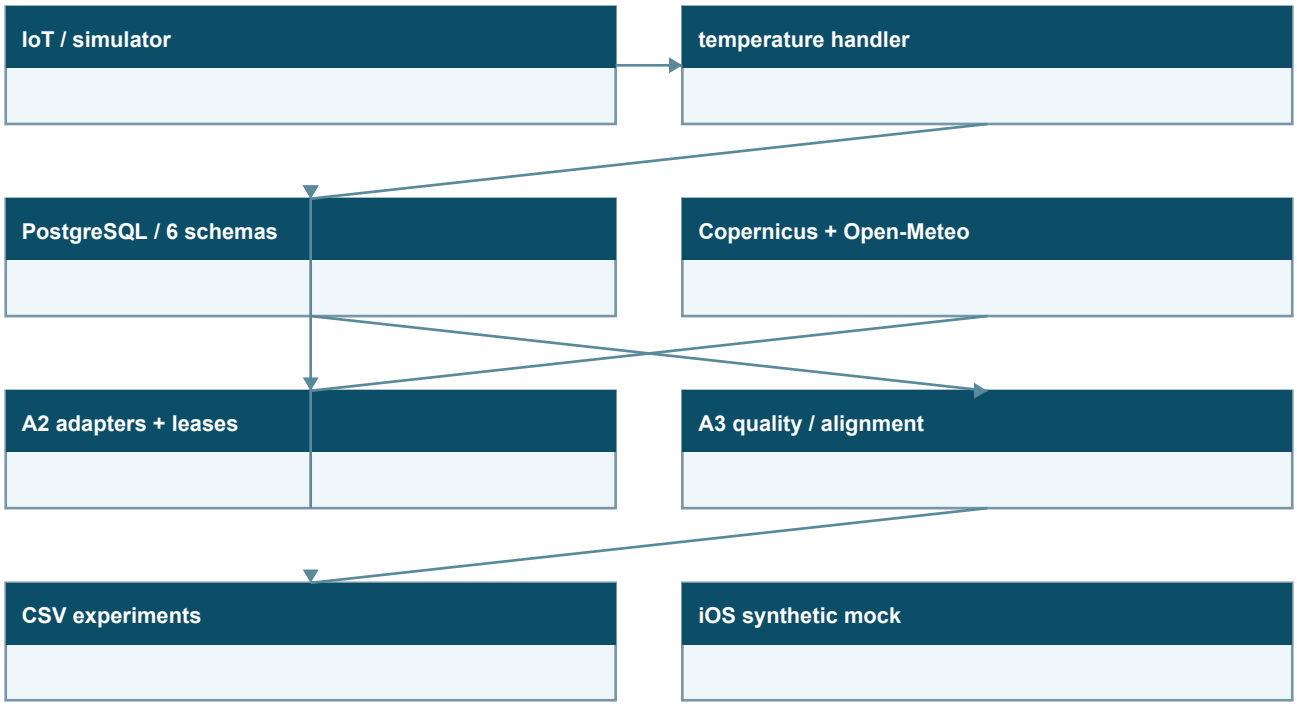
PostgreSQL/Supabase adalah persistence backend yang didefinisikan kode. Schema seagis berisi registri, observasi, forecast dan provenance; seagis_raw menyimpan manifest raw; seagis_api membatasi IoT gateway; seagis_environment mengatur worker regional; seagis_quality menyimpan review QC; seagis_features menyediakan container fitur yang belum dihubungkan ke builder produksi. Schema internal Supabase seperti auth/storage tidak diinventaris sebagai milik aplikasi: akses catalog terkini tidak tersedia dan tidak boleh diasumsikan kosong.

Python melakukan pengambilan provider di luar database. Worker A2/A3 memanggil stored function lewat psycopg; TS handler memanggil hanya fungsi ingest/history/status. Tidak ada ORM yang ditemukan pada jalur ini. Transaksi mutasi berada di fungsi SQL; claim fetch merupakan transaksi terpisah dari persistence agar panggilan jaringan tidak menahan transaksi database.

Diagram memperlihatkan dua komponen yang terputus: mock iOS tidak memiliki edge ke backend. Integrasi masa depan tetap UNVERIFIED. Core ML, cache terdistribusi, warehouse, queue dan search index produksi NOT FOUND dalam file implementasi yang diinspeksi; ini bukan bukti bahwa layanan di luar checkout tidak ada.

Bukti implementasi utama: Seagis-Model/supabase/migrations/0001_initial_schema.sql:1;
Seagis-Model/supabase/migrations/0002_temperature_api.sql:1;
Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1;
Seagis-Model/supabase/migrations/0004_provider_update_status.sql:1;
Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1.

Diagram reproduktif: assets/diagrams/architecture.mmd.



3 Inventaris Database dan Skema

PostgreSQL/Supabase adalah persistence backend yang didefinisikan kode. Schema seagis berisi registri, observasi, forecast dan provenance; seagis_raw menyimpan manifest raw; seagis_api membatasi IoT gateway; seagis_environment mengatur worker regional; seagis_quality menyimpan review QC; seagis_features menyediakan container fitur yang belum dihubungkan ke builder produksi. Schema internal Supabase seperti auth/storage tidak diinventaris sebagai milik aplikasi: akses catalog terkini tidak tersedia dan tidak boleh diasumsikan kosong.

Python melakukan pengambilan provider di luar database. Worker A2/A3 memanggil stored function lewat psycopg; TS handler memanggil hanya fungsi ingest/history/status. Tidak ada ORM yang ditemukan pada jalur ini. Transaksi mutasi berada di fungsi SQL; claim fetch merupakan transaksi terpisah dari persistence agar panggilan jaringan tidak menahan transaksi database.

Diagram memperlihatkan dua komponen yang terputus: mock iOS tidak memiliki edge ke backend. Integrasi masa depan tetap UNVERIFIED. Core ML, cache terdistribusi, warehouse, queue dan search index produksi NOT FOUND dalam file implementasi yang diinspeksi; ini bukan bukti bahwa layanan di luar checkout tidak ada.

Store	PostgreSQL / Supabase
Evidence status	VERIFIED (repository + historical catalog)
Tujuan/lingkungan	Backend; lingkungan deployment terkini UNVERIFIED
Owner	Seagis-Model
Scope	Registri, sensor, regional, QC, fitur
Access	read/write via stored functions
Otoritas	Otoritatif untuk record yang tersimpan; bukan diagnosis
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:1

Store	Supabase Storage
Evidence status	configured / UNVERIFIED usage
Tujuan/lingkungan	Storage enabled lokal; object_uri didukung, bucket aplikasi NOT FOUND
Owner	Supabase
Scope	objects / metadata_only
Access	configured
Otoritas	Tidak dibuktikan sebagai penyimpan raw aktif
Evidence	Seagis-Model/supabase/config.toml:116

Store	UserDefaults / launch arguments
Evidence status	VERIFIED code
Tujuan/lingkungan	Akses preferensi/debug UI; bukan penyimpan plot
Owner	AppPOC
Scope	initialPlot / initialFarmer / skipCaptureSteps / detailDay
Access	read preference/argument values
Otoritas	Tidak otoritatif untuk data backend
Evidence	AppPOC/POCApp/Features/FarmMap/ViewModels/FarmMapViewModel.swift:1

Store	legacy AR files
Evidence status	VERIFIED cleanup code
Tujuan/lingkungan	File ar-site.json/ar-site.worldmap hanya dihapus saat sesi AR baru
Owner	AppPOC
Scope	legacy local AR files

Access	best-effort delete; active writer NOT FOUND
Otoritas	Bukan persistence aktif
Evidence	AppPOC/POCApp/Features/PlotCapture/ViewModels/ARCaptureViewModel.swift:1

Store	JSON / CSV files
Evidence status	VERIFIED code
Tujuan/lingkungan	Fixture sintetis, policy, export eksperimen dan manifest
Owner	Python scripts
Scope	fixtures/config/export
Access	file read/write
Otoritas	Hanya otoritatif untuk capture/fixture versinya
Evidence	Seagis-Model/scripts/export_experiment_csv.py:1

Store	in-memory Swift arrays
Evidence status	VERIFIED code
Tujuan/lingkungan	State prototype; hilang saat proses berakhir
Owner	AppPOC
Scope	Farmer / CultivationPlot
Access	array append/remove
Otoritas	State demo lokal
Evidence	AppPOC/POCApp/Shared/Data/FarmStore.swift:1

Default layanan Supabase auth, vector dan analytics tercantum dalam config lokal, tetapi bucket/index/dataset aplikasi aktif tidak dibuktikan. Ini konfigurasi layanan, bukan database aplikasi tambahan. Jumlah baris seluruh tabel: UNVERIFIED; tidak diestimasi dari fixture.

Schema	seagis
Objek	dataset, dataset_parameter, forecast_run, forecast_value, ingestion_run, observation, parameter, plot, plot_grid_mapping, quality_event, sensor, source, source_grid
Tables	13
Views	0
Batas akses	Akses melalui routine backend; membership live UNVERIFIED

Schema	seagis_api
Objek	credential, packet_receipt, reader_plot, request_event, temperature_policy
Tables	5
Views	0
Batas akses	Akses melalui routine backend; membership live UNVERIFIED

Schema	seagis_environment
Objek	attempt, current_speed, demo_grid_mapping, policy, schedule_state
Tables	4
Views	1
Batas akses	Akses melalui routine backend; membership live UNVERIFIED

Schema	seagis_features

Objek	feature_input, feature_snapshot
Tables	2
Views	0
Batas akses	Akses melalui routine backend; membership live UNVERIFIED

Schema	seagis_quality
Objek	alignment_snapshot, policy, snapshot_input
Tables	3
Views	0
Batas akses	Akses melalui routine backend; membership live UNVERIFIED

Schema	seagis_raw
Objek	raw_payload
Tables	1
Views	0
Batas akses	Akses melalui routine backend; membership live UNVERIFIED

Objek fisik	seagis.dataset
Fungsi	Identitas dataset, versi, cadence, lisensi dan gate ingestion.
Kolom	15
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Objek fisik	seagis.dataset_parameter
Fungsi	Kontrak variabel sumber ke parameter dan aturan konversi.
Kolom	9
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Objek fisik	seagis.forecast_run
Fungsi	Capture/vintage model regional; waktu publikasi yang tidak diketahui tetap null.
Kolom	13
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Objek fisik	seagis.forecast_value
Fungsi	Nilai parameter per run, grid dan valid time; tidak diduplikasi per plot.
Kolom	20
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Objek fisik	seagis.ingestion_run
Fungsi	Audit satu ingestion logis beserta status penyelesaian.
Kolom	9
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Objek fisik	seagis.observation
Fungsi	Pembacaan suhu air sensor lokal, dengan nilai native, QC dan lineage raw.
Kolom	19
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Objek fisik	seagis.parameter
Fungsi	Registri parameter dan satuan normalisasi.
Kolom	3
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36

Objek fisik	seagis.plot
Fungsi	Identitas plot dan status survei; koordinat nullable bukan bukti lokasi lapangan.
Kolom	8
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Objek fisik	seagis.plot_grid_mapping
Fungsi	Pemetaan operasional plot ke grid dengan waktu efektif dan waktu tercatat.
Kolom	10
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Objek fisik	seagis.quality_event
Fungsi	Log QC append-only untuk ingestion atau satu measurement.
Kolom	11
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Objek fisik	seagis.sensor
Fungsi	Identitas deployment sensor suhu, plot, kedalaman dan status aktif.
Kolom	10
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Objek fisik	seagis.source
Fungsi	Registri penyedia asal dataset.
Kolom	3
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11

Objek fisik	seagis.source_grid
Fungsi	Identitas sel regional dan layer kedalaman; bukan plot budidaya.
Kolom	10
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Objek fisik	seagis_api.credential
Fungsi	Identitas aktor API berupa hash token, masa berlaku dan pencabutan.
Kolom	8
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Objek fisik	seagis_api.packet_receipt
Fungsi	Kunci idempotensi paket sensor dan digest semantik.
Kolom	6
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Objek fisik	seagis_api.reader_plot
Fungsi	Daftar plot yang boleh dibaca satu credential reader.

Kolom	2
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39

Objek fisik	seagis_api.request_event
Fungsi	Audit respons API terautentikasi dan dasar rate limit.
Kolom	6
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Objek fisik	seagis_api.temperature_policy
Fungsi	Konfigurasi QC engineering pilot dan batas request; bukan ambang penyakit.
Kolom	8
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Objek fisik	seagis_environment.attempt
Fungsi	Audit satu upaya fetch termasuk duplicate dan abandoned.
Kolom	7
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Objek fisik	seagis_environment.current_speed
Fungsi	View magnitude arus dari pasangan komponen dengan run/grid/waktu yang sama.
Kolom	9
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Objek fisik	seagis_environment.demo_grid_mapping
Fungsi	Bridge khusus dua plot fiktif ke grid nyata untuk preview, bukan otorisasi training.
Kolom	11
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Objek fisik	seagis_environment.policy
Fungsi	Kebijakan fetch, lisensi, cadence scheduler, staleness dan retensi regional.
Kolom	11
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Objek fisik	seagis_environment.schedule_state
Fungsi	State mutable lease, due time, backoff dan keberhasilan terakhir.
Kolom	9
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Objek fisik	seagis_features.feature_input
Fungsi	Lineage FK input feature snapshot dan guard cutoff.
Kolom	4
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

Objek fisik	seagis_features.feature_snapshot
Fungsi	Container fitur draft/sealed dengan cutoff dan hash; builder produksi belum ditemukan.
Kolom	12
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Objek fisik	seagis_quality.alignment_snapshot
Fungsi	Dokumen QC/alignment hourly immutable; selalu training_authorized=false.
Kolom	11
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Objek fisik	seagis_quality.policy
Fungsi	Registri versi kebijakan QC dan hash kode builder yang immutable.
Kolom	5
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

Objek fisik	seagis_quality.snapshot_input
Fungsi	Lineage measurement per jam alignment snapshot.
Kolom	5
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

Objek fisik	seagis_raw.raw_payload
Fungsi	Manifest raw immutable: digest, basis hash, lisensi dan mode retensi.
Kolom	14
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

4 Kamus Data Lengkap

Domain seagis.finite_float adalah double precision yang menolak NaN dan infinity; NULL tetap diatur kolom. value_status present memerlukan angka; status nonpresent memerlukan null; valid_value/valid_support serta guard SQL menetapkan aturan lintas field. Presipitasi mm adalah total satu jam, salinitas 1e-3 tidak otomatis PSU/ppt, wind direction adalah arah dari dalam derajat. Metadata tipe/null/default/constraint/index berasal dari snapshot catalog 22 Oktober; definisi SQL diverifikasi terhadap rantai migrasi lokal 0001–0005. Baris view tidak memiliki PK/FK fisik walaupun field-nya berasal dari measurement. Contoh adalah ilustrasi sintetis dan belum tentu satu row INSERT yang valid. NOT APPLICABLE berarti atribut tidak relevan; UNVERIFIED berarti semantik atau derivasi spesifik belum dibuktikan.

4.1 seagis.dataset

Identitas dataset, versi, cadence, lisensi dan gate ingestion.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (dataset_key)
Unique	UNIQUE (dataset_key, scope)
Rows	UNVERIFIED
Writer	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Reader	A1 ingest/history; A2 persist/preview; A3 source_bundle
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

seagis.dataset.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'sensor_temperature_v1' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.dataset_key

seagis.dataset.source_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND

Atribut	Definisi
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (source_key) REFERENCES seagis.source(source_key)
Arti bisnis / teknis	Kunci penyedia sumber
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'sensor' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.source_key

seagis.dataset.provider_product_id

Atribut	Definisi
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas produk penyedia bila tersedia
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → NULL (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.provider_product_id

seagis.dataset.provider_dataset_id

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas dataset asli penyedia
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'seagis-water-temperature' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.provider_dataset_id

seagis.dataset.dataset_version

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Versi dataset yang dipin
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'v1' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.dataset_version

seagis.dataset.scope

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Asal local_sensor atau regional_model
Valid values / checks	CHECK (scope = ANY (ARRAY['local_sensor'::text, 'regional_model'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'local_sensor' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.scope

seagis.dataset.native_cadence_seconds

Atribut	Definisi
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Jarak waktu asli sampel, bukan interval fetch
Valid values / checks	CHECK (native_cadence_seconds > 0)
Units / time	s; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 3600 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.native_cadence_seconds

seagis.dataset.resolution_lat_deg

Atribut	Definisi
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Resolusi latitude dalam derajat bila diketahui
Valid values / checks	CHECK (resolution_lat_deg::double precision > 0::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → NULL (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.resolution_lat_deg

seagis.dataset.resolution_lon_deg

Atribut	Definisi
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Resolusi longitude dalam derajat bila diketahui
Valid values / checks	CHECK (resolution_lon_deg::double precision > 0::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → NULL (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.resolution_lon_deg

seagis.dataset.source_timezone

Atribut	Definisi
Type / nullable	text / false
Default / generated	'UTC'::text
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Konteks zona waktu asal
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	caller / SQL default → 'UTC'::text (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.source_timezone

seagis.dataset.licence_url

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Referensi lisensi sumber
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'project-owned://sensor-data' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.licence_url

seagis.dataset.attribution

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Teks atribusi yang harus dipertahankan
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → 'SEAGIS; simulated until hardware commissioning' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.attribution

seagis.dataset.raw_storage_permitted

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	false
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Gate izin menyimpan raw, bukan jaminan objek telah disimpan
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE

Atribut	Definisi
Example	true
Asal / transformasi	caller / SQL default → false (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.raw_storage_permitted

seagis.dataset.real_ingestion_enabled

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	false
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Gate ingestion data nonsintetis
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	caller / SQL default → false (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.real_ingestion_enabled

seagis.dataset.metadata_checked_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Waktu metadata sumber diperiksa
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511 → '2026-10-21T00:00:00Z' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17; catalog seagis.dataset.metadata_checked_at

Constraint, indeks dan trigger

constraints

```
UNIQUE (dataset_key, scope);
CHECK (native_cadence_seconds > 0);
PRIMARY KEY (dataset_key);
CHECK (resolution_lat_deg::double precision > 0::double precision);
CHECK (resolution_lon_deg::double precision > 0::double precision);
CHECK (scope = ANY (ARRAY['local_sensor'::text, 'regional_model'::text]));
FOREIGN KEY (source_key) REFERENCES seagis.source(source_key);
```

indexes

```
CREATE UNIQUE INDEX dataset_dataset_key_scope_key ON seagis.dataset USING btree (dataset_key,
scope);
CREATE UNIQUE INDEX dataset_pkey ON seagis.dataset USING btree (dataset_key);
```

triggers

```
CREATE TRIGGER guard_dataset_identity BEFORE UPDATE ON seagis.dataset FOR EACH ROW EXECUTE FUNCTION
seagis.guard_dataset_identity();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.2 seagis.dataset_parameter

Kontrak variabel sumber ke parameter dan aturan konversi.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (dataset_key, parameter_code)
Unique	UNIQUE (dataset_key, parameter_code, native_unit, unit, conversion_code)
Rows	UNVERIFIED
Writer	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Reader	A1 ingest/history; A2 persist/preview; A3 source_bundle
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

seagis.dataset_parameter.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'sensor_temperature_v1' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.dataset_key

seagis.dataset_parameter.parameter_code

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND

Atribut	Definisi
PK / unique	true / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (parameter_code, unit) REFERENCES seagis.parameter(parameter_code, unit)
Arti bisnis / teknis	Identitas parameter lingkungan
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'water_temperature' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.parameter_code

seagis.dataset_parameter.source_variable

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Nama variabel asli provider
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'water_temperature' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.source_variable

seagis.dataset_parameter.native_unit

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Satuan asli sebelum normalisasi
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'degC' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.native_unit

seagis.dataset_parameter.unit

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (parameter_code, unit) REFERENCES seagis.parameter(parameter_code, unit)
Arti bisnis / teknis	Satuan nilai normalisasi menurut registri
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'degC' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.unit

seagis.dataset_parameter.conversion_code

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas metode konversi satuan
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'identity' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.conversion_code

seagis.dataset_parameter.conversion_factor

Atribut	Definisi
Type / nullable	seagis.finite_float / false
Default / generated	1
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Pengali pada value=native_value*factor+offset
Valid values / checks	CHECK (conversion_factor::double precision <> 0::double precision)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	1
Asal / transformasi	caller / SQL default → 1 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.conversion_factor

seagis.dataset_parameter.conversion_offset

Atribut	Definisi
Type / nullable	seagis.finite_float / false
Default / generated	0
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Offset setelah perkalian faktor
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	1
Asal / transformasi	caller / SQL default → 0 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.conversion_offset

seagis.dataset_parameter.temporal_support

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Dukungan waktu instantaneous, mean atau sum
Valid values / checks	CHECK (temporal_support = ANY (ARRAY['instantaneous'::text, 'mean'::text, 'sum'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532 → 'instantaneous' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43; catalog seagis.dataset_parameter.temporal_support

Constraint, indeks dan trigger

constraints

```
CHECK (conversion_factor::double precision <> 0::double precision);
FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key);
UNIQUE (dataset_key, parameter_code, native_unit, unit, conversion_code);
FOREIGN KEY (parameter_code, unit) REFERENCES seagis.parameter(parameter_code, unit);
PRIMARY KEY (dataset_key, parameter_code);
CHECK (temporal_support = ANY (ARRAY['instantaneous'::text, 'mean'::text, 'sum'::text]));
```

indexes

```
CREATE UNIQUE INDEX dataset_parameter_dataset_key_parameter_code_native_unit_un_key ON
seagis.dataset_parameter USING btree (dataset_key, parameter_code, native_unit, unit,
```

conversion_code);
CREATE UNIQUE INDEX dataset_parameter_pkey ON seagis.dataset_parameter USING btree (dataset_key, parameter_code);

triggers

CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis.dataset_parameter FOR EACH ROW EXECUTE FUNCTION seagis.reject_mutation();

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.3 seagis.forecast_run

Capture/vintage model regional; waktu publikasi yang tidak diketahui tetap null.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (dataset_key, run_key); UNIQUE (id, dataset_key, is_synthetic)
Rows	UNVERIFIED
Writer	seagis_environment.persist; A0 fixtures
Reader	seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

seagis.forecast_run.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.id

seagis.forecast_run.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)

Atribut	Definisi
FK	FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope); FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → s.dataset_key (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.dataset_key

seagis.forecast_run.scope

Atribut	Definisi
Type / nullable	text / false
Default / generated	'regional_model'::text
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope)
Arti bisnis / teknis	Asal local_sensor atau regional_model
Valid values / checks	CHECK (scope = 'regional_model'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	caller / SQL default → 'regional_model'::text (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.scope

seagis.forecast_run.ingestion_run_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi ingestion asal
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → ing (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.ingestion_run_id

seagis.forecast_run.raw_payload_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi manifest raw asal
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → raw_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.raw_payload_id

seagis.forecast_run.is_synthetic

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Penanda asal sintetis, terpisah dari QC
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → false (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.is_synthetic

seagis.forecast_run.run_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas capture semantik per dataset
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid

Atribut	Definisi
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → batch->>'run_key' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.run_key

seagis.forecast_run.run_kind

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	forecast, analysis atau analysis_forecast_unspecified
Valid values / checks	CHECK (run_kind = ANY (ARRAY['forecast'::text, 'analysis'::text, 'analysis_forecast_unspecified'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → CASE WHEN d.source_key='copernicus' THEN 'analysis_forecast_unspecified' ELSE 'forecast' END (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.run_kind

seagis.forecast_run.provider_run_id

Atribut	Definisi
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	ID run provider jika benar-benar tersedia
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.provider_run_id

seagis.forecast_run.model_initialized_at

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND

Atribut	Definisi
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan inisialisasi model jika diketahui
Valid values / checks	CHECK (model_initialized_at IS NULL OR model_initialized_at <= ingested_at)
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.model_initialized_at

seagis.forecast_run.issued_at

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan publikasi provider terverifikasi; null jika tidak diketahui
Valid values / checks	CHECK ((issue_time_basis = 'unknown'::text) = (issued_at IS NULL)); CHECK (issued_at IS NULL OR issued_at <= ingested_at)
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.issued_at

seagis.forecast_run.issue_time_basis

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Dasar known/unknown waktu publikasi
Valid values / checks	CHECK ((issue_time_basis = 'unknown'::text) = (issued_at IS NULL)); CHECK (issue_time_basis = ANY (ARRAY['provider_verified'::text, 'unknown'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → 'unknown' (VERIFIED)

Atribut	Definisi
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.issue_time_basis

seagis.forecast_run.ingested_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan data tersedia bagi sistem
Valid values / checks	CHECK (issued_at IS NULL OR issued_at <= ingested_at); CHECK (model_initialized_at IS NULL OR model_initialized_at <= ingested_at)
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168 → t (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216; catalog seagis.forecast_run.ingested_at

Constraint, indeks dan trigger

constraints

```
CHECK ((issue_time_basis = 'unknown'::text) = (issued_at IS NULL));
CHECK (issued_at IS NULL OR issued_at <= ingested_at);
CHECK (model_initialized_at IS NULL OR model_initialized_at <= ingested_at);
UNIQUE (dataset_key, run_key);
FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope);
UNIQUE (id, dataset_key, is_synthetic);
CHECK (issue_time_basis = ANY (ARRAY['provider_verified'::text, 'unknown'::text]));
PRIMARY KEY (id);
FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES
seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic);
CHECK (run_kind = ANY (ARRAY['forecast'::text, 'analysis'::text,
'analysis_forecast_unspecified'::text]));
CHECK (scope = 'regional_model'::text);
```

indexes

```
CREATE INDEX forecast_run_asof ON seagis.forecast_run USING btree (dataset_key, ingested_at DESC);
CREATE UNIQUE INDEX forecast_run_dataset_key_run_key_key ON seagis.forecast_run USING btree
(dataset_key, run_key);
CREATE UNIQUE INDEX forecast_run_id_dataset_key_is_synthetic_key ON seagis.forecast_run USING btree
(id, dataset_key, is_synthetic);
CREATE UNIQUE INDEX forecast_run_pkey ON seagis.forecast_run USING btree (id);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis.forecast_run FOR EACH ROW EXECUTE
FUNCTION seagis.reject_mutation();
CREATE TRIGGER guard_run BEFORE INSERT ON seagis.forecast_run FOR EACH ROW EXECUTE FUNCTION
seagis.guard_run();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.4 seagis.forecast_value

Nilai parameter per run, grid dan valid time; tidak diduplikasi per plot.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (forecast_run_id, source_grid_id, parameter_code, valid_at); UNIQUE (id, dataset_key, is_synthetic)
Rows	UNVERIFIED
Writer	seagis_environment.persist; A0 fixtures
Reader	seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

seagis.forecast_value.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.id

seagis.forecast_value.forecast_run_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (forecast_run_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_run(id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi capture/vintage model regional
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A2 adapter batch → persist run_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.forecast_run_id

seagis.forecast_value.source_grid_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (source_grid_id, dataset_key, is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi sel regional dan layer
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A2 adapter batch → grid lookup by dataset/key/synthetic (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.source_grid_id

seagis.forecast_value.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code); FOREIGN KEY (forecast_run_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_run(id, dataset_key, is_synthetic); FOREIGN KEY (source_grid_id, dataset_key, is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A2 adapter batch → leased dataset (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.dataset_key

seagis.forecast_value.is_synthetic

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)

Atribut	Definisi
FK	FOREIGN KEY (forecast_run_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_run(id, dataset_key, is_synthetic); FOREIGN KEY (source_grid_id, dataset_key, is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic)
Arti bisnis / teknis	Penanda asal sintetis, terpisah dari QC
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	A2 adapter batch → false (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.is_synthetic

seagis.forecast_value.parameter_code

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Arti bisnis / teknis	Identitas parameter lingkungan
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A2 adapter batch → batch.values[].parameter (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.parameter_code

seagis.forecast_value.native_value

Atribut	Definisi
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Nilai fisik dalam satuan sumber
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	Mengikuti unit/native_unit dan dataset_parameter; NOT APPLICABLE
Example	1
Asal / transformasi	A2 adapter batch → batch.values[].value (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.native_value

seagis.forecast_value.native_unit

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Arti bisnis / teknis	Satuan asli sebelum normalisasi
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	registered seagis.dataset_parameter par → par.native_unit (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.native_unit

seagis.forecast_value.value

Atribut	Definisi
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Nilai normalisasi; null mengikuti value_status
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	Mengikuti unit/native_unit dan dataset_parameter; NOT APPLICABLE
Example	1
Asal / transformasi	A2 adapter batch → batch.values[].value (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.value

seagis.forecast_value.unit

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Arti bisnis / teknis	Satuan nilai normalisasi menurut registri
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku

Atribut	Definisi
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	registered seagis.dataset_parameter par → par.unit (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.unit

seagis.forecast_value.conversion_code

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Arti bisnis / teknis	Identitas metode konversi satuan
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	registered seagis.dataset_parameter par → par.conversion_code (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.conversion_code

seagis.forecast_value.value_status

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Ketersediaan angka, terpisah dari kualitas
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A2 adapter batch → v.value_status (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.value_status

seagis.forecast_value.detection_limit

Atribut	Definisi
Type / nullable	seagis.finite_float / true

Atribut	Definisi
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Batas deteksi pada satuan normalisasi bila relevan
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	Mengikuti unit/native_unit dan dataset_parameter; NOT APPLICABLE
Example	1
Asal / transformasi	A2 adapter batch → omitted -> SQL NULL (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.detection_limit

seagis.forecast_value.valid_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Waktu representatif nilai model
Valid values / checks	CHECK (seagis.valid_support(valid_at, temporal_support, interval_status, period_start_at, period_end_at))
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A2 adapter batch → v.valid_at::timestamptz (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.valid_at

seagis.forecast_value.ingested_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan data tersedia bagi sistem
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A2 adapter batch → persist t = clock_timestamp() (VERIFIED)

Atribut	Definisi
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.ingested_at

seagis.forecast_value.temporal_support

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Dukungan waktu instantaneous, mean atau sum
Valid values / checks	CHECK (seagis.valid_support(valid_at, temporal_support, interval_status, period_start_at, period_end_at))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	registered seagis.dataset_parameter par → par.temporal_support (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.temporal_support

seagis.forecast_value.interval_status

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	known, unknown atau not_applicable untuk interval
Valid values / checks	CHECK (seagis.valid_support(valid_at, temporal_support, interval_status, period_start_at, period_end_at))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A2 adapter batch → v.interval_status (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.interval_status

seagis.forecast_value.period_start_at

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Awal interval mean/sum jika diketahui

Atribut	Definisi
Valid values / checks	CHECK (seagis.valid_support(valid_at, temporal_support, interval_status, period_start_at, period_end_at))
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A2 adapter batch → v.period_start_at::timestamptz (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.period_start_at

seagis.forecast_value.period_end_at

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Akhir interval mean/sum jika diketahui
Valid values / checks	CHECK (seagis.valid_support(valid_at, temporal_support, interval_status, period_start_at, period_end_at))
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A2 adapter batch → v.period_end_at::timestamptz (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.period_end_at

seagis.forecast_value.quality_status

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Status QC unreviewed, accepted, suspect atau rejected; bukan diagnosis
Valid values / checks	CHECK (quality_status = ANY (ARRAY['unreviewed'::text, 'accepted'::text, 'suspect'::text, 'rejected'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A2 adapter batch → default unreviewed (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241; catalog seagis.forecast_value.quality_status

Constraint, indeks dan trigger

constraints

CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision));
CHECK (seagis.valid_support(valid_at, temporal_support, interval_status, period_start_at, period_end_at));
FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code);
FOREIGN KEY (forecast_run_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_run(id, dataset_key, is_synthetic);
UNIQUE (forecast_run_id, source_grid_id, parameter_code, valid_at);
UNIQUE (id, dataset_key, is_synthetic);
PRIMARY KEY (id);
CHECK (quality_status = ANY (ARRAY['unreviewed'::text, 'accepted'::text, 'suspect'::text, 'rejected'::text]));
FOREIGN KEY (source_grid_id, dataset_key, is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic);

indexes

CREATE INDEX forecast_grid_time ON seagis.forecast_value USING btree (source_grid_id, parameter_code, valid_at DESC);
CREATE UNIQUE INDEX forecast_value_forecast_run_id_source_grid_id_parameter_cod_key ON seagis.forecast_value USING btree (forecast_run_id, source_grid_id, parameter_code, valid_at);
CREATE UNIQUE INDEX forecast_value_id_dataset_key_is_synthetic_key ON seagis.forecast_value USING btree (id, dataset_key, is_synthetic);
CREATE UNIQUE INDEX forecast_value_pkey ON seagis.forecast_value USING btree (id);

triggers

CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis.forecast_value FOR EACH ROW EXECUTE FUNCTION seagis.reject_mutation();
CREATE TRIGGER guard_forecast_value BEFORE INSERT ON seagis.forecast_value FOR EACH ROW EXECUTE FUNCTION seagis.guard_measurement();

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.5 seagis.ingestion_run

Audit satu ingestion logis beserta status penyelesaian.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (dataset_key, request_key); UNIQUE (id, dataset_key, is_synthetic)
Rows	UNVERIFIED
Writer	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Reader	A3 source_bundle; provenance guards
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

seagis.ingestion_run.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel

Atribut	Definisi
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.id

seagis.ingestion_run.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191 → s.dataset_key (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.dataset_key

seagis.ingestion_run.request_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kunci logis retry ingestion per dataset
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191 → 'a1:'||s.id||':'||key (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.request_key

seagis.ingestion_run.adapter_version

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND

Atribut	Definisi
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Versi implementasi adapter ingestion
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191 → 'temperature-1.0.0' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.adapter_version

seagis.ingestion_run.started_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan dimulainya ingestion atau attempt
Valid values / checks	CHECK (completed_at IS NULL OR completed_at >= started_at)
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191 → received (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.started_at

seagis.ingestion_run.completed_at

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan selesai; null selama running
Valid values / checks	CHECK (completed_at IS NULL OR completed_at >= started_at); CHECK ((status = 'running'::text) = (completed_at IS NULL))
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	owning routine / caller → UPDATE at end of A1/A2: clock_timestamp(); NULL while running (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.completed_at

seagis.ingestion_run.status

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	running/succeeded/failed/partial beserta completed_at
Valid values / checks	CHECK ((status = 'running'::text) = (completed_at IS NULL)); CHECK (status = ANY (ARRAY['running'::text, ' succeeded'::text, 'partial'::text, 'failed'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191 → 'running' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.status

seagis.ingestion_run.is_synthetic

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Penanda asal sintetis, terpisah dari QC
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191 → s.is_synthetic (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.is_synthetic

seagis.ingestion_run.error_code

Atribut	Definisi
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kode kegagalan tersanitasi
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	owning routine / caller → A1/A2 successful rows omit -> NULL; failure codes tracked in environment.attempt (VERIFIED)

Atribut	Definisi
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94; catalog seagis.ingestion_run.error_code

Constraint, indeks dan trigger

constraints

```
CHECK (completed_at IS NULL OR completed_at >= started_at);
CHECK ((status = 'running'::text) = (completed_at IS NULL));
FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key);
UNIQUE (dataset_key, request_key);
UNIQUE (id, dataset_key, is_synthetic);
PRIMARY KEY (id);
CHECK (status = ANY (ARRAY['running'::text, 'succeeded'::text, 'partial'::text, 'failed'::text]));
```

indexes

```
CREATE UNIQUE INDEX ingestion_run_dataset_key_request_key_key ON seagis.ingestion_run USING btree
(dataset_key, request_key);
CREATE UNIQUE INDEX ingestion_run_id_dataset_key_is_synthetic_key ON seagis.ingestion_run USING
btree (id, dataset_key, is_synthetic);
CREATE UNIQUE INDEX ingestion_run_pkey ON seagis.ingestion_run USING btree (id);
```

triggers

```
CREATE TRIGGER guard_ingestion BEFORE INSERT OR UPDATE ON seagis.ingestion_run FOR EACH ROW EXECUTE
FUNCTION seagis.guard_ingestion();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.6 seagis.observation

Pembacaan suhu air sensor lokal, dengan nilai native, QC dan lineage raw.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (id, ingestion_run_id, dataset_key, is_synthetic); UNIQUE (sensor_id, parameter_code, source_record_key)
Rows	UNVERIFIED
Writer	seagis_api.ingest; A0 fixtures
Reader	seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

seagis.observation.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE

Atribut	Definisi
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.id

seagis.observation.plot_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (sensor_id, plot_id, dataset_key, is_synthetic) REFERENCES seagis.sensor(id, plot_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi plot budidaya
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A1 packet + registered identity + server → registered sensor.plot_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.plot_id

seagis.observation.sensor_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (sensor_id, plot_id, dataset_key, is_synthetic) REFERENCES seagis.sensor(id, plot_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi deployment sensor
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A1 packet + registered identity + server → authenticated credential.sensor_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.sensor_id

seagis.observation.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code); FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic); FOREIGN KEY (sensor_id, plot_id, dataset_key, is_synthetic) REFERENCES seagis.sensor(id, plot_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A1 packet + registered identity + server → registered sensor.dataset_key (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.dataset_key

seagis.observation.ingestion_run_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi ingestion asal
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A1 packet + registered identity + server → ingest-generated run_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.ingestion_run_id

seagis.observation.raw_payload_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi manifest raw asal
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE

Atribut	Definisi
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A1 packet + registered identity + server → ingest-generated raw_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.raw_payload_id

seagis.observation.is_synthetic

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic); FOREIGN KEY (sensor_id, plot_id, dataset_key, is_synthetic) REFERENCES seagis.sensor(id, plot_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Penanda asal sintetis, terpisah dari QC
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	A1 packet + registered identity + server → registered sensor.is_synthetic (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.is_synthetic

seagis.observation.source_record_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	boot_id:sequence untuk paket A1
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A1 packet + registered identity + server → boot_id || : || sequence (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.source_record_key

seagis.observation.parameter_code

Atribut	Definisi
Type / nullable	text / false

Atribut	Definisi
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Arti bisnis / teknis	Identitas parameter lingkungan
Valid values / checks	CHECK (parameter_code = 'water_temperature'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A1 packet + registered identity + server → constant water_temperature (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.parameter_code

seagis.observation.native_value

Atribut	Definisi
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Nilai fisik dalam satuan sumber
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	Mengikuti unit/native_unit dan dataset_parameter; NOT APPLICABLE
Example	1
Asal / transformasi	A1 packet + registered identity + server → packet.water_temperature (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.native_value

seagis.observation.native_unit

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Arti bisnis / teknis	Satuan asli sebelum normalisasi
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A1 packet + registered identity + server → constant degC (VERIFIED)

Atribut	Definisi
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.native_unit

seagis.observation.value

Atribut	Definisi
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Nilai normalisasi; null mengikuti value_status
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	Mengikuti unit/native_unit dan dataset_parameter; NOT APPLICABLE
Example	1
Asal / transformasi	A1 packet + registered identity + server → packet.water_temperature (identity) (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.value

seagis.observation.unit

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Arti bisnis / teknis	Satuan nilai normalisasi menurut registri
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A1 packet + registered identity + server → constant degC (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.unit

seagis.observation.conversion_code

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)

Atribut	Definisi
FK	FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code)
Arti bisnis / teknis	Identitas metode konversi satuan
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A1 packet + registered identity + server → constant identity (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.conversion_code

seagis.observation.value_status

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Ketersediaan angka, terpisah dari kualitas
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A1 packet + registered identity + server → packet.value_status (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.value_status

seagis.observation.detection_limit

Atribut	Definisi
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Batas deteksi pada satuan normalisasi bila relevan
Valid values / checks	CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status, detection_limit::double precision))
Units / time	Mengikuti unit/native_unit dan dataset_parameter; NOT APPLICABLE
Example	1
Asal / transformasi	A1 packet + registered identity + server → omitted -> SQL NULL (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.detection_limit

seagis.observation.observed_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan pengukuran sensor, bukan receipt
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A1 packet + registered identity + server → packet.observed_at cast timestamptz (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.observed_at

seagis.observation.ingested_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan data tersedia bagi sistem
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A1 packet + registered identity + server → server received = clock_timestamp() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.ingested_at

seagis.observation.quality_status

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Status QC unreviewed, accepted, suspect atau rejected; bukan diagnosis
Valid values / checks	CHECK (quality_status = ANY (ARRAY['unreviewed'::text, 'accepted'::text, 'suspect'::text, 'rejected'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid

Atribut	Definisi
Asal / transformasi	A1 packet + registered identity + server → pilot policy + flags (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184; catalog seagis.observation.quality_status

Constraint, indeks dan trigger

constraints

```
CHECK (seagis.valid_value(value::double precision, native_value::double precision, value_status,
detection_limit::double precision));
FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES
seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code);
UNIQUE (id, ingestion_run_id, dataset_key, is_synthetic);
CHECK (parameter_code = 'water_temperature'::text);
PRIMARY KEY (id);
CHECK (quality_status = ANY (ARRAY['unreviewed'::text, 'accepted'::text, 'suspect'::text,
'rejected'::text]));
FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES
seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic);
UNIQUE (sensor_id, parameter_code, source_record_key);
FOREIGN KEY (sensor_id, plot_id, dataset_key, is_synthetic) REFERENCES seagis.sensor(id, plot_id,
dataset_key, is_synthetic);
```

indexes

```
CREATE UNIQUE INDEX observation_id_ingestion_run_id_dataset_key_is_synthetic_key ON
seagis.observation USING btree (id, ingestion_run_id, dataset_key, is_synthetic);
CREATE INDEX observation_ingestion ON seagis.observation USING btree (ingestion_run_id);
CREATE UNIQUE INDEX observation_pkey ON seagis.observation USING btree (id);
CREATE INDEX observation_plot_time ON seagis.observation USING btree (plot_id, parameter_code,
observed_at DESC);
CREATE UNIQUE INDEX observation_sensor_id_parameter_code_source_record_key_key ON seagis.observation
 USING btree (sensor_id, parameter_code, source_record_key);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis.observation FOR EACH ROW EXECUTE
FUNCTION seagis.reject_mutation();
CREATE TRIGGER guard_observation BEFORE INSERT ON seagis.observation FOR EACH ROW EXECUTE FUNCTION
seagis.guard_measurement();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.7 seagis.parameter

Registri parameter dan satuan normalisasi.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (parameter_code)
Unique	UNIQUE (parameter_code, unit)
Rows	UNVERIFIED
Writer	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Reader	A1 ingest/history; A2 persist/preview; A3 source_bundle
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36

seagis.parameter.parameter_code

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas parameter lingkungan
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:518 → 'water_temperature' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36; catalog seagis.parameter.parameter_code

seagis.parameter.unit

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Satuan nilai normalisasi menurut registri
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:518 → 'degC' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36; catalog seagis.parameter.unit

seagis.parameter.description

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Penjelasan registri parameter
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:518 → 'Submerged local sensor temperature; distinct from air temperature' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36; catalog seagis.parameter.description

Constraint, indeks dan trigger

constraints

```
UNIQUE (parameter_code, unit);
PRIMARY KEY (parameter_code);
```

indexes

```
CREATE UNIQUE INDEX parameter_parameter_code_unit_key ON seagis.parameter USING btree
(parameter_code, unit);
CREATE UNIQUE INDEX parameter_pkey ON seagis.parameter USING btree (parameter_code);
```

triggers NOT APPLICABLE / NOT FOUND

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.8 seagis.plot

Identitas plot dan status survei; koordinat nullable bukan bukti lokasi lapangan.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (id, is_synthetic); UNIQUE (plot_code)
Rows	UNVERIFIED
Writer	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Reader	A1 ingest/history; A2 persist/preview; A3 source_bundle
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

seagis.plot.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.id

seagis.plot.plot_code

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kode plot unik yang dapat dibaca manusia
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/seed.sql:2 → 'plot-a' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.plot_code

seagis.plot.name

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Nama tampilan plot
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/seed.sql:2 → 'Plot A (synthetic)' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.name

seagis.plot.latitude

Atribut	Definisi
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Latitude WGS84 dalam derajat
Valid values / checks	CHECK ((latitude IS NULL) = (longitude IS NULL)); CHECK (location_status <> 'surveyed'::text OR latitude IS NOT NULL); CHECK (latitude::double precision >= '-90'::integer::double precision AND latitude::double precision <= 90::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Asal / transformasi	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)

Atribut	Definisi
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.latitude

seagis.plot.longitude

Atribut	Definisi
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Longitude WGS84 dalam derajat
Valid values / checks	CHECK ((latitude IS NULL) = (longitude IS NULL)); CHECK (longitude::double precision >= '-180'::integer::double precision AND longitude::double precision <= 180::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Asal / transformasi	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.longitude

seagis.plot.is_synthetic

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Penanda asal sintetis, terpisah dari QC
Valid values / checks	CHECK ((location_status = 'synthetic'::text) = is_synthetic)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	Seagis-Model/supabase/seed.sql:2 → true (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.is_synthetic

seagis.plot.location_status

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Status lokasi synthetic, not_surveyed atau surveyed

Atribut	Definisi
Valid values / checks	CHECK ((location_status = 'synthetic'::text) = is_synthetic); CHECK (location_status <> 'surveyed'::text OR latitude IS NOT NULL); CHECK (location_status = ANY (ARRAY['not_surveyed'::text, 'surveyed'::text, ' synthetic'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/seed.sql:2 → 'synthetic' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.location_status

seagis.plot.created_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	now()
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan pembuatan baris
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	caller / SQL default → now() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58; catalog seagis.plot.created_at

Constraint, indeks dan trigger

constraints

```
CHECK ((latitude IS NULL) = (longitude IS NULL));
CHECK ((location_status = 'synthetic'::text) = is_synthetic);
CHECK (location_status <> 'surveyed'::text OR latitude IS NOT NULL);
UNIQUE (id, is_synthetic);
CHECK (latitude::double precision >= '-90'::integer::double precision AND latitude::double precision
 <= 90::double precision);
CHECK (location_status = ANY (ARRAY['not_surveyed'::text, 'surveyed'::text, 'synthetic'::text]));
CHECK (longitude::double precision >= '-180'::integer::double precision AND longitude::double
precision <= 180::double precision);
PRIMARY KEY (id);
UNIQUE (plot_code);
```

indexes

```
CREATE UNIQUE INDEX plot_id_is_synthetic_key ON seagis.plot USING btree (id, is_synthetic);
CREATE UNIQUE INDEX plot_pkey ON seagis.plot USING btree (id);
CREATE UNIQUE INDEX plot_plot_code_key ON seagis.plot USING btree (plot_code);
```

triggers NOT APPLICABLE / NOT FOUND

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.9 seagis.plot_grid_mapping

Pemetaan operasional plot ke grid dengan waktu efektif dan waktu tercatat.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (plot_id, source_grid_id, effective_from)
Rows	UNVERIFIED
Writer	A0 fixture/admin SQL
Reader	A3 source_bundle; feature_input guards
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

seagis.plot_grid_mapping.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.id

seagis.plot_grid_mapping.plot_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Arti bisnis / teknis	Referensi plot budidaya
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied plot_id subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.plot_id

seagis.plot_grid_mapping.source_grid_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (source_grid_id) REFERENCES seagis.source_grid(id)
Arti bisnis / teknis	Referensi sel regional dan layer
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied source_grid_id subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.source_grid_id

seagis.plot_grid_mapping.is_synthetic

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Arti bisnis / teknis	Penanda asal sintetis, terpisah dari QC
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied is_synthetic subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.is_synthetic

seagis.plot_grid_mapping.sampling_method

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Metode pemilihan grid untuk plot
Valid values / checks	CHECK (sampling_method = ANY (ARRAY['nearest_wet_cell'::text, 'nearest_cell'::text, ' provider_returned_coordinate'::text, 'synthetic'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid

Atribut	Definisi
Asal / transformasi	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied sampling_method subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.sampling_method

seagis.plot_grid_mapping.distance_m

Atribut	Definisi
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Jarak plot ke sel dalam meter
Valid values / checks	CHECK (distance_m::double precision >= 0::double precision)
Units / time	m; NOT APPLICABLE
Example	1
Asal / transformasi	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied distance_m subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.distance_m

seagis.plot_grid_mapping.representativeness

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Status keterwakilan regional, bukan bukti representasi lokal
Valid values / checks	CHECK (representativeness = ANY (ARRAY['unverified'::text, 'regional_only'::text, 'land_mask_issue'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied representativeness subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.representativeness

seagis.plot_grid_mapping.effective_from

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND

Atribut	Definisi
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Awal inklusif rentang pemetaan
Valid values / checks	CHECK (effective_until IS NULL OR effective_until > effective_from)
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied effective_from subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.effective_from

seagis.plot_grid_mapping.effective_until

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Akhir eksklusif pemetaan; null berarti terbuka
Valid values / checks	CHECK (effective_until IS NULL OR effective_until > effective_from)
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A0 fixtures/test SQL; operational creator NOT FOUND → caller-supplied effective_until subject to FK/CHECK/guard_mapping; recorded_at SQL default (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.effective_until

seagis.plot_grid_mapping.recorded_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	clock_timestamp()
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan pemetaan pertama diketahui sistem
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	caller / SQL default → clock_timestamp() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; feature_input guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152; catalog seagis.plot_grid_mapping.recorded_at

Constraint, indeks dan trigger

constraints

```
CHECK (effective_until IS NULL OR effective_until > effective_from);
CHECK (distance_m::double precision >= 0::double precision);
PRIMARY KEY (id);
FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic);
UNIQUE (plot_id, source_grid_id, effective_from);
CHECK (representativeness = ANY (ARRAY['unverified'::text, 'regional_only'::text,
'land_mask_issue'::text]));
CHECK (sampling_method = ANY (ARRAY['nearest_wet_cell'::text, 'nearest_cell'::text,
'provider_returned_coordinate'::text, 'synthetic'::text]));
FOREIGN KEY (source_grid_id) REFERENCES seagis.source_grid(id);
```

indexes

```
CREATE INDEX mapping_grid_plot ON seagis.plot_grid_mapping USING btree (source_grid_id, plot_id);
CREATE UNIQUE INDEX plot_grid_mapping_pkey ON seagis.plot_grid_mapping USING btree (id);
CREATE UNIQUE INDEX plot_grid_mapping_plot_id_source_grid_id_effective_from_key ON
seagis.plot_grid_mapping USING btree (plot_id, source_grid_id, effective_from);
```

triggers

```
CREATE TRIGGER guard_mapping BEFORE INSERT OR UPDATE ON seagis.plot_grid_mapping FOR EACH ROW
EXECUTE FUNCTION seagis.guard_mapping();
CREATE TRIGGER immutable_alignment_mapping BEFORE DELETE OR UPDATE ON seagis.plot_grid_mapping FOR
EACH ROW EXECUTE FUNCTION seagis.reject_mutation();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.10 seagis.quality_event

Log QC append-only untuk ingestion atau satu measurement.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (ingestion_run_id, event_key)
Rows	UNVERIFIED
Writer	seagis_api.ingest; A0 fixtures
Reader	A3 source_bundle/alignment
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

seagis.quality_event.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku

Atribut	Definisi
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.id

seagis.quality_event.ingestion_run_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic); FOREIGN KEY (observation_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.observation(id, ingestion_run_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi ingestion asal
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → run_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.ingestion_run_id

seagis.quality_event.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (forecast_value_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_value(id, dataset_key, is_synthetic); FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic); FOREIGN KEY (observation_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.observation(id, ingestion_run_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → s.dataset_key (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.dataset_key

seagis.quality_event.is_synthetic

Atribut	Definisi
Type / nullable	boolean / false

Atribut	Definisi
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (forecast_value_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_value(id, dataset_key, is_synthetic); FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic); FOREIGN KEY (observation_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.observation(id, ingestion_run_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Penanda asal sintetis, terpisah dari QC
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → s.is_synthetic (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.is_synthetic

seagis.quality_event.event_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas event unik per ingestion
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → flag (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.event_key

seagis.quality_event.observation_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (observation_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.observation(id, ingestion_run_id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi pembacaan sensor lokal
Valid values / checks	CHECK (num_nonnulls(observation_id, forecast_value_id) <= 1)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → observation_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle/alignment

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.observation_id

seagis.quality_event.forecast_value_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (forecast_value_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_value(id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi satu nilai model regional
Valid values / checks	CHECK (num_nonnulls(observation_id, forecast_value_id) <= 1)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.forecast_value_id

seagis.quality_event.event_code

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kode jenis kejadian QC
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → flag (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.event_code

seagis.quality_event.severity

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Tingkat kejadian info, warning atau error; bukan risiko penyakit
Valid values / checks	CHECK (severity = ANY (ARRAY['info'::text, 'warning'::text, 'error'::text]))

Atribut	Definisi
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208 → CASE WHEN quality='rejected' THEN ' error' WHEN flag IN ('gap_before','out_of_order') THEN 'info' ELSE 'warning' END (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.severity

seagis.quality_event.details

Atribut	Definisi
Type / nullable	jsonb / false
Default / generated	'{}'::jsonb
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Rincian QC JSON tanpa rahasia
Valid values / checks	CHECK (jsonb_typeof(details) = 'object'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	{}
Asal / transformasi	caller / SQL default → '{}'::jsonb (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.details

seagis.quality_event.created_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	now()
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan pembuatan baris
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	caller / SQL default → now() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle/alignment
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273; catalog seagis.quality_event.created_at

Constraint, indeks dan trigger

constraints

CHECK (num_nonnulls(observation_id, forecast_value_id) <= 1);
CHECK (jsonb_typeof(details) = 'object'::text);
FOREIGN KEY (forecast_value_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_value(id,
dataset_key, is_synthetic);

FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic);
UNIQUE (ingestion_run_id, event_key);
FOREIGN KEY (observation_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.observation(id, ingestion_run_id, dataset_key, is_synthetic);
PRIMARY KEY (id);
CHECK (severity = ANY (ARRAY['info'::text, 'warning'::text, 'error'::text]));

indexes

CREATE INDEX quality_event_forecast ON seagis.quality_event USING btree (forecast_value_id) WHERE (forecast_value_id IS NOT NULL);
CREATE UNIQUE INDEX quality_event_ingestion_run_id_event_key_key ON seagis.quality_event USING btree (ingestion_run_id, event_key);
CREATE INDEX quality_event_observation ON seagis.quality_event USING btree (observation_id) WHERE (observation_id IS NOT NULL);
CREATE UNIQUE INDEX quality_event_pkey ON seagis.quality_event USING btree (id);

triggers

CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis.quality_event FOR EACH ROW EXECUTE FUNCTION seagis.reject_mutation();
CREATE TRIGGER guard_quality BEFORE INSERT ON seagis.quality_event FOR EACH ROW EXECUTE FUNCTION seagis.guard_quality_event();

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.11 seagis.sensor

Identitas deployment sensor suhu, plot, kedalaman dan status aktif.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (id, plot_id, dataset_key, is_synthetic); UNIQUE (sensor_code)
Rows	UNVERIFIED
Writer	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Reader	A1 ingest/history; A2 persist/preview; A3 source_bundle
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

seagis.sensor.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)

Atribut	Definisi
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.id

seagis.sensor.sensor_code

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kode deployment sensor unik
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/seed.sql:7 → 'simulator-a' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.sensor_code

seagis.sensor.plot_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Arti bisnis / teknis	Referensi plot budidaya
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/seed.sql:7 → '00000000-0000-4000-8000-000000000001' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.plot_id

seagis.sensor.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, parameter_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code); FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku

Atribut	Definisi
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/seed.sql:7 → 'sensor_temperature_v1' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.dataset_key

seagis.sensor.scope

Atribut	Definisi
Type / nullable	text / false
Default / generated	'local_sensor'::text
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope)
Arti bisnis / teknis	Asal local_sensor atau regional_model
Valid values / checks	CHECK (scope = 'local_sensor'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	caller / SQL default → 'local_sensor'::text (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.scope

seagis.sensor.parameter_code

Atribut	Definisi
Type / nullable	text / false
Default / generated	'water_temperature'::text
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, parameter_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code)
Arti bisnis / teknis	Identitas parameter lingkungan
Valid values / checks	CHECK (parameter_code = 'water_temperature'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	caller / SQL default → 'water_temperature'::text (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.parameter_code

seagis.sensor.depth_m

Atribut	Definisi
Type / nullable	seagis.finite_float / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)

Atribut	Definisi
FK	NOT APPLICABLE
Arti bisnis / teknis	Kedalaman positif ke bawah dalam meter
Valid values / checks	CHECK (depth_status <> 'unknown'::text OR depth_m IS NULL); CHECK (depth_status = 'unknown'::text OR depth_m IS NOT NULL); CHECK (depth_m::double precision >= 0::double precision)
Units / time	m; NOT APPLICABLE
Example	1
Asal / transformasi	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.depth_m

seagis.sensor.depth_status

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Status pengetahuan kedalaman sensor
Valid values / checks	CHECK (depth_status <> 'unknown'::text OR depth_m IS NULL); CHECK (depth_status = 'unknown'::text OR depth_m IS NOT NULL); CHECK (depth_status <> 'synthetic'::text OR is_synthetic); CHECK (depth_status = ANY (ARRAY['unknown'::text, 'measured'::text, 'synthetic'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/seed.sql:7 → 'unknown' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.depth_status

seagis.sensor.is_synthetic

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Arti bisnis / teknis	Penanda asal sintetis, terpisah dari QC
Valid values / checks	CHECK (depth_status <> 'synthetic'::text OR is_synthetic)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	Seagis-Model/supabase/seed.sql:7 → true (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.is_synthetic

seagis.sensor.active

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	true
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Apakah deployment sensor masih aktif
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	caller / SQL default → true (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73; catalog seagis.sensor.active

Constraint, indeks dan trigger

constraints

```
CHECK (depth_status <> 'unknown'::text OR depth_m IS NULL);
CHECK (depth_status = 'unknown'::text OR depth_m IS NOT NULL);
CHECK (depth_status <> 'synthetic'::text OR is_synthetic);
FOREIGN KEY (dataset_key, parameter_code) REFERENCES seagis.dataset_parameter(dataset_key,
parameter_code);
FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope);
CHECK (depth_m::double precision >= 0::double precision);
CHECK (depth_status = ANY (ARRAY['unknown'::text, 'measured'::text, 'synthetic'::text]));
UNIQUE (id, plot_id, dataset_key, is_synthetic);
CHECK (parameter_code = 'water_temperature'::text);
PRIMARY KEY (id);
FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic);
CHECK (scope = 'local_sensor'::text);
UNIQUE (sensor_code);
```

indexes

```
CREATE UNIQUE INDEX one_active_temperature_sensor_per_plot ON seagis.sensor USING btree (plot_id)
WHERE active;
CREATE UNIQUE INDEX sensor_id_plot_id_dataset_key_is_synthetic_key ON seagis.sensor USING btree (id,
 plot_id, dataset_key, is_synthetic);
CREATE UNIQUE INDEX sensor_pkey ON seagis.sensor USING btree (id);
CREATE UNIQUE INDEX sensor_sensor_code_key ON seagis.sensor USING btree (sensor_code);
```

triggers

```
CREATE TRIGGER guard_sensor_deployment BEFORE UPDATE ON seagis.sensor FOR EACH ROW EXECUTE FUNCTION
seagis.guard_sensor_deployment();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.12 seagis.source

Registri penyedia asal dataset.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (source_key)
Unique	NOT FOUND

Atribut	Definisi
Rows	UNVERIFIED
Writer	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Reader	A1 ingest/history; A2 persist/preview; A3 source_bundle
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11

seagis.source.source_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kunci penyedia sumber
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:506 → 'sensor' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11; catalog seagis.source.source_key

seagis.source.provider

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Nama penyedia dataset
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:506 → 'SEAGIS local sensor / simulator' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11; catalog seagis.source.provider

seagis.source.documentation_url

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND

Atribut	Definisi
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	URL dokumentasi penyedia
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0001_initial_schema.sql:506 → 'https://github.com/De-Voeux/Seagis-Model' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A1 ingest/history; A2 persist/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11; catalog seagis.source.documentation_url

Constraint, indeks dan trigger

constraints

```
PRIMARY KEY (source_key);
```

indexes

```
CREATE UNIQUE INDEX source_pkey ON seagis.source USING btree (source_key);
```

triggers NOT APPLICABLE / NOT FOUND

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.13 seagis.source_grid

Identitas sel regional dan layer kedalaman; bukan plot budidaya.

Atribut	Definisi
Schema / domain	seagis
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (dataset_key, grid_key, is_synthetic); UNIQUE (id, dataset_key, is_synthetic)
Rows	UNVERIFIED
Writer	seagis_environment.persist; A0 fixtures
Reader	seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

seagis.source_grid.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE

Atribut	Definisi
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.id

seagis.source_grid.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → s.dataset_key (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.dataset_key

seagis.source_grid.scope

Atribut	Definisi
Type / nullable	text / false
Default / generated	'regional_model'::text
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope)
Arti bisnis / teknis	Asal local_sensor atau regional_model
Valid values / checks	CHECK (scope = 'regional_model'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	caller / SQL default → 'regional_model'::text (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.scope

seagis.source_grid.grid_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kunci sel berbasis koordinat/depth atau fixture
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → g->>'key' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.grid_key

seagis.source_grid.grid_version

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Versi identitas grid
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → d.dataset_version (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.grid_version

seagis.source_grid.identity_basis

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Dasar identitas sel asli, koordinat respons atau sintetis
Valid values / checks	CHECK (identity_basis <> 'synthetic'::text OR is_synthetic); CHECK (identity_basis = ANY (ARRAY['provider_cell'::text, 'provider_returned_coordinate'::text, 'synthetic'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → g->>'identity_basis' (VERIFIED)

Atribut	Definisi
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.identity_basis

seagis.source_grid.latitude

Atribut	Definisi
Type / nullable	seagis.finite_float / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Latitude WGS84 dalam derajat
Valid values / checks	CHECK (latitude::double precision >= '-90'::integer::double precision AND latitude::double precision <= 90::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → (g->>'latitude')::float8 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.latitude

seagis.source_grid.longitude

Atribut	Definisi
Type / nullable	seagis.finite_float / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Longitude WGS84 dalam derajat
Valid values / checks	CHECK (longitude::double precision >= '-180'::integer::double precision AND longitude::double precision <= 180::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → (g->>'longitude')::float8 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.longitude

seagis.source_grid.depth_m

Atribut	Definisi
Type / nullable	seagis.finite_float / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)

Atribut	Definisi
FK	NOT APPLICABLE
Arti bisnis / teknis	Kedalaman positif ke bawah dalam meter
Valid values / checks	CHECK (depth_m::double precision >= 0::double precision)
Units / time	m; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → (g->>'depth_m')::float8 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.depth_m

seagis.source_grid.is_synthetic

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Penanda asal sintetis, terpisah dari QC
Valid values / checks	CHECK (identity_basis <> 'synthetic'::text OR is_synthetic)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173 → false (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135; catalog seagis.source_grid.is_synthetic

Constraint, indeks dan trigger

constraints

```
CHECK (identity_basis <> 'synthetic'::text OR is_synthetic);
UNIQUE (dataset_key, grid_key, is_synthetic);
FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope);
CHECK (depth_m::double precision >= 0::double precision);
UNIQUE (id, dataset_key, is_synthetic);
CHECK (identity_basis = ANY (ARRAY['provider_cell'::text, 'provider_returned_coordinate'::text,
'synthetic'::text]));
CHECK (latitude::double precision >= '-90'::integer::double precision AND latitude::double precision
 <= 90::double precision);
CHECK (longitude::double precision >= '-180'::integer::double precision AND longitude::double
precision <= 180::double precision);
PRIMARY KEY (id);
CHECK (scope = 'regional_model'::text);
```

indexes

```
CREATE UNIQUE INDEX source_grid_dataset_key_grid_key_is_synthetic_key ON seagis.source_grid USING
btree (dataset_key, grid_key, is_synthetic);
CREATE UNIQUE INDEX source_grid_id_dataset_key_is_synthetic_key ON seagis.source_grid USING btree
(id, dataset_key, is_synthetic);
CREATE UNIQUE INDEX source_grid_pkey ON seagis.source_grid USING btree (id);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis.source_grid FOR EACH ROW EXECUTE
FUNCTION seagis.reject_mutation();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.14 seagis_api.credential

Identitas aktor API berupa hash token, masa berlaku dan pencabutan.

Atribut	Definisi
Schema / domain	seagis_api
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (token_sha256)
Rows	UNVERIFIED
Writer	scripts/provision_a1.py; migration 0002
Reader	seagis_api.actor/rate_ok/ingest/history/status
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

seagis_api.credential.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.id

seagis_api.credential.label

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Label administratif credential
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid

Atribut	Definisi
Asal / transformasi	owning routine / caller → provision_a1.create(label) CLI --label (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.label

seagis_api.credential.token_sha256

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Hash token Bearer; token asli tidak disimpan di kolom
Valid values / checks	CHECK (token_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	owning routine / caller → sha256((sgi_ + secrets.token_urlsafe(32)).encode()).hexdigest(); original token excluded (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.token_sha256

seagis_api.credential.kind

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Jenis aktor device atau reader
Valid values / checks	CHECK ((kind = 'device'::text) = (sensor_id IS NOT NULL)); CHECK (kind = ANY (ARRAY['device'::text, 'reader'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	owning routine / caller → provision_a1.create(kind), CLI device/reader (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.kind

seagis_api.credential.sensor_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id)

Atribut	Definisi
Arti bisnis / teknis	Referensi deployment sensor
Valid values / checks	CHECK ((kind = 'device'::text) = (sensor_id IS NOT NULL))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	owning routine / caller → validated CLI sensor UUID for device, NULL for reader (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.sensor_id

seagis_api.credential.created_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	now()
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan pembuatan baris
Valid values / checks	CHECK (expires_at > created_at)
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	caller / SQL default → now() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.created_at

seagis_api.credential.expires_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan credential tidak berlaku lagi
Valid values / checks	CHECK (expires_at > created_at)
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	owning routine / caller → now()+make_interval(days=>days); CLI 1..365, default 30 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.expires_at

seagis_api.credential.revoked_at

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan pencabutan; null bila belum dicabut
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	owning routine / caller → NULL initially; revoke command UPDATE revoked_at=now() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27; catalog seagis_api.credential.revoked_at

Constraint, indeks dan trigger

constraints

```
CHECK ((kind = 'device'::text) = (sensor_id IS NOT NULL));
CHECK (expires_at > created_at);
CHECK (kind = ANY (ARRAY['device'::text, 'reader'::text]));
PRIMARY KEY (id);
FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id);
CHECK (token_sha256 ~ '^[0-9a-f]{64}$'::text);
UNIQUE (token_sha256);
```

indexes

```
CREATE UNIQUE INDEX credential_pkey ON seagis_api.credential USING btree (id);
CREATE UNIQUE INDEX credential_token_sha256_key ON seagis_api.credential USING btree (token_sha256);
```

triggers NOT APPLICABLE / NOT FOUND

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.15 seagis_api.packet_receipt

Kunci idempotensi paket sensor dan digest semantik.

Atribut	Definisi
Schema / domain	seagis_api
Kind	table
Primary key	PRIMARY KEY (sensor_id, source_record_key)
Unique	UNIQUE (observation_id)
Rows	UNVERIFIED
Writer	seagis_api.ingest/reply
Reader	seagis_api.actor/rate_ok/ingest/history/status
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

seagis_api.packet_receipt.sensor_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id)
Arti bisnis / teknis	Referensi deployment sensor
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215 → s.id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53; catalog seagis_api.packet_receipt.sensor_id

seagis_api.packet_receipt.source_record_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	boot_id:sequence untuk paket A1
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215 → key (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53; catalog seagis_api.packet_receipt.source_record_key

seagis_api.packet_receipt.semantic_sha256

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Hash paket setelah normalisasi semantik retry
Valid values / checks	CHECK (semantic_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215 → digest (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53; catalog seagis_api.packet_receipt.semantic_sha256

seagis_api.packet_receipt.observation_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / true (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (observation_id) REFERENCES seagis.observation(id)
Arti bisnis / teknis	Referensi pembacaan sensor lokal
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215 → observation_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53; catalog seagis_api.packet_receipt.observation_id

seagis_api.packet_receipt.ingestion_run_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (ingestion_run_id) REFERENCES seagis.ingestion_run(id)
Arti bisnis / teknis	Referensi ingestion asal
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215 → run_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53; catalog seagis_api.packet_receipt.ingestion_run_id

seagis_api.packet_receipt.flags

Atribut	Definisi
Type / nullable	text[] / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kode QC dari penerimaan paket
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku

Atribut	Definisi
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	[]
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215 → flags (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53; catalog seagis_api.packet_receipt.flags

Constraint, indeks dan trigger

constraints

```
FOREIGN KEY (ingestion_run_id) REFERENCES seagis.ingestion_run(id);
FOREIGN KEY (observation_id) REFERENCES seagis.observation(id);
UNIQUE (observation_id);
PRIMARY KEY (sensor_id, source_record_key);
CHECK (semantic_sha256 ~ '^[0-9a-f]{64}$'::text);
FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id);
```

indexes

```
CREATE UNIQUE INDEX packet_receipt_observation_id_key ON seagis_api.packet_receipt USING btree
(observation_id);
CREATE UNIQUE INDEX packet_receipt_pkey ON seagis_api.packet_receipt USING btree (sensor_id,
source_record_key);
```

triggers

```
CREATE TRIGGER receipt_append_only BEFORE DELETE OR UPDATE ON seagis_api.packet_receipt FOR EACH ROW
 EXECUTE FUNCTION seagis.reject_mutation();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.16 seagis_api.reader_plot

Daftar plot yang boleh dibaca satu credential reader.

Atribut	Definisi
Schema / domain	seagis_api
Kind	table
Primary key	PRIMARY KEY (credential_id, plot_id)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	scripts/provision_a1.py; migration 0002
Reader	seagis_api.actor/rate_ok/ingest/history/status
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39

seagis_api.reader_plot.credential_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / false (unique satu kolom; composite di bawah)

Atribut	Definisi
FK	FOREIGN KEY (credential_id) REFERENCES seagis_api.credential(id)
Arti bisnis / teknis	Referensi aktor API terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	owning routine / caller → provision_a1.create generated credential UUID (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39; catalog seagis_api.reader_plot.credential_id

seagis_api.reader_plot.plot_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (plot_id) REFERENCES seagis.plot(id)
Arti bisnis / teknis	Referensi plot budidaya
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	owning routine / caller → CLI plots validated with UUID(), FK checked by database (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39; catalog seagis_api.reader_plot.plot_id

Constraint, indeks dan trigger

constraints

```
FOREIGN KEY (credential_id) REFERENCES seagis_api.credential(id);
PRIMARY KEY (credential_id, plot_id);
FOREIGN KEY (plot_id) REFERENCES seagis.plot(id);
```

indexes

```
CREATE UNIQUE INDEX reader_plot_pkey ON seagis_api.reader_plot USING btree (credential_id, plot_id);
```

triggers NOT APPLICABLE / NOT FOUND

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.17 seagis_api.request_event

Audit respons API terautentikasi dan dasar rate limit.

Atribut	Definisi
Schema / domain	seagis_api
Kind	table
Primary key	PRIMARY KEY (id)
Unique	NOT FOUND

Atribut	Definisi
Rows	UNVERIFIED
Writer	seagis_api.ingest/reply
Reader	seagis_api.actor/rate_ok/ingest/history/status
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

seagis_api.request_event.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44; catalog seagis_api.request_event.id

seagis_api.request_event.credential_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (credential_id) REFERENCES seagis_api.credential(id)
Arti bisnis / teknis	Referensi aktor API terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100 → actor_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44; catalog seagis_api.request_event.credential_id

seagis_api.request_event.sensor_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND

Atribut	Definisi
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id)
Arti bisnis / teknis	Referensi deployment sensor
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100 → sensor (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44; catalog seagis_api.request_event.sensor_id

seagis_api.request_event.http_status

Atribut	Definisi
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Status HTTP respons yang diaudit
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100 → status (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44; catalog seagis_api.request_event.http_status

seagis_api.request_event.code

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kode respons API
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100 → code (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44; catalog seagis_api.request_event.code

seagis_api.request_event.created_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	clock_timestamp()
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan pembuatan baris
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	caller / SQL default → clock_timestamp() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44; catalog seagis_api.request_event.created_at

Constraint, indeks dan trigger

constraints

```
FOREIGN KEY (credential_id) REFERENCES seagis_api.credential(id);
PRIMARY KEY (id);
FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id);
```

indexes

```
CREATE INDEX request_event_actor_time ON seagis_api.request_event USING btree (credential_id,
created_at DESC);
CREATE UNIQUE INDEX request_event_pkey ON seagis_api.request_event USING btree (id);
```

triggers

```
CREATE TRIGGER request_event_append_only BEFORE DELETE OR UPDATE ON seagis_api.request_event FOR
EACH ROW EXECUTE FUNCTION seagis.reject_mutation();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.18 seagis_api.temperature_policy

Konfigurasi QC engineering pilot dan batas request; bukan ambang penyakit.

Atribut	Definisi
Schema / domain	seagis_api
Kind	table
Primary key	PRIMARY KEY (id)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	scripts/provision_a1.py; migration 0002
Reader	seagis_api.actor/rate_ok/ingest/history/status
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

seagis_api.temperature_policy.id

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	true
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Singleton boolean harus true; bukan UUID
Valid values / checks	CHECK (id)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	DB default / explicit UUID → true (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.id

seagis_api.temperature_policy.version

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas versi kebijakan
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → 'pilot-qc-1' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.version

seagis_api.temperature_policy.min_degc

Atribut	Definisi
Type / nullable	double precision / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Batas bawah envelope QC engineering pilot dalam degC
Valid values / checks	CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0 AND stale_after_seconds > 0 AND requests_per_minute > 0)
Units / time	degC; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → -5 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.min_degc

seagis_api.temperature_policy.max_degc

Atribut	Definisi
Type / nullable	double precision / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Batas atas envelope QC engineering pilot dalam degC
Valid values / checks	CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0 AND stale_after_seconds > 0 AND requests_per_minute > 0)
Units / time	degC; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → 60 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.max_degc

seagis_api.temperature_policy.max_step_degc

Atribut	Definisi
Type / nullable	double precision / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Batas perubahan suhu pilot dalam degC
Valid values / checks	CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0 AND stale_after_seconds > 0 AND requests_per_minute > 0)
Units / time	degC; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → 3 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.max_step_degc

seagis_api.temperature_policy.future_tolerance_seconds

Atribut	Definisi
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE

Atribut	Definisi
Arti bisnis / teknis	Toleransi jam sensor ke depan dalam detik
Valid values / checks	CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0 AND stale_after_seconds > 0 AND requests_per_minute > 0)
Units / time	s; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → 300 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.future_tolerance_seconds

seagis_api.temperature_policy.stale_after_seconds

Atribut	Definisi
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Batas keterlambatan sensor dalam detik
Valid values / checks	CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0 AND stale_after_seconds > 0 AND requests_per_minute > 0)
Units / time	s; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → 7200 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.stale_after_seconds

seagis_api.temperature_policy.requests_per_minute

Atribut	Definisi
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Batas request per aktor per jendela satu menit
Valid values / checks	CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0 AND stale_after_seconds > 0 AND requests_per_minute > 0)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80 → 120 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_api.actor/rate_ok/ingest/history/status
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67; catalog seagis_api.temperature_policy.requests_per_minute

Constraint, indeks dan trigger

constraints

```
CHECK (min_degc < max_degc AND max_step_degc > 0::double precision AND future_tolerance_seconds > 0
AND stale_after_seconds > 0 AND requests_per_minute > 0);
CHECK (id);
PRIMARY KEY (id);
```

indexes

```
CREATE UNIQUE INDEX temperature_policy_pkey ON seagis_api.temperature_policy USING btree (id);
```

triggers NOT APPLICABLE / NOT FOUND

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.19 seagis_environment.attempt

Audit satu upaya fetch termasuk duplicate dan abandoned.

Atribut	Definisi
Schema / domain	seagis_environment
Kind	table
Primary key	PRIMARY KEY (id)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	seagis_environment.claim/persist/finish_failure; migration 0003
Reader	A2 status/preview; A3 source_bundle
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

seagis_environment.attempt.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113 → lease (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.id

seagis_environment.attempt.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113 → k (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.dataset_key

seagis_environment.attempt.started_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan dimulainya ingestion atau attempt
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113 → clock_timestamp() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.started_at

seagis_environment.attempt.completed_at

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan selesai; null selama running
Valid values / checks	CHECK ((completed_at IS NULL) = (status = 'running'::text))
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	owning routine / caller → clock_timestamp() at persist/failure/abandon; initially NULL (VERIFIED)

Atribut	Definisi
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.completed_at

seagis_environment.attempt.status

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	running/succeeded/duplicate/failed/abandoned untuk fetch attempt
Valid values / checks	CHECK ((completed_at IS NULL) = (status = 'running'::text)); CHECK (status = ANY (ARRAY['running'::text, ' succeeded'::text, 'duplicate'::text, 'failed'::text, 'abandoned'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113 → 'running' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.status

seagis_environment.attempt.error_code

Atribut	Definisi
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kode kegagalan tersanitasi
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	owning routine / caller → finish_failure sanitized code or lease_expired on abandoned; otherwise NULL (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.error_code

seagis_environment.attempt.forecast_run_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)

Atribut	Definisi
FK	FOREIGN KEY (forecast_run_id) REFERENCES seagis.forecast_run(id)
Arti bisnis / teknis	Referensi capture/vintage model regional
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	owning routine / caller → persist run_id, also on duplicate; NULL before completion (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57; catalog seagis_environment.attempt.forecast_run_id

Constraint, indeks dan trigger

constraints

```
CHECK ((completed_at IS NULL) = (status = 'running'::text));
FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key);
FOREIGN KEY (forecast_run_id) REFERENCES seagis.forecast_run(id);
PRIMARY KEY (id);
CHECK (status = ANY (ARRAY['running'::text, 'succeeded'::text, 'duplicate'::text, 'failed'::text,
'abandoned'::text]));
```

indexes

```
CREATE UNIQUE INDEX attempt_pkey ON seagis_environment.attempt USING btree (id);
```

triggers NOT APPLICABLE / NOT FOUND

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.20 seagis_environment.current_speed

View magnitude arus dari pasangan komponen dengan run/grid/waktu yang sama.

Atribut	Definisi
Schema / domain	seagis_environment
Kind	view
Primary key	NOT APPLICABLE: view
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	NOT APPLICABLE: view
Reader	A2 status/preview; A3 source_bundle
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

seagis_environment.current_speed.eastward_value_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)

Atribut	Definisi
FK	NOT APPLICABLE
Arti bisnis / teknis	ID komponen current_eastward pada view
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	seagis.forecast_value u/v → u.id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.eastward_value_id

seagis_environment.current_speed.northward_value_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	ID komponen current_northward pada view
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	seagis.forecast_value u/v → v.id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.northward_value_id

seagis_environment.current_speed.forecast_run_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Referensi capture/vintage model regional
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	seagis.forecast_value u/v → u.forecast_run_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.forecast_run_id

seagis_environment.current_speed.source_grid_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Referensi sel regional dan layer
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	seagis.forecast_value u/v → u.source_grid_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.source_grid_id

seagis_environment.current_speed.valid_at

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Waktu representatif nilai model
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	seagis.forecast_value u/v → u.valid_at (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.valid_at

seagis_environment.current_speed.ingested_at

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan data tersedia bagi sistem
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	seagis.forecast_value u/v → u.ingested_at (VERIFIED)

Atribut	Definisi
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.ingested_at

seagis_environment.current_speed.available_at

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Waktu konservatif ketersediaan input
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	seagis.forecast_value u/v → greatest(u.ingested_at,v.ingested_at) (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.available_at

seagis_environment.current_speed.speed_m_s

Atribut	Definisi
Type / nullable	double precision / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	sqrt(uo²+vo²), null bila salah satu komponen tidak present
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	m/s; NOT APPLICABLE
Example	1
Asal / transformasi	seagis.forecast_value u/v → CASE WHEN both value_status='present' THEN sqrt(u.value*u.value+v.value*v.value) END (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.speed_m_s

seagis_environment.current_speed.derivation

Atribut	Definisi
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)

Atribut	Definisi
FK	NOT APPLICABLE
Arti bisnis / teknis	Label formula dan versi magnitude arus
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	seagis.forecast_value u/v → 'sqrt(uo^2+vo^2); a2-vector-magnitude-v1' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89; catalog seagis_environment.current_speed.derivation

Constraint, indeks dan trigger

constraints NOT APPLICABLE / NOT FOUND

indexes NOT APPLICABLE / NOT FOUND

triggers NOT APPLICABLE / NOT FOUND

```
SELECT u.id AS eastward_value_id,
    v.id AS northward_value_id,
    u.forecast_run_id,
    u.source_grid_id,
    u.valid_at,
    u.ingested_at,
    GREATEST(u.ingested_at, v.ingested_at) AS available_at,
        CASE
            WHEN ((u.value_status = 'present'::text) AND (v.value_status = 'present'::text)) THEN
sqrt((((u.value)::double precision * (u.value)::double precision) + ((v.value)::double precision *
(v.value)::double precision)))
            ELSE NULL::double precision
        END AS speed_m_s,
    'sqrt(uo^2+vo^2); a2-vector-magnitude-v1'::text AS derivation
   FROM (seagis.forecast_value u
     JOIN seagis.forecast_value v ON (((u.forecast_run_id = v.forecast_run_id) AND (u.source_grid_id
 = v.source_grid_id) AND (u.valid_at = v.valid_at))))
  WHERE ((u.parameter_code = 'current_eastward'::text) AND (v.parameter_code =
'current_northward'::text));
```

4.21 seagis_environment.demo_grid_mapping

Bridge khusus dua plot fiktif ke grid nyata untuk preview, bukan otorisasi training.

Atribut	Definisi
Schema / domain	seagis_environment
Kind	table
Primary key	PRIMARY KEY (plot_id, source_grid_id)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	seagis_environment.claim/persist/finish_failure; migration 0003
Reader	A2 status/preview; A3 source_bundle
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

seagis_environment.demo_grid_mapping.plot_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (plot_id, plot_is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Arti bisnis / teknis	Referensi plot budidaya
Valid values / checks	CHECK (plot_id = ANY (ARRAY['00000000-0000-4000-8000-000000000001'::uuid, '00000000-0000-4000-8000-000000000002'::uuid]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → (m->>'plot_id')::uuid (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.plot_id

seagis_environment.demo_grid_mapping.plot_is_synthetic

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	true
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (plot_id, plot_is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Arti bisnis / teknis	Harus true untuk bridge demo
Valid values / checks	CHECK (plot_is_synthetic)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	caller / SQL default → true (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.plot_is_synthetic

seagis_environment.demo_grid_mapping.source_grid_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (source_grid_id, dataset_key, grid_is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi sel regional dan layer
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → grid_id (VERIFIED)

Atribut	Definisi
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.source_grid_id

seagis_environment.demo_grid_mapping.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key); FOREIGN KEY (source_grid_id, dataset_key, grid_is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → s.dataset_key (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.dataset_key

seagis_environment.demo_grid_mapping.grid_is_synthetic

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	false
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (source_grid_id, dataset_key, grid_is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic)
Arti bisnis / teknis	Harus false untuk grid nyata di bridge demo
Valid values / checks	CHECK (NOT grid_is_synthetic)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	caller / SQL default → false (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.grid_is_synthetic

seagis_environment.demo_grid_mapping.example_latitude

Atribut	Definisi
Type / nullable	seagis.finite_float / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)

Atribut	Definisi
FK	NOT APPLICABLE
Arti bisnis / teknis	Latitude contoh fiktif, bukan survei
Valid values / checks	CHECK (example_latitude::double precision >= '-90'::integer::double precision AND example_latitude::double precision <= 90::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → (m->>'latitude')::float8 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.example_latitude

seagis_environment.demo_grid_mapping.example_longitude

Atribut	Definisi
Type / nullable	seagis.finite_float / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Longitude contoh fiktif, bukan survei
Valid values / checks	CHECK (example_longitude::double precision >= '-180'::integer::double precision AND example_longitude::double precision <= 180::double precision)
Units / time	degree; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → (m->>'longitude')::float8 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.example_longitude

seagis_environment.demo_grid_mapping.coordinate_status

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	example_not_surveyed pada bridge demo
Valid values / checks	CHECK (coordinate_status = 'example_not_surveyed'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → m->>'coordinate_status' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.coordinate_status

seagis_environment.demo_grid_mapping.distance_m

Atribut	Definisi
Type / nullable	seagis.finite_float / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Jarak plot ke sel dalam meter
Valid values / checks	CHECK (distance_m::double precision >= 0::double precision AND distance_m::double precision <= 30000::double precision)
Units / time	m; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → (m->>'distance_m')::float8 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.distance_m

seagis_environment.demo_grid_mapping.sampling_method

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Metode pemilihan grid untuk plot
Valid values / checks	CHECK (sampling_method = ANY (ARRAY['nearest_cell'::text, 'provider_returned_coordinate'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197 → m->>'sampling_method' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.sampling_method

seagis_environment.demo_grid_mapping.created_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	now()
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan pembuatan baris
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z

Atribut	Definisi
Asal / transformasi	caller / SQL default → now() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69; catalog seagis_environment.demo_grid_mapping.created_at

Constraint, indeks dan trigger

constraints

```
CHECK (coordinate_status = 'example_not_surveyed'::text);
FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key);
CHECK (distance_m::double precision >= 0::double precision AND distance_m::double precision <=
30000::double precision);
CHECK (example_latitude::double precision >= '-90'::integer::double precision AND
example_latitude::double precision <= 90::double precision);
CHECK (example_longitude::double precision >= '-180'::integer::double precision AND
example_longitude::double precision <= 180::double precision);
CHECK (NOT grid_is_synthetic);
PRIMARY KEY (plot_id, source_grid_id);
CHECK (plot_id = ANY (ARRAY['00000000-0000-4000-8000-000000000001'::uuid,
'00000000-0000-4000-8000-000000000002'::uuid]));
FOREIGN KEY (plot_id, plot_is_synthetic) REFERENCES seagis.plot(id, is_synthetic);
CHECK (plot_is_synthetic);
CHECK (sampling_method = ANY (ARRAY['nearest_cell'::text, 'provider_returned_coordinate'::text]));
FOREIGN KEY (source_grid_id, dataset_key, grid_is_synthetic) REFERENCES seagis.source_grid(id,
dataset_key, is_synthetic);
```

indexes

```
CREATE UNIQUE INDEX demo_grid_mapping_pkey ON seagis_environment.demo_grid_mapping USING btree
(plot_id, source_grid_id);
```

triggers

```
CREATE TRIGGER immutable_demo_mapping BEFORE DELETE OR UPDATE ON
seagis_environment.demo_grid_mapping FOR EACH ROW EXECUTE FUNCTION seagis.reject_mutation();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.22 seagis_environment.policy

Kebijakan fetch, lisensi, cadence scheduler, staleness dan retensi regional.

Atribut	Definisi
Schema / domain	seagis_environment
Kind	table
Primary key	PRIMARY KEY (dataset_key)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	seagis_environment.claim/persist/finish_failure; migration 0003
Reader	A2 status/preview; A3 source_bundle
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

seagis_environment.policy.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → dataset_key (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.dataset_key

seagis_environment.policy.enabled

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	true
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Apakah policy regional mengizinkan fetch
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	caller / SQL default → true (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.enabled

seagis_environment.policy.purpose

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Tujuan noncommercial_educational_prototype
Valid values / checks	CHECK (purpose = 'noncommercial_educational_prototype'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → ' noncommercial_educational_prototype' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.purpose

seagis_environment.policy.licence_url

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Referensi lisensi sumber
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → licence_url (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.licence_url

seagis_environment.policy.terms_review

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas review ketentuan provider
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → CASE WHEN source_key='copernicus' THEN 'cmems-licence-20261022-sections-2.2-2.4-2.6' ELSE ' open-meteo-terms-20261022-noncommercial-cc-by-4.0' END (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.terms_review

seagis_environment.policy.attribution

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE

Atribut	Definisi
Arti bisnis / teknis	Teks atribusi yang harus dipertahankan
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → attribution (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.attribution

seagis_environment.policy.update_seconds

Atribut	Definisi
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Interval nominal scheduler per provider dalam detik
Valid values / checks	CHECK (release_offset_seconds >= 0 AND release_offset_seconds < update_seconds); CHECK (stale_seconds >= update_seconds); CHECK (update_seconds >= 21600)
Units / time	s; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → CASE WHEN source_key='copernicus' THEN 86400 ELSE 21600 END (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.update_seconds

seagis_environment.policy.release_offset_seconds

Atribut	Definisi
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Offset fase jadwal dalam detik
Valid values / checks	CHECK (release_offset_seconds >= 0 AND release_offset_seconds < update_seconds)
Units / time	s; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → CASE WHEN dataset_key='cmems_nutrients_daily' THEN 45000 WHEN source_key='copernicus' THEN 30600 ELSE 16200 END (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.release_offset_seconds

seagis_environment.policy.stale_seconds

Atribut	Definisi
Type / nullable	integer / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Usia vintage maksimum menurut policy dalam detik
Valid values / checks	CHECK (stale_seconds >= update_seconds)
Units / time	s; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30 → CASE WHEN source_key='copernicus' THEN 172800 ELSE 43200 END (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.stale_seconds

seagis_environment.policy.retention_mode

Atribut	Definisi
Type / nullable	text / false
Default / generated	'metadata_only'::text
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kebijakan metadata_only pada A2
Valid values / checks	CHECK (retention_mode = 'metadata_only'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	caller / SQL default → 'metadata_only'::text (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.retention_mode

seagis_environment.policy.reviewed_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	now()
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan review kebijakan
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z

Atribut	Definisi
Asal / transformasi	caller / SQL default → now() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16; catalog seagis_environment.policy.reviewed_at

Constraint, indeks dan trigger

constraints

```
CHECK (release_offset_seconds >= 0 AND release_offset_seconds < update_seconds);
CHECK (stale_seconds >= update_seconds);
FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key);
PRIMARY KEY (dataset_key);
CHECK (purpose = 'noncommercial_educational_prototype'::text);
CHECK (retention_mode = 'metadata_only'::text);
CHECK (update_seconds >= 21600);
```

indexes

```
CREATE UNIQUE INDEX policy_pkey ON seagis_environment.policy USING btree (dataset_key);
```

triggers NOT APPLICABLE / NOT FOUND

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.23 seagis_environment.schedule_state

State mutable lease, due time, backoff dan keberhasilan terakhir.

Atribut	Definisi
Schema / domain	seagis_environment
Kind	table
Primary key	PRIMARY KEY (dataset_key)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	seagis_environment.claim/persist/finish_failure; migration 0003
Reader	A2 status/preview; A3 source_bundle
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

seagis_environment.schedule_state.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE

Atribut	Definisi
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:56 → dataset_key (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.dataset_key

seagis_environment.schedule_state.lease_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas lease fetch aktif
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	owning routine / caller → claim gen_random_uuid(); cleared after persist/failure (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.lease_id

seagis_environment.schedule_state.lease_until

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan lease tidak lagi valid
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	owning routine / caller → claim clock_timestamp()+interval 20 minutes; cleared after persist/failure (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.lease_until

seagis_environment.schedule_state.next_attempt_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	'-infinity'::timestamp with time zone

Atribut	Definisi
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Waktu fetch paling awal; dapat bernilai -infinity pada awal
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	caller / SQL default → '-infinity'::timestamp with time zone (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.next_attempt_at

seagis_environment.schedule_state.last_success_at

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan fetch/persist terakhir berhasil
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	owning routine / caller → persist t=clock_timestamp() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.last_success_at

seagis_environment.schedule_state.last_attempt_at

Atribut	Definisi
Type / nullable	timestamp with time zone / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan attempt terakhir dimulai
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	owning routine / caller → claim clock_timestamp() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.last_attempt_at

seagis_environment.schedule_state.last_error_code

Atribut	Definisi
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Kode kegagalan attempt terakhir
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	owning routine / caller → finish_failure sanitized code; cleared on success (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.last_error_code

seagis_environment.schedule_state.failures

Atribut	Definisi
Type / nullable	integer / false
Default / generated	0
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Jumlah kegagalan untuk perhitungan backoff
Valid values / checks	CHECK (failures >= 0)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	1
Asal / transformasi	caller / SQL default → 0 (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.failures

seagis_environment.schedule_state.last_forecast_run_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (last_forecast_run_id) REFERENCES seagis.forecast_run(id)
Arti bisnis / teknis	Referensi vintage terakhir berhasil
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku

Atribut	Definisi
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	owning routine / caller → persist run_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A2 status/preview; A3 source_bundle
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45; catalog seagis_environment.schedule_state.last_forecast_run_id

Constraint, indeks dan trigger

constraints

```
FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key);
CHECK (failures >= 0);
FOREIGN KEY (last_forecast_run_id) REFERENCES seagis.forecast_run(id);
PRIMARY KEY (dataset_key);
```

indexes

```
CREATE UNIQUE INDEX schedule_state_pkey ON seagis_environment.schedule_state USING btree
(dataset_key);
```

triggers NOT APPLICABLE / NOT FOUND

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.24 seagis_features.feature_input

Lineage FK input feature snapshot dan guard cutoff.

Atribut	Definisi
Schema / domain	seagis_features
Kind	table
Primary key	PRIMARY KEY (id)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	A0 fixture/test inserts; production builder NOT FOUND
Reader	snapshot/input guards; production ML consumer NOT FOUND
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

seagis_features.feature_input.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku

Atribut	Definisi
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320; catalog seagis_features.feature_input.id

seagis_features.feature_input.snapshot_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (snapshot_id) REFERENCES seagis_features.feature_snapshot(id)
Arti bisnis / teknis	Referensi snapshot induk
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A0 fixtures/tests; production builder NOT FOUND → caller-supplied snapshot_id subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320; catalog seagis_features.feature_input.snapshot_id

seagis_features.feature_input.observation_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (observation_id) REFERENCES seagis.observation(id)
Arti bisnis / teknis	Referensi pembacaan sensor lokal
Valid values / checks	CHECK (num_nonnulls(observation_id, forecast_value_id) = 1)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A0 fixtures/tests; production builder NOT FOUND → caller-supplied observation_id subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320; catalog seagis_features.feature_input.observation_id

seagis_features.feature_input.forecast_value_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (forecast_value_id) REFERENCES seagis.forecast_value(id)
Arti bisnis / teknis	Referensi satu nilai model regional
Valid values / checks	CHECK (num_nonnulls(observation_id, forecast_value_id) = 1)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A0 fixtures/tests; production builder NOT FOUND → caller-supplied forecast_value_id subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320; catalog seagis_features.feature_input.forecast_value_id

Constraint, indeks dan trigger

constraints

```
CHECK (num_nonnulls(observation_id, forecast_value_id) = 1);
FOREIGN KEY (forecast_value_id) REFERENCES seagis.forecast_value(id);
FOREIGN KEY (observation_id) REFERENCES seagis.observation(id);
PRIMARY KEY (id);
FOREIGN KEY (snapshot_id) REFERENCES seagis_features.feature_snapshot(id);
```

indexes

```
CREATE UNIQUE INDEX feature_input_forecast ON seagis_features.feature_input USING btree
(snapshot_id, forecast_value_id) WHERE (forecast_value_id IS NOT NULL);
CREATE UNIQUE INDEX feature_input_observation ON seagis_features.feature_input USING btree
(snapshot_id, observation_id) WHERE (observation_id IS NOT NULL);
CREATE UNIQUE INDEX feature_input_pkey ON seagis_features.feature_input USING btree (id);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis_features.feature_input FOR EACH ROW
EXECUTE FUNCTION seagis.reject_mutation();
CREATE TRIGGER guard_feature_input BEFORE INSERT ON seagis_features.feature_input FOR EACH ROW
EXECUTE FUNCTION seagis_features.guard_input();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.25 seagis_features.feature_snapshot

Container fitur draft/sealed dengan cutoff dan hash; builder produksi belum ditemukan.

Atribut	Definisi
Schema / domain	seagis_features
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (plot_id, as_of, feature_version, input_manifest_sha256)
Rows	UNVERIFIED
Writer	A0 fixture/test inserts; production builder NOT FOUND
Reader	snapshot/input guards; production ML consumer NOT FOUND
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri

Atribut	Definisi
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

seagis_features.feature_snapshot.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.id

seagis_features.feature_snapshot.plot_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic)
Arti bisnis / teknis	Referensi plot budidaya
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A0 fixtures/tests; production builder NOT FOUND → caller-supplied plot_id subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.plot_id

seagis_features.feature_snapshot.is_synthetic

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic)

Atribut	Definisi
Arti bisnis / teknis	Penanda asal sintetis, terpisah dari QC
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	A0 fixtures/tests; production builder NOT FOUND → caller-supplied is_synthetic subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.is_synthetic

seagis_features.feature_snapshot.as_of

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Cutoff informasi untuk feature snapshot
Valid values / checks	CHECK (generated_at >= as_of)
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A0 fixtures/tests; production builder NOT FOUND → caller-supplied as_of subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.as_of

seagis_features.feature_snapshot.generated_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan snapshot fitur dihasilkan
Valid values / checks	CHECK (generated_at >= as_of)
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A0 fixtures/tests; production builder NOT FOUND → caller-supplied generated_at subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.generated_at

seagis_features.feature_snapshot.feature_version

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Versi definisi/builder fitur
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A0 fixtures/tests; production builder NOT FOUND → caller-supplied feature_version subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.feature_version

seagis_features.feature_snapshot.feature_schema_version

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Versi struktur JSON fitur
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A0 fixtures/tests; production builder NOT FOUND → caller-supplied feature_schema_version subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.feature_schema_version

seagis_features.feature_snapshot.features

Atribut	Definisi
Type / nullable	jsonb / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)

Atribut	Definisi
FK	NOT APPLICABLE
Arti bisnis / teknis	Objek JSON fitur; validitas perhitungan tidak dibuktikan container
Valid values / checks	CHECK (jsonb_typeof(features) = 'object'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	{}
Asal / transformasi	A0 fixtures/tests; production builder NOT FOUND → caller-supplied features subject to guard_input/guard_snapshot; no production derivation claimed (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.features

seagis_features.feature_snapshot.missing_context

Atribut	Definisi
Type / nullable	text[] / false
Default / generated	'{}'::text[]
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Daftar konteks yang tidak tersedia
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	[]
Asal / transformasi	caller / SQL default → '{}'::text[] (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.missing_context

seagis_features.feature_snapshot.status

Atribut	Definisi
Type / nullable	text / false
Default / generated	'draft'::text
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	draft sebelum sealing, sealed setelah manifest cocok
Valid values / checks	CHECK ((status = 'draft'::text) = (input_manifest_sha256 IS NULL)); CHECK (status = ANY (ARRAY['draft'::text, 'sealed'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	caller / SQL default → 'draft'::text (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.status

seagis_features.feature_snapshot.input_manifest_sha256

Atribut	Definisi
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Digest daftar input lineage terurut
Valid values / checks	CHECK ((status = 'draft'::text) = (input_manifest_sha256 IS NULL)); CHECK (input_manifest_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	seagis_features.manifest_hash(snapshot) → SHA256 sorted input references at draft -> sealed (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.input_manifest_sha256

seagis_features.feature_snapshot.features_sha256

Atribut	Definisi
Type / nullable	text / true
Default / generated	seagis.jsonb_sha256(features)
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Digest JSONB fitur menurut PostgreSQL
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	DB generated → seagis.jsonb_sha256(features) (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; snapshot/input guards; production ML consumer NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301; catalog seagis_features.feature_snapshot.features_sha256

Constraint, indeks dan trigger

constraints

```
CHECK (generated_at >= as_of);
CHECK ((status = 'draft'::text) = (input_manifest_sha256 IS NULL));
CHECK (jsonb_typeof(features) = 'object'::text);
CHECK (input_manifest_sha256 ~ '^[0-9a-f]{64}$'::text);
PRIMARY KEY (id);
UNIQUE (plot_id, as_of, feature_version, input_manifest_sha256);
FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic);
CHECK (status = ANY (ARRAY['draft'::text, 'sealed'::text]));
```

indexes

```
CREATE UNIQUE INDEX feature_snapshot_pkey ON seagis_features.feature_snapshot USING btree (id);
```

```
CREATE UNIQUE INDEX feature_snapshot_plot_id_as_of_feature_version_input_manife_key ON
seagis_features.feature_snapshot USING btree (plot_id, as_of, feature_version,
input_manifest_sha256);
```

triggers

```
CREATE TRIGGER guard_snapshot BEFORE INSERT OR UPDATE ON seagis_features.feature_snapshot FOR EACH
ROW EXECUTE FUNCTION seagis_features.guard_snapshot();
CREATE TRIGGER no_snapshot_delete BEFORE DELETE ON seagis_features.feature_snapshot FOR EACH ROW
EXECUTE FUNCTION seagis.reject_mutation();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.26 seagis_quality.alignment_snapshot

Dokumen QC/alignment hourly immutable; selalu training_authorized=false.

Atribut	Definisi
Schema / domain	seagis_quality
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (plot_id, mode, start_at, end_at, policy_version, source_sha256)
Rows	UNVERIFIED
Writer	seagis_quality.save_snapshot
Reader	seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

seagis_quality.alignment_snapshot.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.id

seagis_quality.alignment_snapshot.plot_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND

Atribut	Definisi
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (plot_id) REFERENCES seagis.plot(id)
Arti bisnis / teknis	Referensi plot budidaya
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A3 build -> save_snapshot → document.plot_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.plot_id

seagis_quality.alignment_snapshot.mode

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	demo_review atau operational_review sesuai asal plot
Valid values / checks	CHECK (mode = ANY (ARRAY['demo_review'::text, 'operational_review'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A3 build -> save_snapshot → document.mode (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.mode

seagis_quality.alignment_snapshot.start_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Awal inklusif jendela alignment
Valid values / checks	CHECK (end_at > start_at AND (end_at - start_at) <= '7 days'::interval)
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A3 build -> save_snapshot → document.start_at (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.start_at

seagis_quality.alignment_snapshot.end_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Akhir eksklusif jendela alignment
Valid values / checks	CHECK (end_at > start_at AND (end_at - start_at) <= '7 days'::interval)
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A3 build -> save_snapshot → document.end_at (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.end_at

seagis_quality.alignment_snapshot.policy_version

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (policy_version) REFERENCES seagis_quality.policy(version)
Arti bisnis / teknis	Referensi versi policy QC
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A3 build -> save_snapshot → registered policy.version (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.policy_version

seagis_quality.alignment_snapshot.source_sha256

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE

Atribut	Definisi
Arti bisnis / teknis	Digest source bundle untuk deteksi sumber berubah
Valid values / checks	CHECK (source_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A3 build -> save_snapshot → document.source_sha256 checked against fresh source_bundle (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.source_sha256

seagis_quality.alignment_snapshot.input_manifest_sha256

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Digest daftar input lineage terurut
Valid values / checks	CHECK (input_manifest_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A3 build -> save_snapshot → SHA256 sorted distinct at|source_ref lines (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.input_manifest_sha256

seagis_quality.alignment_snapshot.body

Atribut	Definisi
Type / nullable	jsonb / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Dokumen alignment lengkap, termasuk rows dan laporan QC
Valid values / checks	CHECK (jsonb_typeof(body) = 'object'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	{}
Asal / transformasi	A3 build -> save_snapshot → document (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.body

seagis_quality.alignment_snapshot.body_sha256

Atribut	Definisi
Type / nullable	text / true
Default / generated	seagis.jsonb_sha256(body)
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Digest generated atas JSONB body
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	DB generated → seagis.jsonb_sha256(body) (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.body_sha256

seagis_quality.alignment_snapshot.created_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	clock_timestamp()
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan pembuatan baris
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A3 build -> save_snapshot → SQL default clock_timestamp() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21; catalog seagis_quality.alignment_snapshot.created_at

Constraint, indeks dan trigger

constraints

```
CHECK (jsonb_typeof(body) = 'object'::text);
CHECK (end_at > start_at AND (end_at - start_at) <= '7 days'::interval);
CHECK (input_manifest_sha256 ~ '^[0-9a-f]{64}$'::text);
CHECK (mode = ANY (ARRAY['demo_review'::text, 'operational_review'::text]));
PRIMARY KEY (id);
FOREIGN KEY (plot_id) REFERENCES seagis.plot(id);
UNIQUE (plot_id, mode, start_at, end_at, policy_version, source_sha256);
FOREIGN KEY (policy_version) REFERENCES seagis_quality.policy(version);
CHECK (source_sha256 ~ '^[0-9a-f]{64}$'::text);
```

indexes

```
CREATE UNIQUE INDEX alignment_snapshot_pkey ON seagis_quality.alignment_snapshot USING btree (id);
CREATE UNIQUE INDEX alignment_snapshot_plot_id_mode_start_at_end_at_policy_vers_key ON
seagis_quality.alignment_snapshot USING btree (plot_id, mode, start_at, end_at, policy_version,
```

source_sha256);

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis_quality.alignment_snapshot FOR EACH ROW
 EXECUTE FUNCTION seagis.reject_mutation();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.27 seagis_quality.policy

Registri versi kebijakan QC dan hash kode builder yang immutable.

Atribut	Definisi
Schema / domain	seagis_quality
Kind	table
Primary key	PRIMARY KEY (version)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	scripts/quality/store.py::register_policy
Reader	seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

seagis_quality.policy.version

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas versi kebijakan
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	owning routine / caller → register_policy policy[version] (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14; catalog seagis_quality.policy.version

seagis_quality.policy.definition

Atribut	Definisi
Type / nullable	jsonb / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE

Atribut	Definisi
Arti bisnis / teknis	Dokumen kebijakan QC yang diregistrasi
Valid values / checks	CHECK (jsonb_typeof(definition) = 'object'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	{}
Asal / transformasi	owning routine / caller → register_policy JSON policy argument (config/a3_quality.json) (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14; catalog seagis_quality.policy.definition

seagis_quality.policy.policy_sha256

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Digest isi policy
Valid values / checks	CHECK (policy_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	owning routine / caller → register_policy policy_hash from digest(policy) (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14; catalog seagis_quality.policy.policy_sha256

seagis_quality.policy.code_sha256

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Digest source builder alignment
Valid values / checks	CHECK (code_sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	owning routine / caller → register_policy builder_hash from code_hash() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14; catalog seagis_quality.policy.code_sha256

seagis_quality.policy.registered_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	clock_timestamp()
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan registrasi policy immutable
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	caller / SQL default → clock_timestamp() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14; catalog seagis_quality.policy.registered_at

Constraint, indeks dan trigger

constraints

```
CHECK (code_sha256 ~ '^[0-9a-f]{64}$'::text);
CHECK (jsonb_typeof(definition) = 'object'::text);
PRIMARY KEY (version);
CHECK (policy_sha256 ~ '^[0-9a-f]{64}$'::text);
```

indexes

```
CREATE UNIQUE INDEX policy_pkey ON seagis_quality.policy USING btree (version);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis_quality.policy FOR EACH ROW EXECUTE
FUNCTION seagis.reject_mutation();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.28 seagis_quality.snapshot_input

Lineage measurement per jam alignment snapshot.

Atribut	Definisi
Schema / domain	seagis_quality
Kind	table
Primary key	PRIMARY KEY (snapshot_id, aligned_at, source_ref)
Unique	NOT FOUND
Rows	UNVERIFIED
Writer	seagis_quality.save_snapshot
Reader	seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

seagis_quality.snapshot_input.snapshot_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	true / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (snapshot_id) REFERENCES seagis_quality.alignment_snapshot(id)
Arti bisnis / teknis	Referensi snapshot induk
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A3 document.rows[].cells[].lineage → saved snapshot UUID (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35; catalog seagis_quality.snapshot_input.snapshot_id

seagis_quality.snapshot_input.aligned_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	true / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Jam UTC baris alignment
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	A3 document.rows[].cells[].lineage → row.at::timestamptz (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35; catalog seagis_quality.snapshot_input.aligned_at

seagis_quality.snapshot_input.source_ref

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	true / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Referensi typed observation:uuid atau forecast_value:uuid
Valid values / checks	CHECK (source_ref = CASE WHEN observation_id IS NOT NULL THEN 'observation:'::text || observation_id ELSE 'forecast_value:'::text || forecast_value_id END)
Units / time	NOT APPLICABLE; NOT APPLICABLE

Atribut	Definisi
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	A3 document.rows[].cells[].lineage → lineage ref (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35; catalog seagis_quality.snapshot_input.source_ref

seagis_quality.snapshot_input.observation_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (observation_id) REFERENCES seagis.observation(id)
Arti bisnis / teknis	Referensi pembacaan sensor lokal
Valid values / checks	CHECK (num_nonnulls(observation_id, forecast_value_id) = 1); CHECK (source_ref = CASE WHEN observation_id IS NOT NULL THEN 'observation:'::text || observation_id ELSE 'forecast_value:'::text || forecast_value_id END)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A3 document.rows[].cells[].lineage → UUID if ref prefix observation else NULL (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35; catalog seagis_quality.snapshot_input.observation_id

seagis_quality.snapshot_input.forecast_value_id

Atribut	Definisi
Type / nullable	uuid / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (forecast_value_id) REFERENCES seagis.forecast_value(id)
Arti bisnis / teknis	Referensi satu nilai model regional
Valid values / checks	CHECK (num_nonnulls(observation_id, forecast_value_id) = 1); CHECK (source_ref = CASE WHEN observation_id IS NOT NULL THEN 'observation:'::text || observation_id ELSE 'forecast_value:'::text || forecast_value_id END)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	A3 document.rows[].cells[].lineage → UUID if ref prefix forecast_value else NULL (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35; catalog seagis_quality.snapshot_input.forecast_value_id

Constraint, indeks dan trigger

constraints

```
CHECK (num_nonnulls(observation_id, forecast_value_id) = 1);
CHECK (source_ref =
CASE
   WHEN observation_id IS NOT NULL THEN 'observation:'::text || observation_id
   ELSE 'forecast_value:'::text || forecast_value_id
END);
FOREIGN KEY (forecast_value_id) REFERENCES seagis.forecast_value(id);
FOREIGN KEY (observation_id) REFERENCES seagis.observation(id);
PRIMARY KEY (snapshot_id, aligned_at, source_ref);
FOREIGN KEY (snapshot_id) REFERENCES seagis_quality.alignment_snapshot(id);
```

indexes

```
CREATE UNIQUE INDEX snapshot_input_pkey ON seagis_quality.snapshot_input USING btree (snapshot_id,
aligned_at, source_ref);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis_quality.snapshot_input FOR EACH ROW
EXECUTE FUNCTION seagis.reject_mutation();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

4.29 seagis_raw.raw_payload

Manifest raw immutable: digest, basis hash, lisensi dan mode retensi.

Atribut	Definisi
Schema / domain	seagis_raw
Kind	table
Primary key	PRIMARY KEY (id)
Unique	UNIQUE (id, ingestion_run_id, dataset_key, is_synthetic); UNIQUE (ingestion_run_id, sha256)
Rows	UNVERIFIED
Writer	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Reader	A3 source_bundle; provenance guards
Otoritas	Record domain ini; registry/config atau sumber asal tetap ditelusuri
Lifecycle	Lihat trigger di bawah dan Bab 9; tanpa TTL terverifikasi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

seagis_raw.raw_payload.id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	gen_random_uuid()
PK / unique	true / true (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Identitas baris menurut PK tabel
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001

Atribut	Definisi
Asal / transformasi	DB default / explicit UUID → gen_random_uuid() (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.id

seagis_raw.raw_payload.ingestion_run_id

Atribut	Definisi
Type / nullable	uuid / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic)
Arti bisnis / teknis	Referensi ingestion asal
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	00000000-0000-4000-8000-000000000001
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → run_id (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.ingestion_run_id

seagis_raw.raw_payload.dataset_key

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic)
Arti bisnis / teknis	Kunci dataset terdaftar
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → s.dataset_key (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.dataset_key

seagis_raw.raw_payload.is_synthetic

Atribut	Definisi
Type / nullable	boolean / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)

Atribut	Definisi
FK	FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic)
Arti bisnis / teknis	Penanda asal sintetis, terpisah dari QC
Valid values / checks	CHECK (storage_mode <> 'inline'::text OR is_synthetic AND (hash_basis = ANY (ARRAY['synthetic_json_canonical_v1'::text, 'jsonb_utf8_v1'::text])))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	true
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → s.is_synthetic (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.is_synthetic

seagis_raw.raw_payload.sha256

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Digest raw sesuai hash_basis, bukan selalu byte jaringan
Valid values / checks	CHECK (sha256 ~ '^[0-9a-f]{64}$'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → CASE WHEN s.is_synthetic THEN seagis.jsonb_sha256(packet) ELSE wire_sha256 END (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.sha256

seagis_raw.raw_payload.hash_basis

Atribut	Definisi
Type / nullable	text / false
Default / generated	'source_bytes'::text
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Aturan byte/serialisasi yang di-hash
Valid values / checks	CHECK (hash_basis = ANY (ARRAY['source_bytes'::text, 'synthetic_json_canonical_v1'::text, ' jsonb_utf8_v1'::text, 'toolbox_subset_json_v1'::text])); CHECK (storage_mode <> 'inline'::text OR is_synthetic AND (hash_basis = ANY (ARRAY['synthetic_json_canonical_v1'::text, 'jsonb_utf8_v1'::text])))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	caller / SQL default → 'source_bytes'::text (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.hash_basis

seagis_raw.raw_payload.received_at

Atribut	Definisi
Type / nullable	timestamp with time zone / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Instan penerimaan raw oleh worker
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; Instan timestamptz, presisi PostgreSQL mikrodetik; tampilan mengikuti session. Kontrak API A1 wajib UTC Z. next_attempt_at mendukung -infinity.
Example	2026-10-21T12:00:00Z
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → received (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.received_at

seagis_raw.raw_payload.storage_mode

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	inline, object atau metadata_only menentukan lokasi konten
Valid values / checks	CHECK (storage_mode = 'inline'::text AND payload IS NOT NULL AND object_uri IS NULL OR storage_mode = 'object'::text AND payload IS NULL AND object_uri IS NOT NULL AND object_uri ~~ 'storage://%'::text OR storage_mode = 'metadata_only'::text AND payload IS NULL AND object_uri IS NULL); CHECK (storage_mode <> 'inline'::text OR is_synthetic AND (hash_basis = ANY (ARRAY['synthetic_json_canonical_v1'::text, 'jsonb_utf8_v1'::text]))); CHECK (storage_mode = ANY (ARRAY['inline'::text, 'object'::text, 'metadata_only'::text]))
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → CASE WHEN s.is_synthetic THEN 'inline' ELSE 'metadata_only' END (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.storage_mode

seagis_raw.raw_payload.payload

Atribut	Definisi
Type / nullable	jsonb / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Konten JSON raw sintetis bila inline
Valid values / checks	CHECK (storage_mode = 'inline'::text AND payload IS NOT NULL AND object_uri IS NULL OR storage_mode = 'object'::text AND payload IS NULL AND object_uri IS NOT NULL AND object_uri ~~ 'storage://%'::text OR storage_mode = 'metadata_only'::text AND payload IS NULL AND object_uri IS NULL)

Atribut	Definisi
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	{}
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → CASE WHEN s.is_synthetic THEN packet ELSE NULL END (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.payload

seagis_raw.raw_payload.object_uri

Atribut	Definisi
Type / nullable	text / true
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Referensi privat objek bila mode object
Valid values / checks	CHECK (storage_mode = 'inline'::text AND payload IS NOT NULL AND object_uri IS NULL OR storage_mode = 'object'::text AND payload IS NULL AND object_uri IS NOT NULL AND object_uri ~~ 'storage://%'::text OR storage_mode = 'metadata_only'::text AND payload IS NULL AND object_uri IS NULL)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	active loader omits column / nullable schema → NULL in inspected A1/A2 path; other callers may supply under schema guards (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.object_uri

seagis_raw.raw_payload.byte_size

Atribut	Definisi
Type / nullable	bigint / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Ukuran byte yang dilaporkan pada capture
Valid values / checks	CHECK (byte_size >= 0)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	1
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → wire_bytes (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.byte_size

seagis_raw.raw_payload.format

Atribut	Definisi
Type / nullable	text / false

Atribut	Definisi
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Media type/format representasi raw
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → 'application/json' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.format

seagis_raw.raw_payload.licence_snapshot

Atribut	Definisi
Type / nullable	text / false
Default / generated	NOT FOUND
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Keterangan lisensi saat capture
Valid values / checks	Tidak ada CHECK lokal; constraint lintas tabel/trigger tetap berlaku
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	Lihat nilai enum/CHECK di bawah; contoh bebas tidak diasumsikan valid
Asal / transformasi	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193 → 'project-owned; A1 synthetic development / real retention gated' (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.licence_snapshot

seagis_raw.raw_payload.provider_metadata

Atribut	Definisi
Type / nullable	jsonb / false
Default / generated	'{}'::jsonb
PK / unique	false / false (unique satu kolom; composite di bawah)
FK	NOT APPLICABLE
Arti bisnis / teknis	Metadata provider/request tersanitasi
Valid values / checks	CHECK (jsonb_typeof(provider_metadata) = 'object'::text)
Units / time	NOT APPLICABLE; NOT APPLICABLE
Example	{}
Asal / transformasi	caller / SQL default → '{}'::jsonb (VERIFIED)
Mutasi / konsumen	Mengikuti trigger/mutasi tabel, bukan asumsi mutable; A3 source_bundle; provenance guards
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110; catalog seagis_raw.raw_payload.provider_metadata

Constraint, indeks dan trigger

constraints

```
CHECK (byte_size >= 0);
CHECK (storage_mode = 'inline'::text AND payload IS NOT NULL AND object_uri IS NULL OR storage_mode
= 'object'::text AND payload IS NULL AND object_uri IS NOT NULL AND object_uri ~~
'storage://%'::text OR storage_mode = 'metadata_only'::text AND payload IS NULL AND object_uri IS
NULL);
CHECK (hash_basis = ANY (ARRAY['source_bytes'::text, 'synthetic_json_canonical_v1'::text,
'jsonb_utf8_v1'::text, 'toolbox_subset_json_v1'::text]));
UNIQUE (id, ingestion_run_id, dataset_key, is_synthetic);
FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id,
dataset_key, is_synthetic);
UNIQUE (ingestion_run_id, sha256);
CHECK (storage_mode <> 'inline'::text OR is_synthetic AND (hash_basis = ANY
(ARRAY['synthetic_json_canonical_v1'::text, 'jsonb_utf8_v1'::text])));
PRIMARY KEY (id);
CHECK (jsonb_typeof(provider_metadata) = 'object'::text);
CHECK (sha256 ~ '^[0-9a-f]{64}$'::text);
CHECK (storage_mode = ANY (ARRAY['inline'::text, 'object'::text, 'metadata_only'::text]));
```

indexes

```
CREATE UNIQUE INDEX raw_payload_id_ingestion_run_id_dataset_key_is_synthetic_key ON
seagis_raw.raw_payload USING btree (id, ingestion_run_id, dataset_key, is_synthetic);
CREATE UNIQUE INDEX raw_payload_ingestion_run_id_sha256_key ON seagis_raw.raw_payload USING btree
(ingestion_run_id, sha256);
CREATE UNIQUE INDEX raw_payload_pkey ON seagis_raw.raw_payload USING btree (id);
```

triggers

```
CREATE TRIGGER append_only BEFORE DELETE OR UPDATE ON seagis_raw.raw_payload FOR EACH ROW EXECUTE
FUNCTION seagis.reject_mutation();
CREATE TRIGGER guard_raw BEFORE INSERT ON seagis_raw.raw_payload FOR EACH ROW EXECUTE FUNCTION
seagis.guard_raw();
```

RLS pada snapshot: True; FORCE RLS: False. Membership/grant terkini UNVERIFIED.

5 Hubungan Antarentitas

Semua relasi di bawah adalah FK fisik dari catalog lama dan migrasi. Parent memiliki 0..N child kecuali child join key unik, yang membatasi 0..1 child; child merujuk tepat satu parent bila seluruh kolom NOT NULL, atau 0..1 bila nullable dengan MATCH SIMPLE. FK composite mengikat dataset/flag sintetis agar lineage tidak tercampur. Keberadaan FK tidak membuktikan parent wajib mempunyai child.

FK1 seagis.source → seagis.dataset

Atribut	Definisi
Constraint	dataset_source_key_fkey
Join	(source_key) → (source_key)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Identitas dataset, versi, cadence, lisensi dan gate ingestion.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

FOREIGN KEY (source_key) REFERENCES seagis.source(source_key);

FK2 seagis.dataset → seagis.dataset_parameter

Atribut	Definisi
Constraint	dataset_parameter_dataset_key_fkey
Join	(dataset_key) → (dataset_key)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Kontrak variabel sumber ke parameter dan aturan konversi.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key);

FK3 seagis.parameter → seagis.dataset_parameter

Atribut	Definisi
Constraint	dataset_parameter_parameter_code_unit_fkey
Join	(parameter_code,unit) → (parameter_code,unit)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Kontrak variabel sumber ke parameter dan aturan konversi.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

FOREIGN KEY (parameter_code, unit) REFERENCES seagis.parameter(parameter_code, unit);

FK4 seagis.dataset → seagis.forecast_run

Atribut	Definisi
Constraint	forecast_run_dataset_key_scope_fkey
Join	(dataset_key,scope) → (dataset_key,scope)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Capture/vintage model regional; waktu publikasi yang tidak diketahui tetap null.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope);

FK5 seagis_raw.raw_payload → seagis.forecast_run

Atribut	Definisi
Constraint	forecast_run_raw_payload_id_ingestion_run_id_dataset_key_i_fkey
Join	(raw_payload_id,ingestion_run_id,dataset_key,is_synthetic) → (id,ingestion_run_id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Capture/vintage model regional; waktu publikasi yang tidak diketahui tetap null.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic);

FK6 seagis.dataset_parameter → seagis.forecast_value

Atribut	Definisi
Constraint	forecast_value_dataset_key_parameter_code_native_unit_unit_fkey
Join	(dataset_key,parameter_code,native_unit,unit,conversion_code) → (dataset_key,parameter_code,native_unit,unit,conversion_code)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Nilai parameter per run, grid dan valid time; tidak diduplikasi per plot.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code);

FK7 seagis.forecast_run → seagis.forecast_value

Atribut	Definisi
Constraint	forecast_value_forecast_run_id_dataset_key_is_synthetic_fkey
Join	(forecast_run_id,dataset_key,is_synthetic) → (id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Nilai parameter per run, grid dan valid time; tidak diduplikasi per plot.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

FOREIGN KEY (forecast_run_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_run(id, dataset_key, is_synthetic);

FK8 seagis.source_grid → seagis.forecast_value

Atribut	Definisi
Constraint	forecast_value_source_grid_id_dataset_key_is_synthetic_fkey
Join	(source_grid_id,dataset_key,is_synthetic) → (id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Nilai parameter per run, grid dan valid time; tidak diduplikasi per plot.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

FOREIGN KEY (source_grid_id, dataset_key, is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic);

FK9 seagis.dataset → seagis.ingestion_run

Atribut	Definisi
Constraint	ingestion_run_dataset_key_fkey
Join	(dataset_key) → (dataset_key)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Audit satu ingestion logis beserta status penyelesaian.

Atribut	Definisi
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key);

FK10 seagis.dataset_parameter → seagis.observation

Atribut	Definisi
Constraint	observation_dataset_key_parameter_code_native_unit_unit_co_fkey
Join	(dataset_key,parameter_code,native_unit,unit,conversion_code) → (dataset_key,parameter_code,native_unit,unit,conversion_code)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Pembacaan suhu air sensor lokal, dengan nilai native, QC dan lineage raw.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

FOREIGN KEY (dataset_key, parameter_code, native_unit, unit, conversion_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code, native_unit, unit, conversion_code);

FK11 seagis_raw.raw_payload → seagis.observation

Atribut	Definisi
Constraint	observation_raw_payload_id_ingestion_run_id_dataset_key_is_fkey
Join	(raw_payload_id,ingestion_run_id,dataset_key,is_synthetic) → (id,ingestion_run_id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Pembacaan suhu air sensor lokal, dengan nilai native, QC dan lineage raw.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

FOREIGN KEY (raw_payload_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis_raw.raw_payload(id, ingestion_run_id, dataset_key, is_synthetic);

FK12 seagis.sensor → seagis.observation

Atribut	Definisi
Constraint	observation_sensor_id_plot_id_dataset_key_is_synthetic_fkey
Join	(sensor_id,plot_id,dataset_key,is_synthetic) → (id,plot_id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)

Atribut	Definisi
Meaning	Pembacaan suhu air sensor lokal, dengan nilai native, QC dan lineage raw.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

FOREIGN KEY (sensor_id, plot_id, dataset_key, is_synthetic) REFERENCES seagis.sensor(id, plot_id, dataset_key, is_synthetic);

FK13 seagis.plot → seagis.plot_grid_mapping

Atribut	Definisi
Constraint	plot_grid_mapping_plot_id_is_synthetic_fkey
Join	(plot_id,is_synthetic) → (id,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Pemetaan operasional plot ke grid dengan waktu efektif dan waktu tercatat.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic);

FK14 seagis.source_grid → seagis.plot_grid_mapping

Atribut	Definisi
Constraint	plot_grid_mapping_source_grid_id_fkey
Join	(source_grid_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Pemetaan operasional plot ke grid dengan waktu efektif dan waktu tercatat.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

FOREIGN KEY (source_grid_id) REFERENCES seagis.source_grid(id);

FK15 seagis.forecast_value → seagis.quality_event

Atribut	Definisi
Constraint	quality_event_forecast_value_id_dataset_key_is_synthetic_fkey
Join	(forecast_value_id,dataset_key,is_synthetic) → (id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)

Atribut	Definisi
Meaning	Log QC append-only untuk ingestion atau satu measurement.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

FOREIGN KEY (forecast_value_id, dataset_key, is_synthetic) REFERENCES seagis.forecast_value(id, dataset_key, is_synthetic);

FK16 seagis.ingestion_run → seagis.quality_event

Atribut	Definisi
Constraint	quality_event_ingestion_run_id_dataset_key_is_synthetic_fkey
Join	(ingestion_run_id,dataset_key,is_synthetic) → (id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Log QC append-only untuk ingestion atau satu measurement.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic);

FK17 seagis.observation → seagis.quality_event

Atribut	Definisi
Constraint	quality_event_observation_id_ingestion_run_id_dataset_key__fkey
Join	(observation_id,ingestion_run_id,dataset_key,is_synthetic) → (id,ingestion_run_id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Log QC append-only untuk ingestion atau satu measurement.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

FOREIGN KEY (observation_id, ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.observation(id, ingestion_run_id, dataset_key, is_synthetic);

FK18 seagis.dataset_parameter → seagis.sensor

Atribut	Definisi
Constraint	sensor_dataset_key_parameter_code_fkey
Join	(dataset_key,parameter_code) → (dataset_key,parameter_code)
Cardinality	1:0..N; child → parent 1
Optional	false

Atribut	Definisi
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Identitas deployment sensor suhu, plot, kedalaman dan status aktif.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

FOREIGN KEY (dataset_key, parameter_code) REFERENCES seagis.dataset_parameter(dataset_key, parameter_code);

FK19 seagis.dataset → seagis.sensor

Atribut	Definisi
Constraint	sensor_dataset_key_scope_fkey
Join	(dataset_key,scope) → (dataset_key,scope)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Identitas deployment sensor suhu, plot, kedalaman dan status aktif.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope);

FK20 seagis.plot → seagis.sensor

Atribut	Definisi
Constraint	sensor_plot_id_is_synthetic_fkey
Join	(plot_id,is_synthetic) → (id,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Identitas deployment sensor suhu, plot, kedalaman dan status aktif.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic);

FK21 seagis.dataset → seagis.source_grid

Atribut	Definisi
Constraint	source_grid_dataset_key_scope_fkey
Join	(dataset_key,scope) → (dataset_key,scope)
Cardinality	1:0..N; child → parent 1
Optional	false

Atribut	Definisi
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Identitas sel regional dan layer kedalaman; bukan plot budidaya.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

FOREIGN KEY (dataset_key, scope) REFERENCES seagis.dataset(dataset_key, scope);

FK22 seagis.sensor → seagis_api.credential

Atribut	Definisi
Constraint	credential_sensor_id_fkey
Join	(sensor_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Identitas aktor API berupa hash token, masa berlaku dan pencabutan.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id);

FK23 seagis.ingestion_run → seagis_api.packet_receipt

Atribut	Definisi
Constraint	packet_receipt_ingestion_run_id_fkey
Join	(ingestion_run_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Kunci idempotensi paket sensor dan digest semantik.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

FOREIGN KEY (ingestion_run_id) REFERENCES seagis.ingestion_run(id);

FK24 seagis.observation → seagis_api.packet_receipt

Atribut	Definisi
Constraint	packet_receipt_observation_id_fkey
Join	(observation_id) → (id)
Cardinality	1:0..1; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)

Atribut	Definisi
Meaning	Kunci idempotensi paket sensor dan digest semantik.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

FOREIGN KEY (observation_id) REFERENCES seagis.observation(id);

FK25 seagis.sensor → seagis_api.packet_receipt

Atribut	Definisi
Constraint	packet_receipt_sensor_id_fkey
Join	(sensor_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Kunci idempotensi paket sensor dan digest semantik.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id);

FK26 seagis_api.credential → seagis_api.reader_plot

Atribut	Definisi
Constraint	reader_plot_credential_id_fkey
Join	(credential_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Daftar plot yang boleh dibaca satu credential reader.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39

FOREIGN KEY (credential_id) REFERENCES seagis_api.credential(id);

FK27 seagis.plot → seagis_api.reader_plot

Atribut	Definisi
Constraint	reader_plot_plot_id_fkey
Join	(plot_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Daftar plot yang boleh dibaca satu credential reader.

Atribut	Definisi
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39

FOREIGN KEY (plot_id) REFERENCES seagis.plot(id);

FK28 seagis_api.credential → seagis_api.request_event

Atribut	Definisi
Constraint	request_event_credential_id_fkey
Join	(credential_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Audit respons API terautentikasi dan dasar rate limit.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

FOREIGN KEY (credential_id) REFERENCES seagis_api.credential(id);

FK29 seagis.sensor → seagis_api.request_event

Atribut	Definisi
Constraint	request_event_sensor_id_fkey
Join	(sensor_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Audit respons API terautentikasi dan dasar rate limit.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

FOREIGN KEY (sensor_id) REFERENCES seagis.sensor(id);

FK30 seagis_environment.policy → seagis_environment.attempt

Atribut	Definisi
Constraint	attempt_dataset_key_fkey
Join	(dataset_key) → (dataset_key)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Audit satu upaya fetch termasuk duplicate dan abandoned.

Atribut	Definisi
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key);

FK31 seagis.forecast_run → seagis_environment.attempt

Atribut	Definisi
Constraint	attempt_forecast_run_id_fkey
Join	(forecast_run_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Audit satu upaya fetch termasuk duplicate dan abandoned.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

FOREIGN KEY (forecast_run_id) REFERENCES seagis.forecast_run(id);

FK32 seagis_environment.policy → seagis_environment.demo_grid_mapping

Atribut	Definisi
Constraint	demo_grid_mapping_dataset_key_fkey
Join	(dataset_key) → (dataset_key)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Bridge khusus dua plot fiktif ke grid nyata untuk preview, bukan otorisasi training.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key);

FK33 seagis.plot → seagis_environment.demo_grid_mapping

Atribut	Definisi
Constraint	demo_grid_mapping_plot_id_plot_is_synthetic_fkey
Join	(plot_id,plot_is_synthetic) → (id,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Bridge khusus dua plot fiktif ke grid nyata untuk preview, bukan otorisasi training.

Atribut	Definisi
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

FOREIGN KEY (plot_id, plot_is_synthetic) REFERENCES seagis.plot(id, is_synthetic);

FK34 seagis.source_grid → seagis_environment.demo_grid_mapping

Atribut	Definisi
Constraint	demo_grid_mapping_source_grid_id_dataset_key_grid_is_synth_fkey
Join	(source_grid_id,dataset_key,grid_is_synthetic) → (id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Bridge khusus dua plot fiktif ke grid nyata untuk preview, bukan otorisasi training.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

FOREIGN KEY (source_grid_id, dataset_key, grid_is_synthetic) REFERENCES seagis.source_grid(id, dataset_key, is_synthetic);

FK35 seagis.dataset → seagis_environment.policy

Atribut	Definisi
Constraint	policy_dataset_key_fkey
Join	(dataset_key) → (dataset_key)
Cardinality	1:0..1; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Kebijakan fetch, lisensi, cadence scheduler, staleness dan retensi regional.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

FOREIGN KEY (dataset_key) REFERENCES seagis.dataset(dataset_key);

FK36 seagis_environment.policy → seagis_environment.schedule_state

Atribut	Definisi
Constraint	schedule_state_dataset_key_fkey
Join	(dataset_key) → (dataset_key)
Cardinality	1:0..1; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	State mutable lease, due time, backoff dan keberhasilan terakhir.

Atribut	Definisi
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

FOREIGN KEY (dataset_key) REFERENCES seagis_environment.policy(dataset_key);

FK37 seagis.forecast_run → seagis_environment.schedule_state

Atribut	Definisi
Constraint	schedule_state_last_forecast_run_id_fkey
Join	(last_forecast_run_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	State mutable lease, due time, backoff dan keberhasilan terakhir.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

FOREIGN KEY (last_forecast_run_id) REFERENCES seagis.forecast_run(id);

FK38 seagis.forecast_value → seagis_features.feature_input

Atribut	Definisi
Constraint	feature_input_forecast_value_id_fkey
Join	(forecast_value_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Lineage FK input feature snapshot dan guard cutoff.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

FOREIGN KEY (forecast_value_id) REFERENCES seagis.forecast_value(id);

FK39 seagis.observation → seagis_features.feature_input

Atribut	Definisi
Constraint	feature_input_observation_id_fkey
Join	(observation_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Lineage FK input feature snapshot dan guard cutoff.

Atribut	Definisi
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

FOREIGN KEY (observation_id) REFERENCES seagis.observation(id);

FK40 seagis_features.feature_snapshot → seagis_features.feature_input

Atribut	Definisi
Constraint	feature_input_snapshot_id_fkey
Join	(snapshot_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Lineage FK input feature snapshot dan guard cutoff.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

FOREIGN KEY (snapshot_id) REFERENCES seagis_features.feature_snapshot(id);

FK41 seagis.plot → seagis_features.feature_snapshot

Atribut	Definisi
Constraint	feature_snapshot_plot_id_is_synthetic_fkey
Join	(plot_id,is_synthetic) → (id,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Container fitur draft/sealed dengan cutoff dan hash; builder produksi belum ditemukan.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

FOREIGN KEY (plot_id, is_synthetic) REFERENCES seagis.plot(id, is_synthetic);

FK42 seagis.plot → seagis_quality.alignment_snapshot

Atribut	Definisi
Constraint	alignment_snapshot_plot_id_fkey
Join	(plot_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Dokumen QC/alignment hourly immutable; selalu training_authorized=false.

Atribut	Definisi
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

FOREIGN KEY (plot_id) REFERENCES seagis.plot(id);

FK43 seagis_quality.policy → seagis_quality.alignment_snapshot

Atribut	Definisi
Constraint	alignment_snapshot_policy_version_fkey
Join	(policy_version) → (version)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Dokumen QC/alignment hourly immutable; selalu training_authorized=false.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

FOREIGN KEY (policy_version) REFERENCES seagis_quality.policy(version);

FK44 seagis.forecast_value → seagis_quality.snapshot_input

Atribut	Definisi
Constraint	snapshot_input_forecast_value_id_fkey
Join	(forecast_value_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Lineage measurement per jam alignment snapshot.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

FOREIGN KEY (forecast_value_id) REFERENCES seagis.forecast_value(id);

FK45 seagis.observation → seagis_quality.snapshot_input

Atribut	Definisi
Constraint	snapshot_input_observation_id_fkey
Join	(observation_id) → (id)
Cardinality	1:0..N; child → parent 0..1
Optional	true
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Lineage measurement per jam alignment snapshot.

Atribut	Definisi
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

FOREIGN KEY (observation_id) REFERENCES seagis.observation(id);

FK46 seagis_quality.alignment_snapshot → seagis_quality.snapshot_input

Atribut	Definisi
Constraint	snapshot_input_snapshot_id_fkey
Join	(snapshot_id) → (id)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Lineage measurement per jam alignment snapshot.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

FOREIGN KEY (snapshot_id) REFERENCES seagis_quality.alignment_snapshot(id);

FK47 seagis.ingestion_run → seagis_raw.raw_payload

Atribut	Definisi
Constraint	raw_payload_ingestion_run_id_dataset_key_is_synthetic_fkey
Join	(ingestion_run_id,dataset_key,is_synthetic) → (id,dataset_key,is_synthetic)
Cardinality	1:0..N; child → parent 1
Optional	false
ON DELETE / ON UPDATE	NO ACTION / NO ACTION (default)
Meaning	Manifest raw immutable: digest, basis hash, lisensi dan mode retensi.
Risiko integritas	FK menolak orphan; nullable composite memakai MATCH SIMPLE. Semantik QC/waktu memerlukan guard tambahan.
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

FOREIGN KEY (ingestion_run_id, dataset_key, is_synthetic) REFERENCES seagis.ingestion_run(id, dataset_key, is_synthetic);

Relasi logis tambahan: dua alias forecast_value di current_speed bergabung pada forecast_run_id, source_grid_id, valid_at; parameter_code membedakan u/v. Ini join view, bukan FK baru. source_ref memakai prefix typed dan CHECK serta dua FK alternatif; tidak dibuat FK ke string polimorfik. Tidak ditemukan self-reference FK. reader_plot, dataset_parameter, feature_input dan snapshot_input berperan sebagai association/lineage, dengan composite key/unique yang tercantum pada dictionary. Relasi Farmer→CultivationPlot berada di memori AppPOC dan tidak masuk 47 FK PostgreSQL.

ERD lengkap dan per schema terdapat pada modul architecture/ERD.md.

ERD lengkap memuat seluruh kolom tabel fisik; PK/FK/UK mempertahankan nama fisik. FK1–FK47 dijelaskan pada Bab 5. Nama node menggunakan underscore hanya untuk sintaks Mermaid; label menyimpan schema.table. View current_speed didokumentasikan terpisah karena tidak memiliki FK fisik.

seagis.dataset

dataset_key : text [PK]
source_key : text [FK]
provider_product_id : text
provider_dataset_id : text
dataset_version : text
scope : text
native_cadence_seconds : integer
resolution_lat_deg : finite_float
resolution_lon_deg : finite_float
source_timezone : text
licence_url : text
attribution : text
raw_storage_permitted : boolean
real_ingestion_enabled : boolean
metadata_checked_at : timestamptz

→ **seagis.source**
1 parent / 0..N child

seagis.dataset_parameter

dataset_key : text [PK,FK]
parameter_code : text [PK,FK]
source_variable : text
native_unit : text
unit : text [FK]
conversion_code : text
conversion_factor : finite_float
conversion_offset : finite_float
temporal_support : text

→ **seagis.dataset**
1 parent / 0..N child

→ **seagis.parameter**
1 parent / 0..N child

seagis.forecast_run

id : uuid [PK]
dataset_key : text [FK]
scope : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
run_key : text
run_kind : text
provider_run_id : text
model_initialized_at : timestamptz
issued_at : timestamptz
issue_time_basis : text
ingested_at : timestamptz

→ **seagis.dataset**
1 parent / 0..N child

→ **seagis_raw.raw_payload**
1 parent / 0..N child

seagis.forecast_value

id : uuid [PK]
forecast_run_id : uuid [FK]
source_grid_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
valid_at : timestamptz
ingested_at : timestamptz
temporal_support : text
interval_status : text
period_start_at : timestamptz
period_end_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis.forecast_run**
1 parent / 0..N child

→ **seagis.source_grid**
1 parent / 0..N child

seagis.ingestion_run

id : uuid [PK]
dataset_key : text [FK]
request_key : text
adapter_version : text
started_at : timestamptz
completed_at : timestamptz
status : text
is_synthetic : boolean
error_code : text

→ **seagis.dataset**
1 parent / 0..N child

seagis.observation

id : uuid [PK]
plot_id : uuid [FK]
sensor_id : uuid [FK]
dataset_key : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
source_record_key : text
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
observed_at : timestamptz
ingested_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis_raw.raw_payload**
1 parent / 0..N child

→ **seagis.sensor**
1 parent / 0..N child

seagis.parameter

parameter_code : text [PK]
unit : text
description : text

No outgoing physical FK

seagis.plot

id : uuid [PK]
plot_code : text
name : text
latitude : finite_float
longitude : finite_float
is_synthetic : boolean
location_status : text
created_at : timestamptz

No outgoing physical FK

seagis.plot_grid_mapping

id : uuid [PK]
plot_id : uuid [FK]
source_grid_id : uuid [FK]
is_synthetic : boolean [FK]
sampling_method : text
distance_m : finite_float
representativeness : text
effective_from : timestamptz
effective_until : timestamptz
recorded_at : timestamptz

→ **seagis.plot**
1 parent / 0..N child

→ **seagis.source_grid**
1 parent / 0..N child

seagis.quality_event

id : uuid [PK]
ingestion_run_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
event_key : text
observation_id : uuid [FK]
forecast_value_id : uuid [FK]
event_code : text
severity : text
details : jsonb
created_at : timestamptz

→ **seagis.forecast_value**
0..1 parent / 0..N child

→ **seagis.ingestion_run**
1 parent / 0..N child

→ **seagis.observation**
0..1 parent / 0..N child

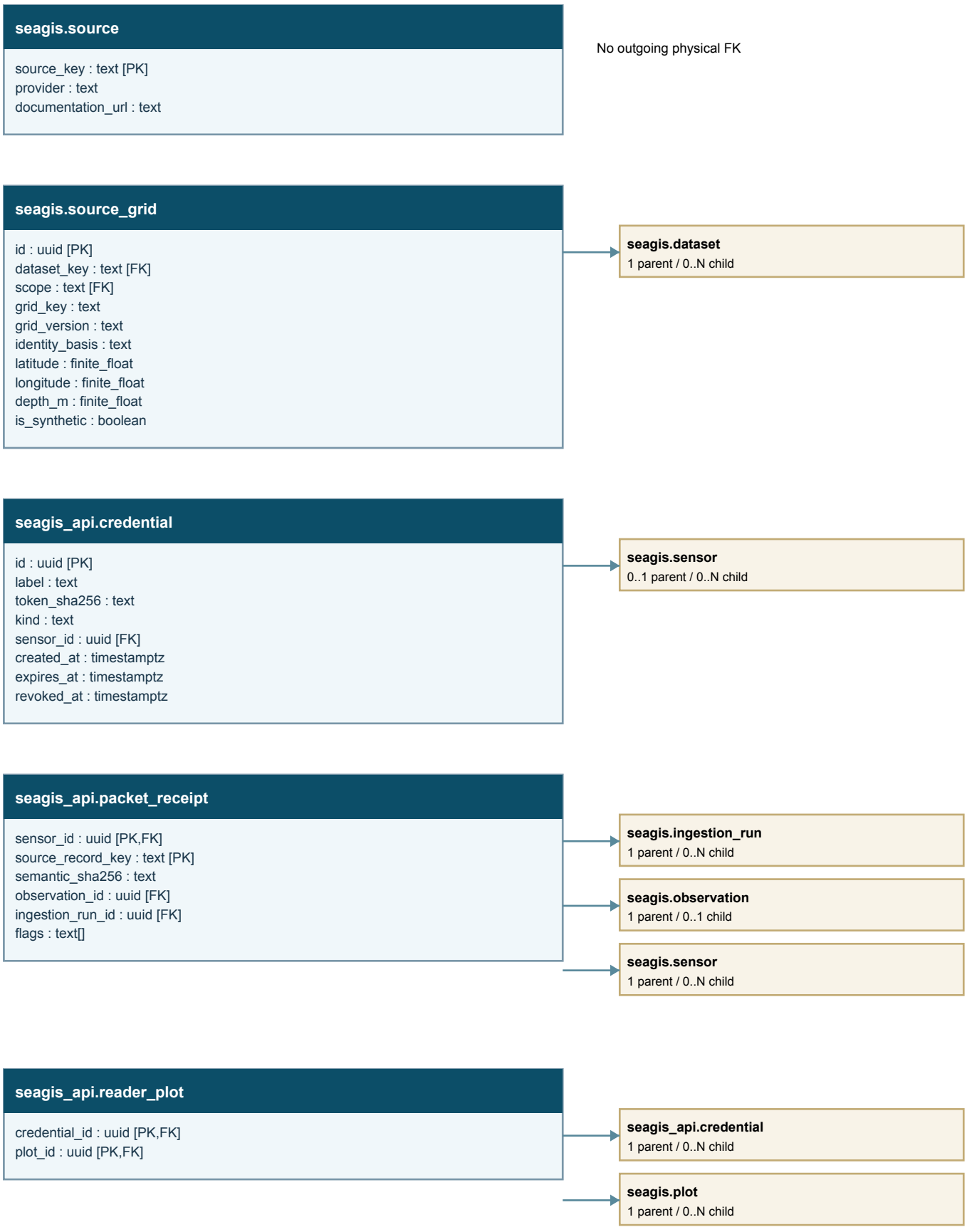
seagis.sensor

id : uuid [PK]
sensor_code : text
plot_id : uuid [FK]
dataset_key : text [FK]
scope : text [FK]
parameter_code : text [FK]
depth_m : finite_float
depth_status : text
is_synthetic : boolean [FK]
active : boolean

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis.dataset**
1 parent / 0..N child

→ **seagis.plot**
1 parent / 0..N child


seagis.source
source_key : text [PK]
provider : text
documentation_url : text
No outgoing physical FK
seagis.source_grid
id : uuid [PK]
dataset_key : text [FK]
scope : text [FK]
grid_key : text
grid_version : text
identity_basis : text
latitude : finite_float
longitude : finite_float
depth_m : finite_float
is_synthetic : boolean
seagis.dataset
1 parent / 0..N child
seagis_api.credential
id : uuid [PK]
label : text
token_sha256 : text
kind : text
sensor_id : uuid [FK]
created_at : timestamptz
expires_at : timestamptz
revoked_at : timestamptz
seagis.sensor
0..1 parent / 0..N child
seagis_api.packet_receipt
sensor_id : uuid [PK,FK]
source_record_key : text [PK]
semantic_sha256 : text
observation_id : uuid [FK]
ingestion_run_id : uuid [FK]
flags : text[]
seagis.ingestion_run
1 parent / 0..N child
seagis.observation
1 parent / 0..1 child
seagis.sensor
1 parent / 0..N child
seagis_api.reader_plot
credential_id : uuid [PK,FK]
plot_id : uuid [PK,FK]
seagis_api.credential
1 parent / 0..N child
seagis.plot
1 parent / 0..N child

seagis_api.request_event

id : uuid [PK]
credential_id : uuid [FK]
sensor_id : uuid [FK]
http_status : integer
code : text
created_at : timestamptz

→ **seagis_api.credential**
1 parent / 0..N child

→ **seagis.sensor**
0..1 parent / 0..N child

seagis_api.temperature_policy

id : boolean [PK]
version : text
min_degc : float8
max_degc : float8
max_step_degc : float8
future_tolerance_seconds : integer
stale_after_seconds : integer
requests_per_minute : integer

No outgoing physical FK

seagis_environment.attempt

id : uuid [PK]
dataset_key : text [FK]
started_at : timestamptz
completed_at : timestamptz
status : text
error_code : text
forecast_run_id : uuid [FK]

→ **seagis_environment.policy**
1 parent / 0..N child

→ **seagis.forecast_run**
0..1 parent / 0..N child

seagis_environment.demo_grid_mapping

plot_id : uuid [PK,FK]
plot_is_synthetic : boolean [FK]
source_grid_id : uuid [PK,FK]
dataset_key : text [FK]
grid_is_synthetic : boolean [FK]
example_latitude : finite_float
example_longitude : finite_float
coordinate_status : text
distance_m : finite_float
sampling_method : text
created_at : timestamptz

→ **seagis_environment.policy**
1 parent / 0..N child

→ **seagis.plot**
1 parent / 0..N child

→ **seagis.source_grid**
1 parent / 0..N child

seagis_environment.policy

dataset_key : text [PK,FK]
enabled : boolean
purpose : text
licence_url : text
terms_review : text
attribution : text
update_seconds : integer
release_offset_seconds : integer
stale_seconds : integer
retention_mode : text
reviewed_at : timestamptz

→ **seagis.dataset**
1 parent / 0..1 child

seagis_environment.schedule_state

dataset_key : text [PK,FK]
lease_id : uuid
lease_until : timestamptz
next_attempt_at : timestamptz
last_success_at : timestamptz
last_attempt_at : timestamptz
last_error_code : text
failures : integer
last_forecast_run_id : uuid [FK]

→ **seagis_environment.policy**
1 parent / 0..1 child

→ **seagis.forecast_run**
0..1 parent / 0..N child

seagis_features.feature_input

id : uuid [PK]
snapshot_id : uuid [FK]
observation_id : uuid [FK]
forecast_value_id : uuid [FK]

→ **seagis.forecast_value**
0..1 parent / 0..N child

→ **seagis.observation**
0..1 parent / 0..N child

→ **seagis_features.feature_snapshot**
1 parent / 0..N child

seagis_features.feature_snapshot

id : uuid [PK]
plot_id : uuid [FK]
is_synthetic : boolean [FK]
as_of : timestamptz
generated_at : timestamptz
feature_version : text
feature_schema_version : text
features : jsonb
missing_context : text[]
status : text
input_manifest_sha256 : text
features_sha256 : text

→ **seagis.plot**
1 parent / 0..N child

seagis_quality.alignment_snapshot

id : uuid [PK]
plot_id : uuid [FK]
mode : text
start_at : timestamptz
end_at : timestamptz
policy_version : text [FK]
source_sha256 : text
input_manifest_sha256 : text
body : jsonb
body_sha256 : text
created_at : timestamptz

→ **seagis.plot**
1 parent / 0..N child

→ **seagis_quality.policy**
1 parent / 0..N child

seagis_quality.policy

version : text [PK]
definition : jsonb
policy_sha256 : text
code_sha256 : text
registered_at : timestamptz

No outgoing physical FK

seagis_quality.snapshot_input

snapshot_id : uuid [PK,FK]
aligned_at : timestamptz [PK]
source_ref : text [PK]
observation_id : uuid [FK]
forecast_value_id : uuid [FK]

→ **seagis.forecast_value**
0..1 parent / 0..N child

→ **seagis.observation**
0..1 parent / 0..N child

→ **seagis_quality.alignment_snapshot**
1 parent / 0..N child

seagis_raw.raw_payload

id : uuid [PK]
ingestion_run_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
sha256 : text
hash_basis : text
received_at : timestamptz
storage_mode : text
payload : jsonb
object_uri : text
byte_size : bigint
format : text
licence_snapshot : text
provider_metadata : jsonb

→ **seagis.ingestion_run**
1 parent / 0..N child

seagis

seagis.dataset

dataset_key : text [PK]
source_key : text [FK]
provider_product_id : text
provider_dataset_id : text
dataset_version : text
scope : text
native_cadence_seconds : integer
resolution_lat_deg : finite_float
resolution_lon_deg : finite_float
source_timezone : text
licence_url : text
attribution : text
raw_storage_permitted : boolean
real_ingestion_enabled : boolean
metadata_checked_at : timestamptz

→ **seagis.source**
1 parent / 0..N child

seagis.dataset_parameter

dataset_key : text [PK,FK]
parameter_code : text [PK,FK]
source_variable : text
native_unit : text
unit : text [FK]
conversion_code : text
conversion_factor : finite_float
conversion_offset : finite_float
temporal_support : text

seagis.dataset
1 parent / 0..N child

seagis.parameter
1 parent / 0..N child

seagis.forecast_run

id : uuid [PK]
dataset_key : text [FK]
scope : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
run_key : text
run_kind : text
provider_run_id : text
model_initialized_at : timestamptz
issued_at : timestamptz
issue_time_basis : text
ingested_at : timestamptz

seagis.dataset
1 parent / 0..N child

seagis_raw.raw_payload
1 parent / 0..N child

seagis.forecast_value

id : uuid [PK]
forecast_run_id : uuid [FK]
source_grid_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
valid_at : timestamptz
ingested_at : timestamptz
temporal_support : text
interval_status : text
period_start_at : timestamptz
period_end_at : timestamptz
quality_status : text

seagis.dataset_parameter
1 parent / 0..N child

seagis.forecast_run
1 parent / 0..N child

seagis.source_grid
1 parent / 0..N child

seagis.ingestion_run

id : uuid [PK]
dataset_key : text [FK]
request_key : text
adapter_version : text
started_at : timestamptz
completed_at : timestamptz
status : text
is_synthetic : boolean
error_code : text

→ **seagis.dataset**
1 parent / 0..N child

seagis.observation

id : uuid [PK]
plot_id : uuid [FK]
sensor_id : uuid [FK]
dataset_key : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
source_record_key : text
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
observed_at : timestamptz
ingested_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis_raw.raw_payload**
1 parent / 0..N child

→ **seagis.sensor**
1 parent / 0..N child

seagis.parameter

parameter_code : text [PK]
unit : text
description : text

No outgoing physical FK

seagis.plot

id : uuid [PK]
plot_code : text
name : text
latitude : finite_float
longitude : finite_float
is_synthetic : boolean
location_status : text
created_at : timestamptz

No outgoing physical FK

seagis.plot_grid_mapping

id : uuid [PK]
plot_id : uuid [FK]
source_grid_id : uuid [FK]
is_synthetic : boolean [FK]
sampling_method : text
distance_m : finite_float
representativeness : text
effective_from : timestamptz
effective_until : timestamptz
recorded_at : timestamptz

→ **seagis.plot**
1 parent / 0..N child

→ **seagis.source_grid**
1 parent / 0..N child

seagis.quality_event

id : uuid [PK]
ingestion_run_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
event_key : text
observation_id : uuid [FK]
forecast_value_id : uuid [FK]
event_code : text
severity : text
details : jsonb
created_at : timestamptz

→ **seagis.forecast_value**
0..1 parent / 0..N child

→ **seagis.ingestion_run**
1 parent / 0..N child

→ **seagis.observation**
0..1 parent / 0..N child

seagis.sensor

id : uuid [PK]
sensor_code : text
plot_id : uuid [FK]
dataset_key : text [FK]
scope : text [FK]
parameter_code : text [FK]
depth_m : finite_float
depth_status : text
is_synthetic : boolean [FK]
active : boolean

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis.dataset**
1 parent / 0..N child

→ **seagis.plot**
1 parent / 0..N child

seagis.source

source_key : text [PK]
provider : text
documentation_url : text

No outgoing physical FK

seagis.source_grid

id : uuid [PK]
dataset_key : text [FK]
scope : text [FK]
grid_key : text
grid_version : text
identity_basis : text
latitude : finite_float
longitude : finite_float
depth_m : finite_float
is_synthetic : boolean

seagis.dataset
1 parent / 0..N child

seagis_raw.raw_payload

id : uuid [PK]
ingestion_run_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
sha256 : text
hash_basis : text
received_at : timestamptz
storage_mode : text
payload : jsonb
object_uri : text
byte_size : bigint
format : text
licence_snapshot : text
provider_metadata : jsonb

seagis.ingestion_run
1 parent / 0..N child

seagis_api

seagis.ingestion_run

id : uuid [PK]
dataset_key : text [FK]
request_key : text
adapter_version : text
started_at : timestamptz
completed_at : timestamptz
status : text
is_synthetic : boolean
error_code : text

seagis.dataset
1 parent / 0..N child

seagis.observation

id : uuid [PK]
plot_id : uuid [FK]
sensor_id : uuid [FK]
dataset_key : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
source_record_key : text
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
observed_at : timestamptz
ingested_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis_raw.raw_payload**
1 parent / 0..N child

→ **seagis.sensor**
1 parent / 0..N child

seagis.plot

id : uuid [PK]
plot_code : text
name : text
latitude : finite_float
longitude : finite_float
is_synthetic : boolean
location_status : text
created_at : timestamptz

No outgoing physical FK

seagis.sensor

id : uuid [PK]
sensor_code : text
plot_id : uuid [FK]
dataset_key : text [FK]
scope : text [FK]
parameter_code : text [FK]
depth_m : finite_float
depth_status : text
is_synthetic : boolean [FK]
active : boolean

→ **seagis.dataset_parameter**
1 parent / 0..N child

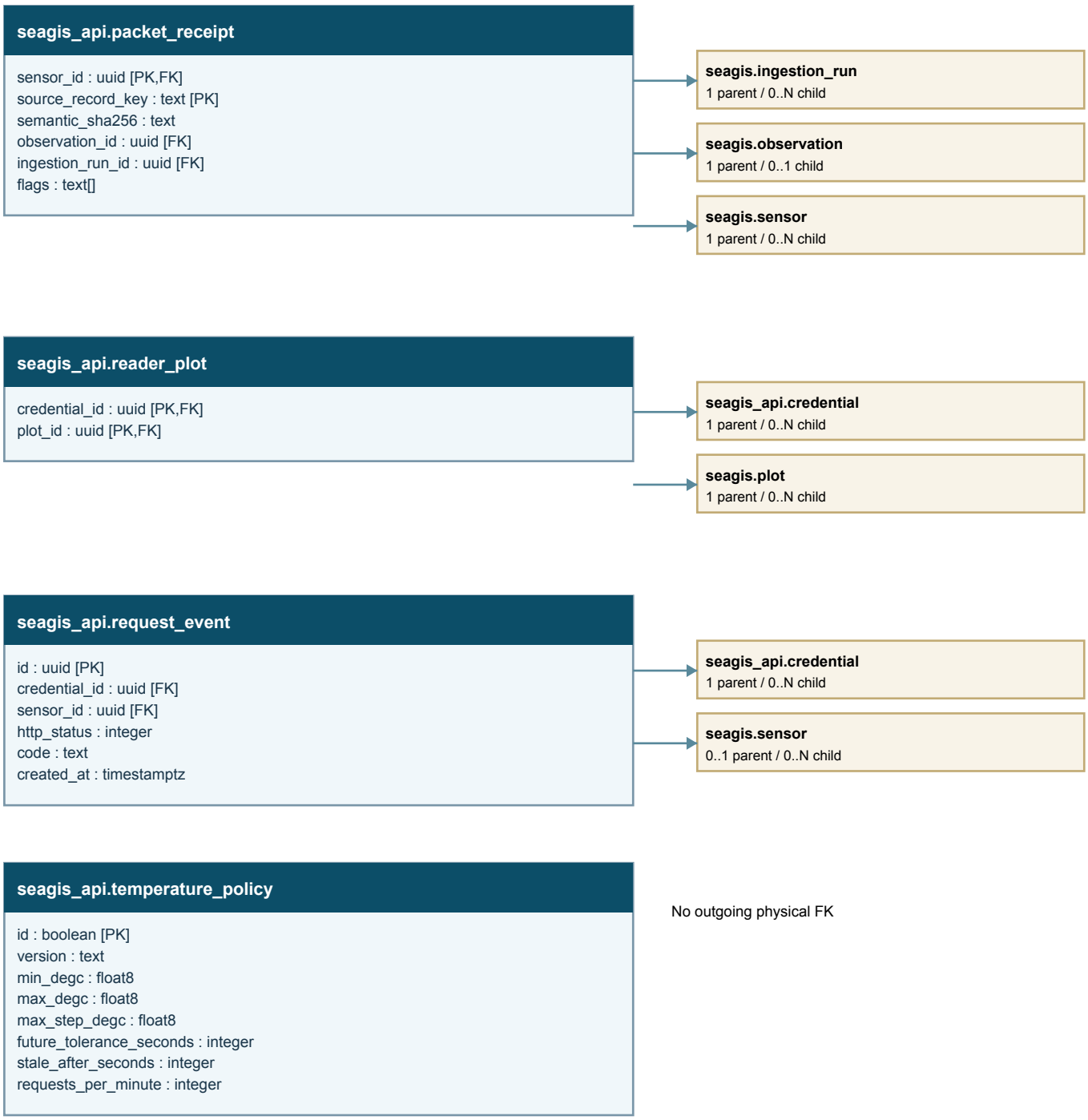
→ **seagis.dataset**
1 parent / 0..N child

→ **seagis.plot**
1 parent / 0..N child

seagis_api.credential

id : uuid [PK]
label : text
token_sha256 : text
kind : text
sensor_id : uuid [FK]
created_at : timestamptz
expires_at : timestamptz
revoked_at : timestamptz

→ **seagis.sensor**
0..1 parent / 0..N child



seagis_environment

seagis.dataset

dataset_key : text [PK]
source_key : text [FK]
provider_product_id : text
provider_dataset_id : text
dataset_version : text
scope : text
native_cadence_seconds : integer
resolution_lat_deg : finite_float
resolution_lon_deg : finite_float
source_timezone : text
licence_url : text
attribution : text
raw_storage_permitted : boolean
real_ingestion_enabled : boolean
metadata_checked_at : timestamptz

→ **seagis.source**
1 parent / 0..N child

seagis.forecast_run

id : uuid [PK]
dataset_key : text [FK]
scope : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
run_key : text
run_kind : text
provider_run_id : text
model_initialized_at : timestamptz
issued_at : timestamptz
issue_time_basis : text
ingested_at : timestamptz

→ **seagis.dataset**
1 parent / 0..N child

→ **seagis_raw.raw_payload**
1 parent / 0..N child

seagis.plot

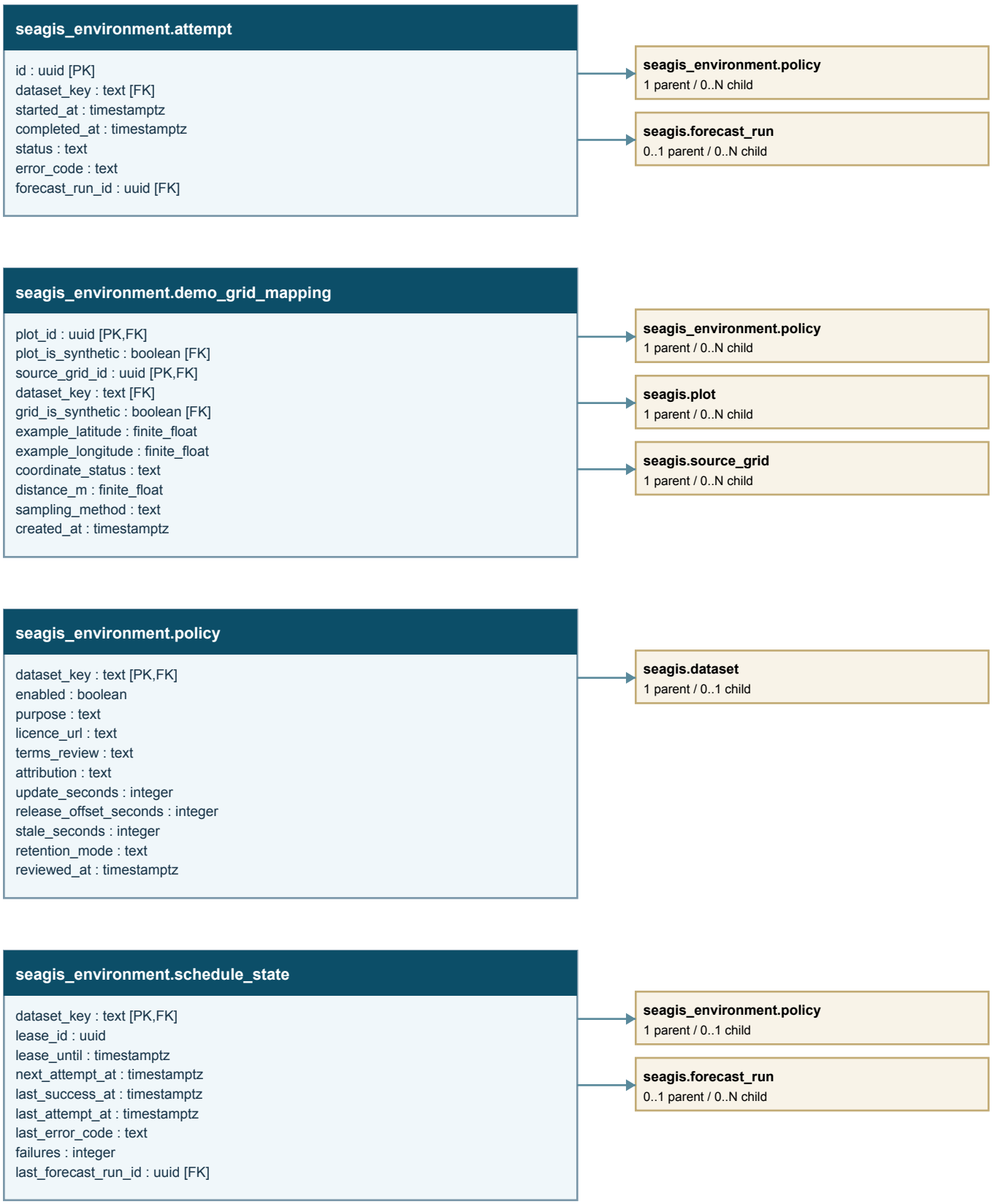
id : uuid [PK]
plot_code : text
name : text
latitude : finite_float
longitude : finite_float
is_synthetic : boolean
location_status : text
created_at : timestamptz

No outgoing physical FK

seagis.source_grid

id : uuid [PK]
dataset_key : text [FK]
scope : text [FK]
grid_key : text
grid_version : text
identity_basis : text
latitude : finite_float
longitude : finite_float
depth_m : finite_float
is_synthetic : boolean

→ **seagis.dataset**
1 parent / 0..N child



seagis_features

seagis.forecast_value

id : uuid [PK]
forecast_run_id : uuid [FK]
source_grid_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
valid_at : timestamptz
ingested_at : timestamptz
temporal_support : text
interval_status : text
period_start_at : timestamptz
period_end_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis.forecast_run**
1 parent / 0..N child

→ **seagis.source_grid**
1 parent / 0..N child

seagis.observation

id : uuid [PK]
plot_id : uuid [FK]
sensor_id : uuid [FK]
dataset_key : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
source_record_key : text
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
observed_at : timestamptz
ingested_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

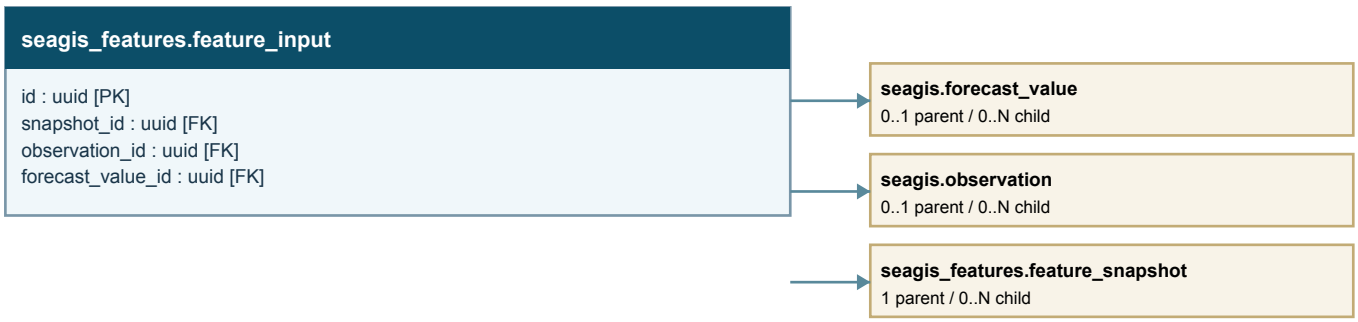
→ **seagis_raw.raw_payload**
1 parent / 0..N child

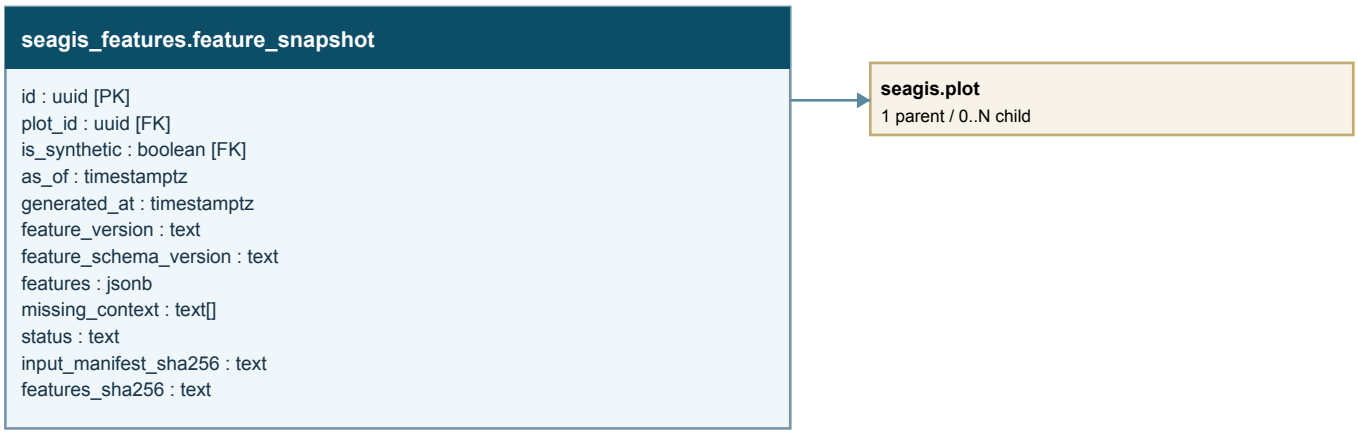
→ **seagis.sensor**
1 parent / 0..N child

seagis.plot

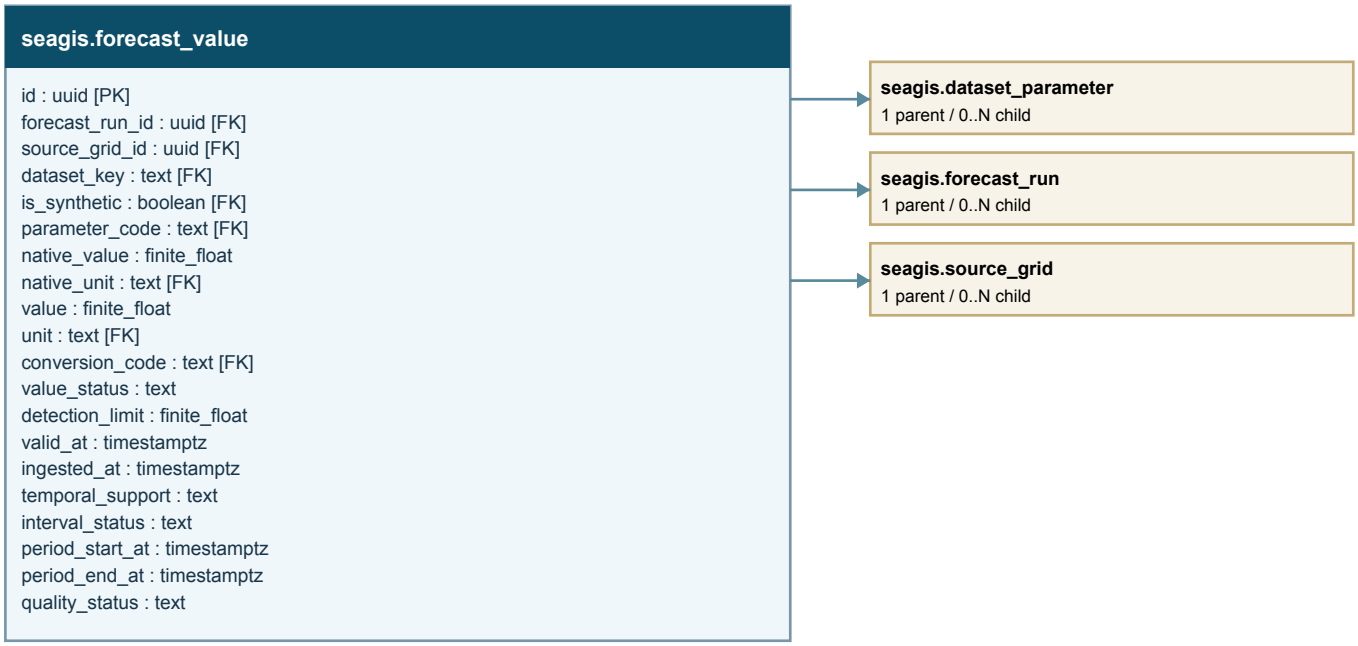
id : uuid [PK]
plot_code : text
name : text
latitude : finite_float
longitude : finite_float
is_synthetic : boolean
location_status : text
created_at : timestamptz

No outgoing physical FK





seagis_quality



seagis.observation

id : uuid [PK]
plot_id : uuid [FK]
sensor_id : uuid [FK]
dataset_key : text [FK]
ingestion_run_id : uuid [FK]
raw_payload_id : uuid [FK]
is_synthetic : boolean [FK]
source_record_key : text
parameter_code : text [FK]
native_value : finite_float
native_unit : text [FK]
value : finite_float
unit : text [FK]
conversion_code : text [FK]
value_status : text
detection_limit : finite_float
observed_at : timestamptz
ingested_at : timestamptz
quality_status : text

→ **seagis.dataset_parameter**
1 parent / 0..N child

→ **seagis_raw.raw_payload**
1 parent / 0..N child

→ **seagis.sensor**
1 parent / 0..N child

seagis.plot

id : uuid [PK]
plot_code : text
name : text
latitude : finite_float
longitude : finite_float
is_synthetic : boolean
location_status : text
created_at : timestamptz

No outgoing physical FK

seagis_quality.alignment_snapshot

id : uuid [PK]
plot_id : uuid [FK]
mode : text
start_at : timestamptz
end_at : timestamptz
policy_version : text [FK]
source_sha256 : text
input_manifest_sha256 : text
body : jsonb
body_sha256 : text
created_at : timestamptz

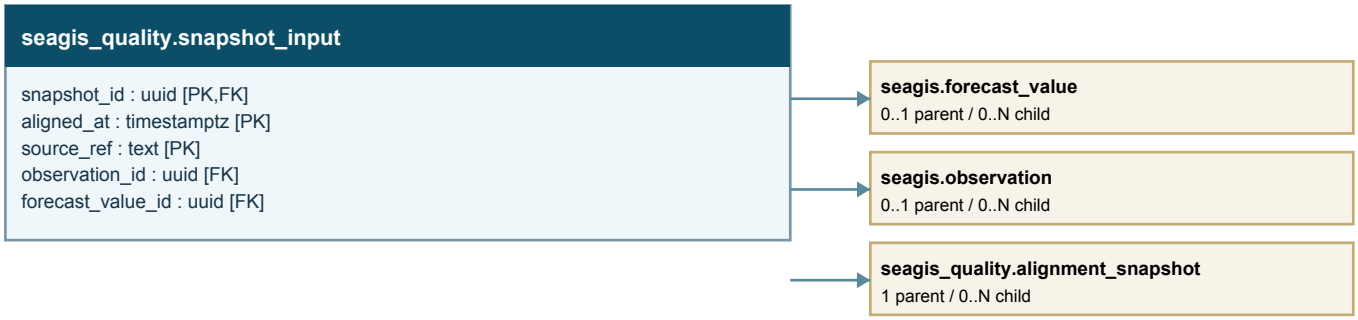
→ **seagis.plot**
1 parent / 0..N child

→ **seagis_quality.policy**
1 parent / 0..N child

seagis_quality.policy

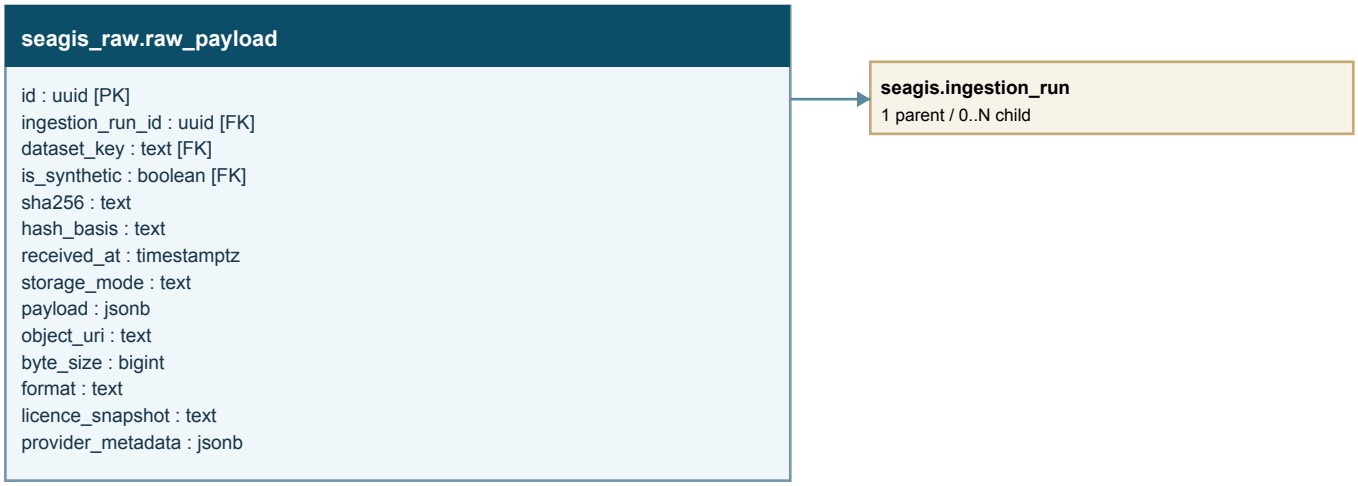
version : text [PK]
definition : jsonb
policy_sha256 : text
code_sha256 : text
registered_at : timestamptz

No outgoing physical FK


seagis_quality.snapshot_input
snapshot_id : uuid [PK,FK]
aligned_at : timestamptz [PK]
source_ref : text [PK]
observation_id : uuid [FK]
forecast_value_id : uuid [FK]
seagis.forecast_value
0..1 parent / 0..N child
seagis.observation
0..1 parent / 0..N child
seagis_quality.alignment_snapshot
1 parent / 0..N child

seagis_raw


seagis.ingestion_run
id : uuid [PK]
dataset_key : text [FK]
request_key : text
adapter_version : text
started_at : timestamptz
completed_at : timestamptz
status : text
is_synthetic : boolean
error_code : text
seagis.dataset
1 parent / 0..N child


seagis_raw.raw_payload
id : uuid [PK]
ingestion_run_id : uuid [FK]
dataset_key : text [FK]
is_synthetic : boolean [FK]
sha256 : text
hash_basis : text
received_at : timestamptz
storage_mode : text
payload : jsonb
object_uri : text
byte_size : bigint
format : text
licence_snapshot : text
provider_metadata : jsonb
seagis.ingestion_run
1 parent / 0..N child

6 Sumber dan Asal-Usul Data

A1: simulator/perangkat → byte JSON → validasi HTTP → autentikasi dan identitas sensor → validasi SQL → digest semantik → raw manifest + observation + quality_event + packet_receipt → history/status. sensor_id, plot dan flag sintetis tidak dipercaya dari input bebas: identitas diverifikasi terhadap credential dan deployment. Paket hilang tidak menciptakan observasi nol.

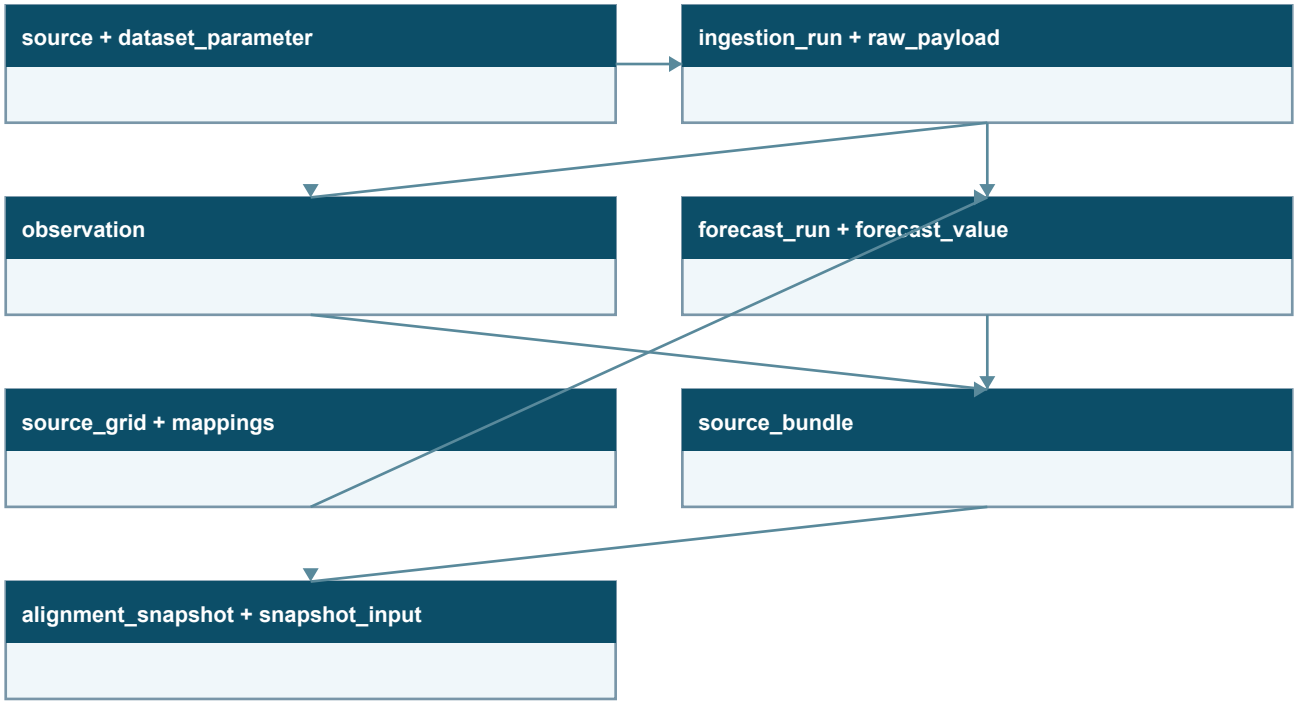
A2: registry/config → claim lease → fetch provider → pemeriksaan metadata/unit/cadence/koordinat → satu sel/layer Copernicus atau koordinat respons Open-Meteo → deduplikasi → run_key semantik → manifest metadata_only → forecast_run + source_grid + forecast_value + bridge demo → preview atau A3. Source bytes Open-Meteo dan canonical logical subset Copernicus memiliki hash_basis berbeda. STAC update dan generationtime_ms tidak dijadikan issued_at.

A3: source_bundle menggabungkan sensor, parameter, pemetaan, record dan event QC; jendela maksimum tujuh hari dengan lookback 72 jam dan sentinel sensor lebih lama bila diperlukan. Builder memfilter source_time dan available_at pada setiap cutoff, memilih konteks, mencatat spatial metadata, flags dan lineage. Penyimpanan membangun ulang source bundle, memeriksa hash/kebijakan/kode, lalu menyimpan snapshot dan input per jam. Perubahan sumber memaksa rebuild.

Ekspor CSV membaca source_bundle dalam REPEATABLE READ READ ONLY, atau capture JSON offline. Dua plot demo diperbolehkan. CSV regional memilih vintage terbaru pada waktu capture, sedangkan alignment JSON menyimpan cutoff historis. Keduanya tidak boleh saling menggantikan untuk backtest. Semua output exporter menandai training_authorized=false.

Bukti implementasi utama: Seagis-Model/supabase/functions/temperature/handler.ts:1; Seagis-Model/scripts/environmental/copernicus.py:1; Seagis-Model/scripts/environmental/open_meteo.py:1; Seagis-Model/scripts/quality/alignment.py:1; Seagis-Model/scripts/export_experiment_csv.py:1.

Diagram reproduktif: assets/diagrams/lineage.mmd.



Dataset	sensor_temperature_v1
Owner / source	device/simulator
Field	water_temperature
Native unit	degC
Unit	degC

Cadence s	3600
Support	instantaneous
Target	seagis.observation.value

Dataset	cmems_salinity_6h
Owner / source	copernicus
Field	so
Native unit	1e-3
Unit	1e-3
Cadence s	21600
Support	instantaneous
Target	seagis.forecast_value.value (salinity)

Dataset	cmems_currents_6h
Owner / source	copernicus
Field	uo
Native unit	m s-1
Unit	m/s
Cadence s	21600
Support	instantaneous
Target	seagis.forecast_value.value (current_eastward)

Dataset	cmems_currents_6h
Owner / source	copernicus
Field	vo
Native unit	m s-1
Unit	m/s
Cadence s	21600
Support	instantaneous
Target	seagis.forecast_value.value (current_northward)

Dataset	cmems_nutrients_daily
Owner / source	copernicus
Field	no3
Native unit	mmol m-3
Unit	mmol/m3
Cadence s	86400
Support	mean
Target	seagis.forecast_value.value (nitrate)

Dataset	cmems_nutrients_daily
Owner / source	copernicus
Field	po4
Native unit	mmol m-3

Unit	mmol/m3
Cadence s	86400
Support	mean
Target	seagis.forecast_value.value (phosphate)

Dataset	open_meteo_gfs
Owner / source	open_meteo
Field	precipitation
Native unit	mm
Unit	mm
Cadence s	3600
Support	sum
Target	seagis.forecast_value.value (precipitation)

Dataset	open_meteo_gfs
Owner / source	open_meteo
Field	wind_speed_10m
Native unit	m/s
Unit	m/s
Cadence s	3600
Support	instantaneous
Target	seagis.forecast_value.value (wind_speed_10m)

Dataset	open_meteo_gfs
Owner / source	open_meteo
Field	wind_direction_10m
Native unit	°
Unit	degree
Cadence s	3600
Support	instantaneous
Target	seagis.forecast_value.value (wind_direction_10m)

Dataset	open_meteo_gfs
Owner / source	open_meteo
Field	cloud_cover
Native unit	%
Unit	percent
Cadence s	3600
Support	instantaneous
Target	seagis.forecast_value.value (cloud_cover)

Dataset	open_meteo_gfs
Owner / source	open_meteo
Field	uv_index

Native unit	
Unit	1
Cadence s	3600
Support	instantaneous
Target	seagis.forecast_value.value (uv_index)

Dataset	open_meteo_gfs
Owner / source	open_meteo
Field	temperature_2m
Native unit	°C
Unit	degC
Cadence s	3600
Support	instantaneous
Target	seagis.forecast_value.value (air_temperature_2m)

Sensor cadence 3600 berasal dari seed kontrak pilot, bukan bukti setiap perangkat selalu mengirim tepat waktu. Keandalan provider tidak diukur ulang; adapter memiliki retry/validasi tetapi tidak menjamin SLA.

Bukti implementasi utama: Seagis-Model/supabase/seed.sql:1; Seagis-Model/config/a2_sources.json:1.

7 Ingesti dan Transformasi Data

A1 menjalankan satu function call SQL per request. Handler menerima application/json maksimum 8192 byte; byte di-hash SHA-256 sebelum JSON dikirim ke SQL. SQL memerlukan delapan field tepat, schema_version=1.0.0, degC, sequence uint32 dan UTC Z dengan presisi maksimal enam digit. Advisory transaction lock per aktor melindungi rate limit, lalu lock per sensor melindungi deduplikasi dan QC. Retry sama menghasilkan 200 duplicate, konten berbeda pada boot/sequence yang sama menghasilkan 409 packet_conflict. Record baru menghasilkan 201 stored. QC dapat menyimpan reading rejected; rejected bukan respons penolakan HTTP otomatis.

A2 berjalan per dataset setelah claim mengambil row lock pada schedule_state, membuat lease 20 menit dan attempt running. File installer launchd menetapkan tick 900 detik, tetapi ini konfigurasi installer, bukan bukti job aktif. Policy fetch nominal Copernicus 86400 detik, Open-Meteo 21600 detik; cadence nilai asli tetap 6 jam/harian/jam. Offset salinity/currents 30600 detik, nutrients 45000 detik, weather 16200 detik. Worker wrapper timeout 900 detik; HTTP timeout 40 detik; response maksimum 4 juta byte; retry tiga attempt pada timeout/network/429/5xx tertentu dengan backoff dan jitter. Retry-After panjang diserahkan pada backoff scheduler.

Open-Meteo dipin gfs_global, UTC, wind_speed_unit=ms dan forecast_days=3. Parser memeriksa unit, array sejajar dan timezone sebelum menambahkan UTC pada label tanpa suffix. Sel respons yang sama dideduplikasi. Copernicus memeriksa product/dataset/version, STAC cadence/grid/depth/unit, memuat nearest cell tetap untuk centroid dua plot contoh, memeriksa revisi katalog sebelum/sesudah, lalu decode melalui Toolbox/xarray. Nilai nonfinite/fill ekstrem menjadi null missing; daily mean tidak diubah menjadi 24 sampel independen.

Adapter memvalidasi 1–4 grid, tepat dua mapping, nilai tidak kosong dan key unik; SQL membatasi batch dua juta byte dan 1–5000 nilai. persist memeriksa lease, policy, versi dataset dan receipt time sebelum seluruh insert. Run_key sudah ada berarti duplicate; perubahan konten yang mengubah identitas semantik membuat vintage baru. Grid dan demo mapping memakai ON CONFLICT DO NOTHING dengan pemeriksaan identitas grid. Kegagalan statement mengembalikan seluruh transaksi persistence; claim/attempt tetap tercatat dan finish_failure dijalankan terpisah. Hilangnya commit acknowledgement memerlukan pemeriksaan status sebelum retry manual.

A3 tidak fetch provider dan tidak mengubah observation/forecast. build menghasilkan konteks hourly, mendeteksi unit mismatch, missing, rapid change, flatline candidate, staleness dan ketidakpastian spatial. Policy pilot-quality-2 adalah batas engineering. SQL memeriksa lineage, cutoff, nilai terpilih, pemetaan dan determinisme saat conflict; ia tidak menjalankan ulang seluruh keputusan QC Python. Tidak ada imputation atau training otomatis pada jalur ini.

Bukti implementasi utama: Seagis-Model/supabase/migrations/0001_initial_schema.sql:1;
Seagis-Model/supabase/migrations/0002_temperature_api.sql:1;
Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1;
Seagis-Model/supabase/migrations/0004_provider_update_status.sql:1;
Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1.

Bukti implementasi utama: Seagis-Model/scripts/ingest_environment.py:1; Seagis-Model/scripts/environmental/common.py:1;
Seagis-Model/scripts/install_a2_scheduler.py:1; Seagis-Model/scripts/run_a2_worker.py:1;
Seagis-Model/scripts/curate_environment.py:1.

Diagram reproduktif: assets/diagrams/a1.mmd.

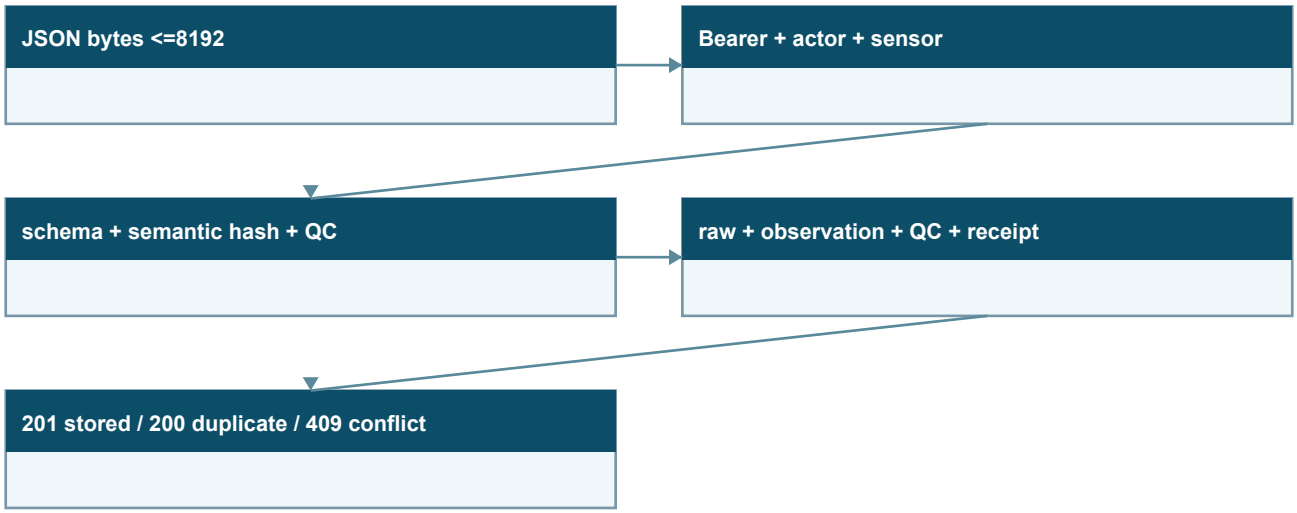


Diagram reproduktif: assets/diagrams/a2.mmd.

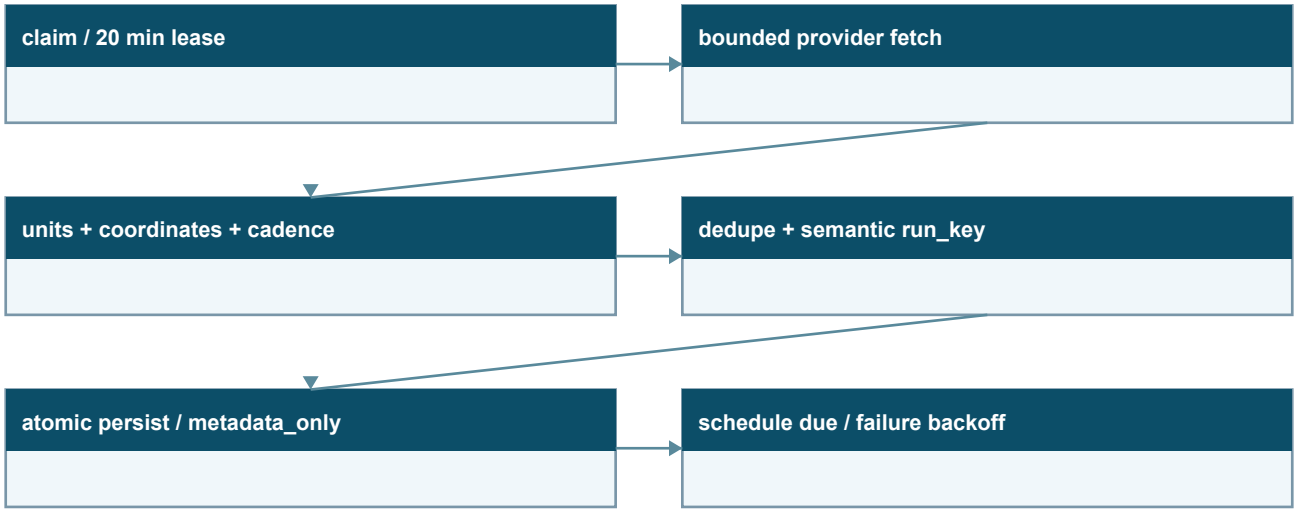
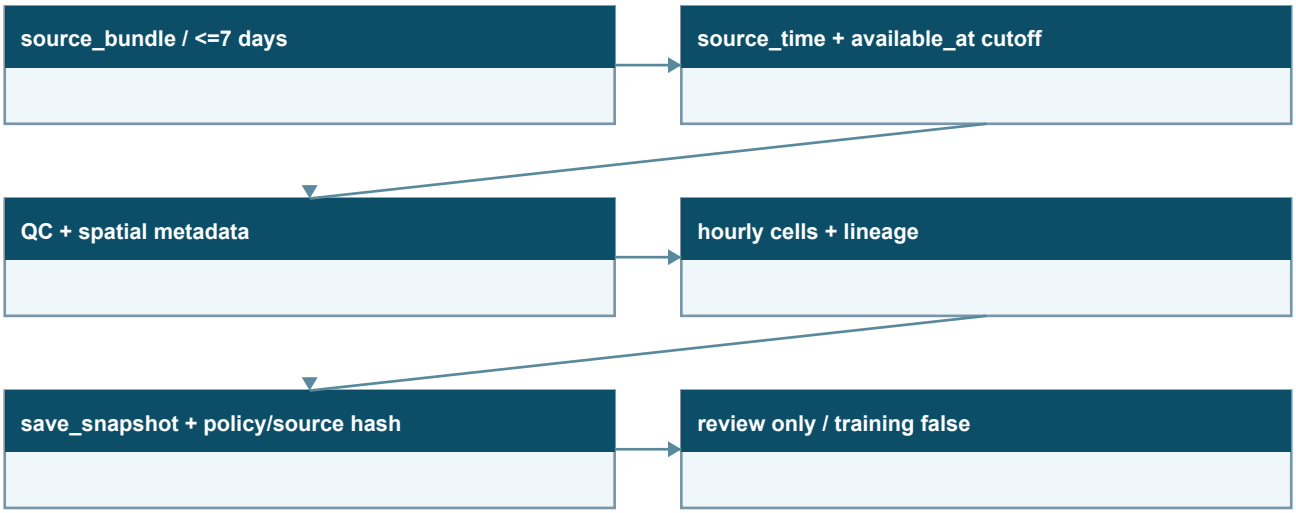


Diagram reproduktif: assets/diagrams/a3.mmd.



7.1 Pemetaan Sumber–Tujuan

Setiap kolom tujuan memiliki satu baris pemetaan. VERIFIED berarti ekspresi/default/join memiliki bukti; pemetaan administrasi/fixture bukan klaim loader produksi. UNVERIFIED menandai derivasi yang belum ditutup dan dihitung sebagai gap. Lihat kamus data untuk tipe, arti dan guard; ekspresi SQL mempertahankan identifier asli.

Destination table	seagis.dataset
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformasi	'sensor_temperature_v1'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	source_key
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformasi	'sensor'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	provider_product_id
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformasi	NULL
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset

Column	provider_dataset_id
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformasi	'seagis-water-temperature'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	dataset_version
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformasi	'v1'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	scope
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformasi	'local_sensor'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	native_cadence_seconds
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformasi	3600
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	resolution_lat_deg
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformasi	NULL
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	resolution_lon_deg
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformasi	NULL
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	source_timezone
Source / field	caller / SQL default
Transformasi	'UTC'::text
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	licence_url
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformasi	'project-owned://sensor-data'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	attribution
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformasi	'SEAGIS; simulated until hardware commissioning'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	raw_storage_permitted
Source / field	caller / SQL default
Transformasi	false
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	real_ingestion_enabled
Source / field	caller / SQL default
Transformasi	false
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset
Column	metadata_checked_at

Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:511
Transformasi	'2026-10-21T00:00:00Z'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:17

Destination table	seagis.dataset_parameter
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformasi	'sensor_temperature_v1'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	parameter_code
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformasi	'water_temperature'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	source_variable
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformasi	'water_temperature'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	native_unit
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformasi	'degC'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	unit
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformasi	'degC'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	conversion_code
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformasi	'identity'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	conversion_factor
Source / field	caller / SQL default
Transformasi	1
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	conversion_offset
Source / field	caller / SQL default
Transformasi	0
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.dataset_parameter
Column	temporal_support
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:532
Transformasi	'instantaneous'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:43

Destination table	seagis.forecast_run
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168

Transformasi	s.dataset_key
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	scope
Source / field	caller / SQL default
Transformasi	'regional_model'::text
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	ingestion_run_id
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformasi	ing
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	raw_payload_id
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformasi	raw_id
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	is_synthetic
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformasi	false
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	run_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformasi	batch->>'run_key'
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	run_kind
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformasi	CASE WHEN d.source_key='copernicus' THEN 'analysis_forecast_unspecified' ELSE 'forecast' END
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	provider_run_id
Source / field	active loader omits column / nullable schema
Transformasi	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	model_initialized_at
Source / field	active loader omits column / nullable schema
Transformasi	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	issued_at
Source / field	active loader omits column / nullable schema
Transformasi	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	issue_time_basis
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformasi	'unknown'
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_run
Column	ingested_at
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:168
Transformasi	t

Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:216

Destination table	seagis.forecast_value
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	forecast_run_id
Source / field	A2 adapter batch
Transformasi	persist run_id
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	source_grid_id
Source / field	A2 adapter batch
Transformasi	grid lookup by dataset/key/synthetic
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	dataset_key
Source / field	A2 adapter batch
Transformasi	leased dataset
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	is_synthetic
Source / field	A2 adapter batch
Transformasi	false
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value

Column	parameter_code
Source / field	A2 adapter batch
Transformasi	batch.values[].parameter
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	native_value
Source / field	A2 adapter batch
Transformasi	batch.values[].value
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	native_unit
Source / field	registered seagis.dataset_parameter par
Transformasi	par.native_unit
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	value
Source / field	A2 adapter batch
Transformasi	batch.values[].value
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	unit
Source / field	registered seagis.dataset_parameter par
Transformasi	par.unit
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	conversion_code
Source / field	registered seagis.dataset_parameter par
Transformasi	par.conversion_code
Loader / trigger	seagis_environment.persist; A0 fixtures

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	value_status
Source / field	A2 adapter batch
Transformasi	v.value_status
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	detection_limit
Source / field	A2 adapter batch
Transformasi	omitted -> SQL NULL
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	valid_at
Source / field	A2 adapter batch
Transformasi	v.valid_at::timestamptz
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	ingested_at
Source / field	A2 adapter batch
Transformasi	persist t = clock_timestamp()
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	temporal_support
Source / field	registered seagis.dataset_parameter par
Transformasi	par.temporal_support
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	interval_status

Source / field	A2 adapter batch
Transformasi	v.interval_status
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	period_start_at
Source / field	A2 adapter batch
Transformasi	v.period_start_at::timestamptz
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	period_end_at
Source / field	A2 adapter batch
Transformasi	v.period_end_at::timestamptz
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.forecast_value
Column	quality_status
Source / field	A2 adapter batch
Transformasi	default unreviewed
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:241

Destination table	seagis.ingestion_run
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191
Transformasi	s.dataset_key
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	request_key
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191
Transformasi	'a1:'||s.id||':'||key
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	adapter_version
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191
Transformasi	'temperature-1.0.0'
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	started_at
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191
Transformasi	received
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	completed_at
Source / field	owning routine / caller
Transformasi	UPDATE at end of A1/A2: clock_timestamp(); NULL while running
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	status
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191
Transformasi	'running'
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	is_synthetic
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:191

Transformasi	s.is_synthetic
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.ingestion_run
Column	error_code
Source / field	owning routine / caller
Transformasi	A1/A2 successful rows omit -> NULL; failure codes tracked in environment.attempt
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:94

Destination table	seagis.observation
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	plot_id
Source / field	A1 packet + registered identity + server
Transformasi	registered sensor.plot_id
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	sensor_id
Source / field	A1 packet + registered identity + server
Transformasi	authenticated credential.sensor_id
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	dataset_key
Source / field	A1 packet + registered identity + server
Transformasi	registered sensor.dataset_key
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	ingestion_run_id
Source / field	A1 packet + registered identity + server
Transformasi	ingest-generated run_id
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	raw_payload_id
Source / field	A1 packet + registered identity + server
Transformasi	ingest-generated raw_id
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	is_synthetic
Source / field	A1 packet + registered identity + server
Transformasi	registered sensor.is_synthetic
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	source_record_key
Source / field	A1 packet + registered identity + server
Transformasi	boot_id || : || sequence
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	parameter_code
Source / field	A1 packet + registered identity + server
Transformasi	constant water_temperature
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	native_value
Source / field	A1 packet + registered identity + server
Transformasi	packet.water_temperature

Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	native_unit
Source / field	A1 packet + registered identity + server
Transformasi	constant degC
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	value
Source / field	A1 packet + registered identity + server
Transformasi	packet.water_temperature (identity)
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	unit
Source / field	A1 packet + registered identity + server
Transformasi	constant degC
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	conversion_code
Source / field	A1 packet + registered identity + server
Transformasi	constant identity
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	value_status
Source / field	A1 packet + registered identity + server
Transformasi	packet.value_status
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation

Column	detection_limit
Source / field	A1 packet + registered identity + server
Transformasi	omitted -> SQL NULL
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	observed_at
Source / field	A1 packet + registered identity + server
Transformasi	packet.observed_at cast timestamptz
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	ingested_at
Source / field	A1 packet + registered identity + server
Transformasi	server received = clock_timestamp()
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.observation
Column	quality_status
Source / field	A1 packet + registered identity + server
Transformasi	pilot policy + flags
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:184

Destination table	seagis.parameter
Column	parameter_code
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:518
Transformasi	'water_temperature'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36

Destination table	seagis.parameter
Column	unit
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:518
Transformasi	'degC'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36

Destination table	seagis.parameter
Column	description
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:518
Transformasi	'Submerged local sensor temperature; distinct from air temperature'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:36

Destination table	seagis.plot
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	plot_code
Source / field	Seagis-Model/supabase/seed.sql:2
Transformasi	'plot-a'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	name
Source / field	Seagis-Model/supabase/seed.sql:2
Transformasi	'Plot A (synthetic)'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	latitude
Source / field	active loader omits column / nullable schema
Transformasi	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	longitude

Source / field	active loader omits column / nullable schema
Transformasi	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	is_synthetic
Source / field	Seagis-Model/supabase/seed.sql:2
Transformasi	true
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	location_status
Source / field	Seagis-Model/supabase/seed.sql:2
Transformasi	'synthetic'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot
Column	created_at
Source / field	caller / SQL default
Transformasi	now()
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:58

Destination table	seagis.plot_grid_mapping
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	plot_id
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformasi	caller-supplied plot_id subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	source_grid_id
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformasi	caller-supplied source_grid_id subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	is_synthetic
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformasi	caller-supplied is_synthetic subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	sampling_method
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformasi	caller-supplied sampling_method subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	distance_m
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformasi	caller-supplied distance_m subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	representativeness
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformasi	caller-supplied representativeness subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	effective_from
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND

Transformasi	caller-supplied effective_from subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	effective_until
Source / field	A0 fixtures/test SQL; operational creator NOT FOUND
Transformasi	caller-supplied effective_until subject to FK/CHECK/guard_mapping; recorded_at SQL default
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.plot_grid_mapping
Column	recorded_at
Source / field	caller / SQL default
Transformasi	clock_timestamp()
Loader / trigger	A0 fixture/admin SQL
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:152

Destination table	seagis.quality_event
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	ingestion_run_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformasi	run_id
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformasi	s.dataset_key
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	is_synthetic
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformasi	s.is_synthetic
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	event_key
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformasi	flag
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	observation_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformasi	observation_id
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	forecast_value_id
Source / field	active loader omits column / nullable schema
Transformasi	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	event_code
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformasi	flag
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	severity
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:208
Transformasi	CASE WHEN quality='rejected' THEN 'error' WHEN flag IN ('gap_before','out_of_order') THEN 'info' ELSE 'warning' END

Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	details
Source / field	caller / SQL default
Transformasi	'{}'::jsonb
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.quality_event
Column	created_at
Source / field	caller / SQL default
Transformasi	now()
Loader / trigger	seagis_api.ingest; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:273

Destination table	seagis.sensor
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	sensor_code
Source / field	Seagis-Model/supabase/seed.sql:7
Transformasi	'simulator-a'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	plot_id
Source / field	Seagis-Model/supabase/seed.sql:7
Transformasi	'00000000-0000-4000-8000-000000000001'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor

Column	dataset_key
Source / field	Seagis-Model/supabase/seed.sql:7
Transformasi	'sensor_temperature_v1'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	scope
Source / field	caller / SQL default
Transformasi	'local_sensor'::text
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	parameter_code
Source / field	caller / SQL default
Transformasi	'water_temperature'::text
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	depth_m
Source / field	active loader omits column / nullable schema
Transformasi	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	depth_status
Source / field	Seagis-Model/supabase/seed.sql:7
Transformasi	'unknown'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	is_synthetic
Source / field	Seagis-Model/supabase/seed.sql:7
Transformasi	true
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.sensor
Column	active
Source / field	caller / SQL default
Transformasi	true
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:73

Destination table	seagis.source
Column	source_key
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:506
Transformasi	'sensor'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11

Destination table	seagis.source
Column	provider
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:506
Transformasi	'SEAGIS local sensor / simulator'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11

Destination table	seagis.source
Column	documentation_url
Source / field	Seagis-Model/supabase/migrations/0001_initial_schema.sql:506
Transformasi	'https://github.com/De-Voeux/Seagis-Model'
Loader / trigger	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:11

Destination table	seagis.source_grid
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	dataset_key

Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformasi	s.dataset_key
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	scope
Source / field	caller / SQL default
Transformasi	'regional_model'::text
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	grid_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformasi	g->>'key'
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	grid_version
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformasi	d.dataset_version
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	identity_basis
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformasi	g->>'identity_basis'
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	latitude
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformasi	(g->>'latitude')::float8
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	longitude
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformasi	(g->>'longitude')::float8
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	depth_m
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformasi	(g->>'depth_m')::float8
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis.source_grid
Column	is_synthetic
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:173
Transformasi	false
Loader / trigger	seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:135

Destination table	seagis_api.credential
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	label
Source / field	owning routine / caller
Transformasi	provision_a1.create(label) CLI --label
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	token_sha256
Source / field	owning routine / caller

Transformasi	sha256((sgi_ + secrets.token_urlsafe(32)).encode()).hexdigest(); original token excluded
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	kind
Source / field	owning routine / caller
Transformasi	provision_a1.create(kind), CLI device/reader
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	sensor_id
Source / field	owning routine / caller
Transformasi	validated CLI sensor UUID for device, NULL for reader
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	created_at
Source / field	caller / SQL default
Transformasi	now()
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	expires_at
Source / field	owning routine / caller
Transformasi	now()+make_interval(days=>days); CLI 1..365, default 30
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.credential
Column	revoked_at
Source / field	owning routine / caller
Transformasi	NULL initially; revoke command UPDATE revoked_at=now()
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:27

Destination table	seagis_api.packet_receipt
Column	sensor_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215
Transformasi	s.id
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Destination table	seagis_api.packet_receipt
Column	source_record_key
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215
Transformasi	key
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Destination table	seagis_api.packet_receipt
Column	semantic_sha256
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215
Transformasi	digest
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Destination table	seagis_api.packet_receipt
Column	observation_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215
Transformasi	observation_id
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Destination table	seagis_api.packet_receipt
Column	ingestion_run_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215
Transformasi	run_id
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Destination table	seagis_api.packet_receipt
Column	flags
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:215
Transformasi	flags

Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:53

Destination table	seagis_api.reader_plot
Column	credential_id
Source / field	owning routine / caller
Transformasi	provision_a1.create generated credential UUID
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39

Destination table	seagis_api.reader_plot
Column	plot_id
Source / field	owning routine / caller
Transformasi	CLI plots validated with UUID(), FK checked by database
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:39

Destination table	seagis_api.request_event
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Destination table	seagis_api.request_event
Column	credential_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100
Transformasi	actor_id
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Destination table	seagis_api.request_event
Column	sensor_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100
Transformasi	sensor
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Destination table	seagis_api.request_event

Column	http_status
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100
Transformasi	status
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Destination table	seagis_api.request_event
Column	code
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:100
Transformasi	code
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Destination table	seagis_api.request_event
Column	created_at
Source / field	caller / SQL default
Transformasi	clock_timestamp()
Loader / trigger	seagis_api.ingest/reply
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:44

Destination table	seagis_api.temperature_policy
Column	id
Source / field	DB default / explicit UUID
Transformasi	true
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	version
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformasi	'pilot-qc-1'
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	min_degc
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformasi	-5
Loader / trigger	scripts/provision_a1.py; migration 0002

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	max_degc
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformasi	60
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	max_step_degc
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformasi	3
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	future_tolerance_seconds
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformasi	300
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	stale_after_seconds
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformasi	7200
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_api.temperature_policy
Column	requests_per_minute
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:80
Transformasi	120
Loader / trigger	scripts/provision_a1.py; migration 0002
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:67

Destination table	seagis_environment.attempt
Column	id

Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113
Transformasi	lease
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.attempt
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113
Transformasi	k
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.attempt
Column	started_at
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113
Transformasi	clock_timestamp()
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.attempt
Column	completed_at
Source / field	owning routine / caller
Transformasi	clock_timestamp() at persist/failure/abandon; initially NULL
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.attempt
Column	status
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:113
Transformasi	'running'
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.attempt
Column	error_code
Source / field	owning routine / caller
Transformasi	finish_failure sanitized code or lease_expired on abandoned; otherwise NULL
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.attempt
Column	forecast_run_id
Source / field	owning routine / caller
Transformasi	persist run_id, also on duplicate; NULL before completion
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:57

Destination table	seagis_environment.current_speed
Column	eastward_value_id
Source / field	seagis.forecast_value u/v
Transformasi	u.id
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	northward_value_id
Source / field	seagis.forecast_value u/v
Transformasi	v.id
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	forecast_run_id
Source / field	seagis.forecast_value u/v
Transformasi	u.forecast_run_id
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	source_grid_id
Source / field	seagis.forecast_value u/v
Transformasi	u.source_grid_id
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	valid_at
Source / field	seagis.forecast_value u/v

Transformasi	u.valid_at
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	ingested_at
Source / field	seagis.forecast_value u/v
Transformasi	u.ingested_at
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	available_at
Source / field	seagis.forecast_value u/v
Transformasi	greatest(u.ingested_at,v.ingested_at)
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	speed_m_s
Source / field	seagis.forecast_value u/v
Transformasi	CASE WHEN both value_status='present' THEN sqrt(u.value*u.value+v.value*v.value) END
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.current_speed
Column	derivation
Source / field	seagis.forecast_value u/v
Transformasi	'sqrt(uo^2+vo^2); a2-vector-magnitude-v1'
Loader / trigger	NOT APPLICABLE: view
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:89

Destination table	seagis_environment.demo_grid_mapping
Column	plot_id
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformasi	(m->>'plot_id')::uuid
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	plot_is_synthetic
Source / field	caller / SQL default
Transformasi	true
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	source_grid_id
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformasi	grid_id
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformasi	s.dataset_key
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	grid_is_synthetic
Source / field	caller / SQL default
Transformasi	false
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	example_latitude
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformasi	(m->>'latitude')::float8
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	example_longitude
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformasi	(m->>'longitude')::float8

Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	coordinate_status
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformasi	m->>'coordinate_status'
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	distance_m
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformasi	(m->>'distance_m')::float8
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	sampling_method
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:197
Transformasi	m->>'sampling_method'
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.demo_grid_mapping
Column	created_at
Source / field	caller / SQL default
Transformasi	now()
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:69

Destination table	seagis_environment.policy
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformasi	dataset_key
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy

Column	enabled
Source / field	caller / SQL default
Transformasi	true
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	purpose
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformasi	'noncommercial_educational_prototype'
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	licence_url
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformasi	licence_url
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	terms_review
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformasi	CASE WHEN source_key='copernicus' THEN 'cmems-licence-20261022-sections-2.2-2.4-2.6' ELSE ' open-meteo-terms-20261022-noncommercial-cc-by-4.0' END
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	attribution
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformasi	attribution
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	update_seconds
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformasi	CASE WHEN source_key='copernicus' THEN 86400 ELSE 21600 END
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	release_offset_seconds
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformasi	CASE WHEN dataset_key='cmems_nutrients_daily' THEN 45000 WHEN source_key='copernicus' THEN 30600 ELSE 16200 END
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	stale_seconds
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:30
Transformasi	CASE WHEN source_key='copernicus' THEN 172800 ELSE 43200 END
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	retention_mode
Source / field	caller / SQL default
Transformasi	'metadata_only'::text
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.policy
Column	reviewed_at
Source / field	caller / SQL default
Transformasi	now()
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:16

Destination table	seagis_environment.schedule_state
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:56
Transformasi	dataset_key
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state

Column	lease_id
Source / field	owning routine / caller
Transformasi	claim gen_random_uuid(); cleared after persist/failure
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	lease_until
Source / field	owning routine / caller
Transformasi	claim clock_timestamp()+interval 20 minutes; cleared after persist/failure
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	next_attempt_at
Source / field	caller / SQL default
Transformasi	'-infinity'::timestamp with time zone
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	last_success_at
Source / field	owning routine / caller
Transformasi	persist t=clock_timestamp()
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	last_attempt_at
Source / field	owning routine / caller
Transformasi	claim clock_timestamp()
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	last_error_code
Source / field	owning routine / caller
Transformasi	finish_failure sanitized code; cleared on success
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	failures
Source / field	caller / SQL default
Transformasi	0
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_environment.schedule_state
Column	last_forecast_run_id
Source / field	owning routine / caller
Transformasi	persist run_id
Loader / trigger	seagis_environment.claim/persist/finish_failure; migration 0003
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:45

Destination table	seagis_features.feature_input
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

Destination table	seagis_features.feature_input
Column	snapshot_id
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformasi	caller-supplied snapshot_id subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

Destination table	seagis_features.feature_input
Column	observation_id
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformasi	caller-supplied observation_id subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

Destination table	seagis_features.feature_input
Column	forecast_value_id

Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformasi	caller-supplied forecast_value_id subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:320

Destination table	seagis_features.feature_snapshot
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	plot_id
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformasi	caller-supplied plot_id subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	is_synthetic
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformasi	caller-supplied is_synthetic subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	as_of
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformasi	caller-supplied as_of subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	generated_at
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformasi	caller-supplied generated_at subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	feature_version
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformasi	caller-supplied feature_version subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	feature_schema_version
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformasi	caller-supplied feature_schema_version subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	features
Source / field	A0 fixtures/tests; production builder NOT FOUND
Transformasi	caller-supplied features subject to guard_input/guard_snapshot; no production derivation claimed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	missing_context
Source / field	caller / SQL default
Transformasi	'{}'::text[]
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	status
Source / field	caller / SQL default
Transformasi	'draft'::text
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	input_manifest_sha256
Source / field	seagis_features.manifest_hash(snapshot)

Transformasi	SHA256 sorted input references at draft -> sealed
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_features.feature_snapshot
Column	features_sha256
Source / field	DB generated
Transformasi	seagis.jsonb_sha256(features)
Loader / trigger	A0 fixture/test inserts; production builder NOT FOUND
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:301

Destination table	seagis_quality.alignment_snapshot
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	plot_id
Source / field	A3 build -> save_snapshot
Transformasi	document.plot_id
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	mode
Source / field	A3 build -> save_snapshot
Transformasi	document.mode
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	start_at
Source / field	A3 build -> save_snapshot
Transformasi	document.start_at
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	end_at
Source / field	A3 build -> save_snapshot
Transformasi	document.end_at
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	policy_version
Source / field	A3 build -> save_snapshot
Transformasi	registered policy.version
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	source_sha256
Source / field	A3 build -> save_snapshot
Transformasi	document.source_sha256 checked against fresh source_bundle
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	input_manifest_sha256
Source / field	A3 build -> save_snapshot
Transformasi	SHA256 sorted distinct at|source_ref lines
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	body
Source / field	A3 build -> save_snapshot
Transformasi	document
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	body_sha256
Source / field	DB generated
Transformasi	seagis.jsonb_sha256(body)

Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.alignment_snapshot
Column	created_at
Source / field	A3 build -> save_snapshot
Transformasi	SQL default clock_timestamp()
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:21

Destination table	seagis_quality.policy
Column	version
Source / field	owning routine / caller
Transformasi	register_policy policy[version]
Loader / trigger	scripts/quality/store.py::register_policy
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

Destination table	seagis_quality.policy
Column	definition
Source / field	owning routine / caller
Transformasi	register_policy JSON policy argument (config/a3_quality.json)
Loader / trigger	scripts/quality/store.py::register_policy
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

Destination table	seagis_quality.policy
Column	policy_sha256
Source / field	owning routine / caller
Transformasi	register_policy policy_hash from digest(policy)
Loader / trigger	scripts/quality/store.py::register_policy
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

Destination table	seagis_quality.policy
Column	code_sha256
Source / field	owning routine / caller
Transformasi	register_policy builder_hash from code_hash()
Loader / trigger	scripts/quality/store.py::register_policy
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

Destination table	seagis_quality.policy

Column	registered_at
Source / field	caller / SQL default
Transformasi	clock_timestamp()
Loader / trigger	scripts/quality/store.py::register_policy
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:14

Destination table	seagis_quality.snapshot_input
Column	snapshot_id
Source / field	A3 document.rows[].cells[].lineage
Transformasi	saved snapshot UUID
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

Destination table	seagis_quality.snapshot_input
Column	aligned_at
Source / field	A3 document.rows[].cells[].lineage
Transformasi	row.at::timestamptz
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

Destination table	seagis_quality.snapshot_input
Column	source_ref
Source / field	A3 document.rows[].cells[].lineage
Transformasi	lineage ref
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

Destination table	seagis_quality.snapshot_input
Column	observation_id
Source / field	A3 document.rows[].cells[].lineage
Transformasi	UUID if ref prefix observation else NULL
Loader / trigger	seagis_quality.save_snapshot
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

Destination table	seagis_quality.snapshot_input
Column	forecast_value_id
Source / field	A3 document.rows[].cells[].lineage
Transformasi	UUID if ref prefix forecast_value else NULL
Loader / trigger	seagis_quality.save_snapshot

Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:35

Destination table	seagis_raw.raw_payload
Column	id
Source / field	DB default / explicit UUID
Transformasi	gen_random_uuid()
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	ingestion_run_id
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformasi	run_id
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	dataset_key
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformasi	s.dataset_key
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	is_synthetic
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformasi	s.is_synthetic
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	sha256
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformasi	CASE WHEN s.is_synthetic THEN seagis.jsonb_sha256(packet) ELSE wire_sha256 END
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	hash_basis

Source / field	caller / SQL default
Transformasi	'source_bytes'::text
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	received_at
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformasi	received
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	storage_mode
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformasi	CASE WHEN s.is_synthetic THEN 'inline' ELSE 'metadata_only' END
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	payload
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformasi	CASE WHEN s.is_synthetic THEN packet ELSE NULL END
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	object_uri
Source / field	active loader omits column / nullable schema
Transformasi	NULL in inspected A1/A2 path; other callers may supply under schema guards
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	byte_size
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformasi	wire_bytes
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED

Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	format
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformasi	'application/json'
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	licence_snapshot
Source / field	Seagis-Model/supabase/migrations/0002_temperature_api.sql:193
Transformasi	'project-owned; A1 synthetic development / real retention gated'
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

Destination table	seagis_raw.raw_payload
Column	provider_metadata
Source / field	caller / SQL default
Transformasi	'{}'::jsonb
Loader / trigger	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Status	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:110

8 Kontrak Data

Kontrak eksplisit berupa empat JSON Schema; tujuh kontrak lain diturunkan dari implementasi. Klasifikasi tidak menyatakan semuanya endpoint publik. Tidak ditemukan version negotiation umum; versi/schema konstan ditolak bila tidak sesuai. Perubahan breaking memerlukan versi baru dan consumer review, bukan kompatibilitas otomatis. Error JSON yang dicontohkan di bawah berasal dari handler/SQL, bukan error standar yang diasumsikan.

ID	C01
Contract	A0 source envelope
Producer	fixture/provider logical extract
Consumer	validate_source
Interface	JSON / Draft 2020-12
Class	explicit
Database	seagis.ingestion_run + seagis_raw.raw_payload
Evidence	Seagis-Model/scripts/validate_contracts.py:1

ID	C02
Contract	A0 normalized envelope
Producer	fixture normalization
Consumer	validate_normalized + SQL guards
Interface	JSON / Draft 2020-12
Class	explicit
Database	observation / forecast_run / forecast_value / source_grid
Evidence	Seagis-Model/scripts/validate_contracts.py:1

ID	C03
Contract	A1 device packet
Producer	device/simulator
Consumer	temperature handler + seagis_api.ingest
Interface	JSON / HTTP POST
Class	explicit
Database	observation + raw_payload + packet_receipt + quality_event
Evidence	Seagis-Model/supabase/functions/temperature/handler.ts:1

ID	C04
Contract	A3 alignment document
Producer	scripts/quality/alignment.py::build
Consumer	seagis_quality.save_snapshot
Interface	JSON / SQL jsonb
Class	explicit
Database	seagis_quality.alignment_snapshot + snapshot_input
Evidence	Seagis-Model/scripts/curate_environment.py:1

ID	C05
Contract	A1 history/status/replies
Producer	seagis_api.history/status/reply

Consumer	authorized reader
Interface	JSON / HTTP GET
Class	implementation-derived
Database	observation + sensor + packet_receipt
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:1

ID	C06
Contract	A2 adapter batch/lease
Producer	Adapter.envelope + claim
Consumer	seagis_environment.persist
Interface	JSON / SQL jsonb
Class	implementation-derived
Database	raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
Evidence	Seagis-Model/scripts/environmental/common.py:1

ID	C07
Contract	A3 source_bundle
Producer	seagis_quality.source_bundle
Consumer	build / CSV exporter
Interface	JSON / SQL jsonb
Class	implementation-derived
Database	plot + sensor + dataset_parameter + mappings + observation + forecast_value
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1

ID	C08
Contract	CSV experiment-csv-2
Producer	export_experiment_csv.py::export
Consumer	offline experiments / developer
Interface	CSV + JSON manifest
Class	implementation-derived
Database	derived files; no database writes
Evidence	Seagis-Model/scripts/export_experiment_csv.py:1

ID	C09
Contract	iOS Assessment domain
Producer	MockAssessmentRepository
Consumer	LoadDemoAssessment / ViewModel
Interface	Swift structs
Class	implementation-derived / mock
Database	NOT APPLICABLE: no backend mapping
Evidence	Seagis/Seagis/Models/Shared/Assessment.swift:1

ID	C10
Contract	Open-Meteo response

Producer	Open-Meteo forecast API
Consumer	OpenMeteo.parse/normalize
Interface	JSON hourly arrays
Class	implementation-derived
Database	forecast_value + grid + raw manifest
Evidence	Seagis-Model/scripts/environmental/open_meteo.py:1

ID	C11
Contract	Copernicus subset
Producer	Copernicus Toolbox/xarray + STAC
Consumer	Copernicus.parse/normalize
Interface	xarray + canonical logical JSON subset
Class	implementation-derived
Database	forecast_value + grid + raw manifest
Evidence	Seagis-Model/scripts/environmental/copernicus.py:1

C01 A0 source envelope

Atribut	Definisi
Producer → consumer	fixture/provider logical extract → validate_source
Interface / format	JSON / Draft 2020-12
Classification	explicit
Database mapping	seagis.ingestion_run + seagis_raw.raw_payload
Compatibility	Versi konstan bila ditetapkan; negosiasi umum NOT FOUND
Evidence	Seagis-Model/scripts/validate_contracts.py:1

Field	schema_version
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Versi struktur kontrak; bukan versi model biologis
Validation / allowed values	{"const": "0.1.0"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"0.1.0"
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Field	source_type
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Jenis adapter asal yang menentukan payload fixture

Validation / allowed values	{"enum": ["sensor", "copernicus", "open_meteo"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"sensor"
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Field	dataset_key
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Kunci dataset terdaftar
Validation / allowed values	{"enum": ["sensor_temperature_v1", "cmems_salinity_6h", "cmems_currents_6h", "cmems_nutrients_daily", "open_meteo_gfs"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"sensor_temperature_v1"
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Field	is_synthetic
Type / ref	"boolean"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Penanda asal sintetis, terpisah dari QC
Validation / allowed values	{"type": "boolean"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	false
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Field	received_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Instan penerimaan raw oleh worker
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Field	provider_metadata
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND

Arti	Metadata provider/request tersanitasi
Validation / allowed values	{"type": "object"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Field	payload
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Konten JSON raw sintetis bila inline
Validation / allowed values	{"type": "object"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis.ingestion_run + seagis_raw.raw_payload

Schema lengkap untuk aturan kondisional dan struktur polimorfik (required di atas berlaku dalam parent masing-masing):


```
{
  "type": "object",
  "additionalProperties": false,
  "properties": {
    "schema_version": {
      "const": "0.1.0"
    },
    "source_type": {
      "enum": [
        "sensor",
        "copernicus",
        "open_meteo"
      ]
    },
    "dataset_key": {
      "enum": [
        "sensor_temperature_v1",
        "cmems_salinity_6h",
        "cmems_currents_6h",
        "cmems_nutrients_daily",
        "open_meteo_gfs"
      ]
    },
    "is_synthetic": {
      "type": "boolean"
    },
    "received_at": {
      "type": "string",
      "format": "date-time",
      "pattern": "Z$"
    },
    "provider_metadata": {
      "type": "object"
    },
    "payload": {
      "type": "object"
    }
  },
  "required": [
    "schema_version",
    "source_type",
    "dataset_key",
    "is_synthetic",
    "received_at",
    "provider_metadata",
    "payload"
  ],
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://seagis.local/contracts/source.schema.json",
```

```
"title": "SEAGIS source envelopes v0.1.0",
"allOf": [
 {
  "if": {
   "properties": {
    "source_type": {
     "const": "sensor"
    }
   }
  },
  "then": {
   "properties": {
    "payload": {
     "type": "object",
     "additionalProperties": false,
     "properties": {
      "sensor_id": {
       "type": "string",
       "format": "uuid"
      },
      "boot_id": {
       "type": "string",
       "minLength": 1
      },
      "sequence": {
       "type": "integer",
       "minimum": 0
      },
      "observed_at": {
       "type": "string",
       "format": "date-time",
       "pattern": "Z$"
      },
      "water_temperature": {
       "type": [
        "number",
        "null"
       ]
      },
      "unit": {
       "const": "degC"
      },
      "value_status": {
       "enum": [
        "present",
        "missing",
        "below_detection_limit",
        "not_available"
       ]
      },
      "detection_limit": {
       "type": [
        "number",
        "null"
       ]
      }
     },
     "required": [
      "sensor_id",
      "boot_id",
      "sequence",
      "observed_at",
      "water_temperature",
      "unit",
      "value_status",
      "detection_limit"
     ]
    },
    "dataset_key": {
     "enum": [
      "sensor_temperature_v1"
     ]
    }
   }
  }
 },
 {
  "if": {
   "properties": {
    "source_type": {
     "const": "copernicus"
    }
   }
```

```
    },
    "then": {
      "properties": {
        "payload": {
          "type": "object",
          "additionalProperties": false,
          "properties": {
            "representation": {
              "const": "netcdf-logical-extract"
            },
            "coordinates": {
              "type": "object",
              "additionalProperties": false,
              "properties": {
                "time": {
                  "type": "array",
                  "minItems": 1,
                  "items": {
                    "type": "string",
                    "format": "date-time",
                    "pattern": "Z$"
                  }
                },
                "depth": {
                  "type": "array",
                  "minItems": 1,
                  "items": {
                    "type": "number",
                    "minimum": 0
                  }
                },
                "latitude": {
                  "type": "array",
                  "minItems": 1,
                  "items": {
                    "type": "number",
                    "minimum": -90,
                    "maximum": 90
                  }
                },
                "longitude": {
                  "type": "array",
                  "minItems": 1,
                  "items": {
                    "type": "number",
                    "minimum": -180,
                    "maximum": 180
                  }
                }
              },
              "required": [
                "time",
                "depth",
                "latitude",
                "longitude"
              ]
            },
            "variables": {
              "type": "object",
              "additionalProperties": false,
              "properties": {
                "so": {
                  "type": "object",
                  "additionalProperties": false,
                  "properties": {
                    "unit": {
                      "const": "1e-3"
                    },
                    "fill_value": {
                      "type": "number"
                    },
                    "values": {
                      "type": "array",
                      "minItems": 1,
                      "items": {
                        "type": [
                          "number",
                          "null"
                        ]
                      }
                    }
                  },
                  "required": [
```

```
      "unit",
      "fill_value",
      "values"
    ]
  },
  "uo": {
    "type": "object",
    "additionalProperties": false,
    "properties": {
      "unit": {
        "const": "m s-1"
      },
      "fill_value": {
        "type": "number"
      },
      "values": {
        "type": "array",
        "minItems": 1,
        "items": {
          "type": [
            "number",
            "null"
          ]
        }
      }
    },
    "required": [
      "unit",
      "fill_value",
      "values"
    ]
  },
  "vo": {
    "type": "object",
    "additionalProperties": false,
    "properties": {
      "unit": {
        "const": "m s-1"
      },
      "fill_value": {
        "type": "number"
      },
      "values": {
        "type": "array",
        "minItems": 1,
        "items": {
          "type": [
            "number",
            "null"
          ]
        }
      }
    },
    "required": [
      "unit",
      "fill_value",
      "values"
    ]
  },
  "no3": {
    "type": "object",
    "additionalProperties": false,
    "properties": {
      "unit": {
        "const": "mmol m-3"
      },
      "fill_value": {
        "type": "number"
      },
      "values": {
        "type": "array",
        "minItems": 1,
        "items": {
          "type": [
            "number",
            "null"
          ]
        }
      }
    },
    "required": [
      "unit",
      "fill_value",
```

```
              "values"
             ]
            },
            "po4": {
             "type": "object",
             "additionalProperties": false,
             "properties": {
              "unit": {
               "const": "mmol m-3"
              },
              "fill_value": {
               "type": "number"
              },
              "values": {
               "type": "array",
               "minItems": 1,
               "items": {
                "type": [
                 "number",
                 "null"
                ]
               }
              }
             },
             "required": [
              "unit",
              "fill_value",
              "values"
             ]
            }
           },
           "minProperties": 1
          }
         },
         "required": [
          "representation",
          "coordinates",
          "variables"
         ]
        },
        "dataset_key": {
         "enum": [
          "cmems_salinity_6h",
          "cmems_currents_6h",
          "cmems_nutrients_daily"
         ]
        }
       }
      }
     },
     {
      "if": {
       "properties": {
        "source_type": {
         "const": "open_meteo"
        }
       }
      },
      "then": {
       "properties": {
        "payload": {
         "type": "object",
         "additionalProperties": false,
         "properties": {
          "latitude": {
           "type": "number"
          },
          "longitude": {
           "type": "number"
          },
          "generationtime_ms": {
           "type": "number"
          },
          "utc_offset_seconds": {
           "const": 0
          },
          "timezone": {
           "enum": [
            "GMT",
            "UTC"
           ]
          },
          "timezone_abbreviation": {
```

```
      "enum": [
        "GMT",
        "UTC"
      ]
    },
    "hourly_units": {
      "type": "object",
      "additionalProperties": false,
      "properties": {
        "time": {
          "const": "iso8601"
        },
        "precipitation": {
          "const": "mm"
        },
        "wind_speed_10m": {
          "const": "m/s"
        },
        "wind_direction_10m": {
          "const": "°"
        },
        "cloud_cover": {
          "const": "%"
        },
        "uv_index": {
          "const": ""
        },
        "temperature_2m": {
          "const": "°C"
        }
      },
      "required": [
        "time",
        "precipitation",
        "wind_speed_10m",
        "wind_direction_10m",
        "cloud_cover",
        "uv_index",
        "temperature_2m"
      ]
    },
    "hourly": {
      "type": "object",
      "additionalProperties": false,
      "properties": {
        "time": {
          "type": "array",
          "minItems": 1,
          "items": {
            "type": "string",
            "pattern": "^\\d{4}-\\d{2}-\\d{2}T\\d{2}:\\d{2}$"
          }
        },
        "precipitation": {
          "type": "array",
          "minItems": 1,
          "items": {
            "type": [
              "number",
              "null"
            ]
          }
        },
        "wind_speed_10m": {
          "type": "array",
          "minItems": 1,
          "items": {
            "type": [
              "number",
              "null"
            ]
          }
        },
        "wind_direction_10m": {
          "type": "array",
          "minItems": 1,
          "items": {
            "type": [
              "number",
              "null"
            ]
          }
        },
```

```
            "cloud_cover": {
              "type": "array",
              "minItems": 1,
              "items": {
                "type": [
                  "number",
                  "null"
                ]
              }
            },
            "uv_index": {
              "type": "array",
              "minItems": 1,
              "items": {
                "type": [
                  "number",
                  "null"
                ]
              }
            },
            "temperature_2m": {
              "type": "array",
              "minItems": 1,
              "items": {
                "type": [
                  "number",
                  "null"
                ]
              }
            }
          },
          "required": [
            "time",
            "precipitation",
            "wind_speed_10m",
            "wind_direction_10m",
            "cloud_cover",
            "uv_index",
            "temperature_2m"
          ]
        }
      },
      "required": [
        "latitude",
        "longitude",
        "generationtime_ms",
        "utc_offset_seconds",
        "timezone",
        "timezone_abbreviation",
        "hourly_units",
        "hourly"
      ]
    },
    "dataset_key": {
      "enum": [
        "open_meteo_gfs"
      ]
    }
  }
 }
 ]
}
```


C02 A0 normalized envelope

Atribut	Definisi
Producer → consumer	fixture normalization → validate_normalized + SQL guards
Interface / format	JSON / Draft 2020-12
Classification	explicit
Database mapping	observation / forecast_run / forecast_value / source_grid
Compatibility	Versi konstan bila ditetapkan; negosiasi umum NOT FOUND
Evidence	Seagis-Model/scripts/validate_contracts.py:1

Field	schema_version
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Versi struktur kontrak; bukan versi model biologis
Validation / allowed values	{"const": "0.1.0"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"0.1.0"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	dataset_key
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Kunci dataset terdaftar
Validation / allowed values	{"enum": ["sensor_temperature_v1", "cmems_salinity_6h", "cmems_currents_6h", "cmems_nutrients_daily", "open_meteo_gfs"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"sensor_temperature_v1"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "object", "additionalProperties": false, "required": ["id", "request_key", "adapter_version", "started_at", "completed_at", "status", "is_synthetic"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Identitas baris menurut PK tabel
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"

DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.request_key
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Kunci logis retry ingestion per dataset
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.adapter_version
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Versi implementasi adapter ingestion
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.started_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Instan dimulainya ingestion atau attempt
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.completed_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Instan selesai; null selama running
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"

DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"const": "succeeded"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"succeeded"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	ingestion_run.is_synthetic
Type / ref	"boolean"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Penanda asal sintetis, terpisah dari QC
Validation / allowed values	{"type": "boolean"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	false
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	raw_payload
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "object", "additionalProperties": false, "required": ["id", "sha256", "received_at", "format", " licence_snapshot"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	raw_payload.id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Identitas baris menurut PK tabel
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	raw_payload.sha256
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Digest raw sesuai hash_basis, bukan selalu byte jaringan
Validation / allowed values	{"type": "string", "pattern": "^[0-9a-f]{64}$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	raw_payload.received_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Instan penerimaan raw oleh worker
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	raw_payload.format
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Media type/format representasi raw
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	raw_payload.licence_snapshot
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Keterangan lisensi saat capture
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0
Type / ref	"object"
Required	False
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "object", "additionalProperties": false, "required": ["id", "run_key", "run_kind", "provider_run_id", " model_initialized_at", "issued_at", "issue_time_basis", "ingested_at"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Identitas baris menurut PK tabel
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.run_key
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Identitas capture semantik per dataset
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.run_kind
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	forecast, analysis atau analysis_forecast_unspecified
Validation / allowed values	{"enum": ["forecast", "analysis"]}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"forecast"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.provider_run_id
Type / ref	["string", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Arti	ID run provider jika benar-benar tersedia
Validation / allowed values	{"type": ["string", "null"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.model_initialized_at/anyOf/0
Type / ref	"string"
Required	False
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.model_initialized_at/anyOf/1
Type / ref	"null"
Required	False
Nullable	True
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "null"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.issued_at/anyOf/0
Type / ref	"string"
Required	False
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.issued_at/anyOf/1
Type / ref	"null"
Required	False
Nullable	True
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "null"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.issue_time_basis
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Dasar known/unknown waktu publikasi
Validation / allowed values	{"enum": ["provider_verified", "unknown"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"provider_verified"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/0.ingested_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Instan data tersedia bagi sistem
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	forecast_run/anyOf/1
Type / ref	"null"
Required	False
Nullable	True
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "null"}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0
Type / ref	"object"
Required	False
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "object", "additionalProperties": false, "required": ["id", "grid_key", "grid_version", "identity_basis", " latitude", "longitude", "depth_m"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Identitas baris menurut PK tabel
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.grid_key
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Kunci sel berbasis koordinat/depth atau fixture
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.grid_version
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Versi identitas grid

Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.identity_basis
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Dasar identitas sel asli, koordinat respons atau sintetis
Validation / allowed values	{"enum": ["provider_cell", "provider_returned_coordinate", "synthetic"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"provider_cell"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.latitude
Type / ref	"number"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Latitude WGS84 dalam derajat
Validation / allowed values	{"type": "number", "minimum": -90, "maximum": 90}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	0
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.longitude
Type / ref	"number"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Longitude WGS84 dalam derajat
Validation / allowed values	{"type": "number", "minimum": -180, "maximum": 180}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	0
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/0.depth_m
Type / ref	"number"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Kedalaman positif ke bawah dalam meter

Validation / allowed values	{"type": "number", "minimum": 0}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	0
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	source_grid/anyOf/1
Type / ref	"null"
Required	False
Nullable	True
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "null"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Daftar observation/forecast_value yang divalidasi lintas field
Validation / allowed values	{"type": "array", "minItems": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0
Type / ref	"object"
Required	False
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "object", "additionalProperties": false, "required": ["id", "parameter_code", "native_value", "native_unit", "value", "unit", "conversion_code", "value_status", "detection_limit", "ingested_at", "quality_status", "kind", "plot_id", "sensor_id", "source_record_key", "observed_at"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND

Arti	Identitas baris menurut PK tabel
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.parameter_code
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Identitas parameter lingkungan
Validation / allowed values	{"enum": ["water_temperature", "salinity", "current_eastward", "current_northward", "nitrate", "phosphate", " precipitation", "wind_speed_10m", "wind_direction_10m", "cloud_cover", "uv_index", "air_temperature_2m"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"water_temperature"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.native_value
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Arti	Nilai fisik dalam satuan sumber
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.native_unit
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Satuan asli sebelum normalisasi
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.value
Type / ref	["number", "null"]
Required	True
Nullable	True

Default	NOT FOUND
Arti	Nilai normalisasi; null mengikuti value_status
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.unit
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Satuan nilai normalisasi menurut registri
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.conversion_code
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Identitas metode konversi satuan
Validation / allowed values	{"enum": ["identity", "unit_notation"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"identity"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.value_status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Ketersediaan angka, terpisah dari kualitas
Validation / allowed values	{"enum": ["present", "missing", "below_detection_limit", "not_available"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"present"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.detection_limit
Type / ref	["number", "null"]
Required	True
Nullable	True

Default	NOT FOUND
Arti	Batas deteksi pada satuan normalisasi bila relevan
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.ingested_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Instan data tersedia bagi sistem
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.quality_status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Status QC unreviewed, accepted, suspect atau rejected; bukan diagnosis
Validation / allowed values	{"enum": ["unreviewed", "accepted", "suspect", "rejected"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"unreviewed"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.kind
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Jenis aktor device atau reader
Validation / allowed values	{"const": "observation"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"observation"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.plot_id
Type / ref	"string"
Required	True
Nullable	False

Default	NOT FOUND
Arti	Referensi plot budidaya
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.sensor_id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Referensi deployment sensor
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.source_record_key
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	boot_id:sequence untuk paket A1
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/0.observed_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Instan pengukuran sensor, bukan receipt
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1
Type / ref	"object"
Required	False
Nullable	False

Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "object", "additionalProperties": false, "required": ["id", "parameter_code", "native_value", "native_unit", " value", "unit", "conversion_code", "value_status", "detection_limit", "ingested_at", "quality_status", "kind", " valid_at", "temporal_support", "interval_status", "period_start_at", "period_end_at"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Identitas baris menurut PK tabel
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.parameter_code
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Identitas parameter lingkungan
Validation / allowed values	{"enum": ["water_temperature", "salinity", "current_eastward", "current_northward", "nitrate", "phosphate", " precipitation", "wind_speed_10m", "wind_direction_10m", "cloud_cover", "uv_index", "air_temperature_2m"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"water_temperature"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.native_value
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Arti	Nilai fisik dalam satuan sumber
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.native_unit
Type / ref	"string"

Required	True
Nullable	False
Default	NOT FOUND
Arti	Satuan asli sebelum normalisasi
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.value
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Arti	Nilai normalisasi; null mengikuti value_status
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.unit
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Satuan nilai normalisasi menurut registri
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.conversion_code
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Identitas metode konversi satuan
Validation / allowed values	{"enum": ["identity", "unit_notation"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"identity"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.value_status
Type / ref	"const/enum"

Required	True
Nullable	False
Default	NOT FOUND
Arti	Ketersediaan angka, terpisah dari kualitas
Validation / allowed values	{"enum": ["present", "missing", "below_detection_limit", "not_available"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"present"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.detection_limit
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Arti	Batas deteksi pada satuan normalisasi bila relevan
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.ingested_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Instan data tersedia bagi sistem
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.quality_status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Status QC unreviewed, accepted, suspect atau rejected; bukan diagnosis
Validation / allowed values	{"enum": ["unreviewed", "accepted", "suspect", "rejected"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"unreviewed"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.kind
Type / ref	"const/enum"

Required	True
Nullable	False
Default	NOT FOUND
Arti	Jenis aktor device atau reader
Validation / allowed values	{"const": "forecast_value"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"forecast_value"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.valid_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Waktu representatif nilai model
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.temporal_support
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Dukungan waktu instantaneous, mean atau sum
Validation / allowed values	{"enum": ["instantaneous", "mean", "sum"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"instantaneous"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.interval_status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	known, unknown atau not_applicable untuk interval
Validation / allowed values	{"enum": ["not_applicable", "known", "unknown"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"not_applicable"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.period_start_at/anyOf/0
Type / ref	"string"

Required	False
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.period_start_at/anyOf/1
Type / ref	"null"
Required	False
Nullable	True
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "null"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.period_end_at/anyOf/0
Type / ref	"string"
Required	False
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "Z$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Field	measurements[]/oneOf/1.period_end_at/anyOf/1
Type / ref	"null"
Required	False
Nullable	True
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "null"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation / forecast_run / forecast_value / source_grid

Schema lengkap untuk aturan kondisional dan struktur polimorfik (required di atas berlaku dalam parent masing-masing):

{

```
 "type": "object",
 "additionalProperties": false,
 "properties": {
  "schema_version": {
   "const": "0.1.0"
  },
  "dataset_key": {
   "enum": [
    "sensor_temperature_v1",
    "cmems_salinity_6h",
    "cmems_currents_6h",
    "cmems_nutrients_daily",
    "open_meteo_gfs"
   ]
  },
  "ingestion_run": {
   "type": "object",
   "additionalProperties": false,
   "properties": {
    "id": {
     "type": "string",
     "format": "uuid"
    },
    "request_key": {
     "type": "string",
     "minLength": 1
    },
    "adapter_version": {
     "type": "string",
     "minLength": 1
    },
    "started_at": {
     "type": "string",
     "format": "date-time",
     "pattern": "Z$"
    },
    "completed_at": {
     "type": "string",
     "format": "date-time",
     "pattern": "Z$"
    },
    "status": {
     "const": "succeeded"
    },
    "is_synthetic": {
     "type": "boolean"
    }
   },
   "required": [
    "id",
    "request_key",
    "adapter_version",
    "started_at",
    "completed_at",
    "status",
    "is_synthetic"
   ]
  },
  "raw_payload": {
   "type": "object",
   "additionalProperties": false,
   "properties": {
    "id": {
     "type": "string",
     "format": "uuid"
    },
    "sha256": {
     "type": "string",
     "pattern": "^[0-9a-f]{64}$"
    },
    "received_at": {
     "type": "string",
     "format": "date-time",
     "pattern": "Z$"
    },
    "format": {
     "type": "string",
     "minLength": 1
    },
    "licence_snapshot": {
     "type": "string",
     "minLength": 1
    }
```

```
    },
    "required": [
      "id",
      "sha256",
      "received_at",
      "format",
      "licence_snapshot"
    ]
  },
  "forecast_run": {
    "anyOf": [
      {
        "type": "object",
        "additionalProperties": false,
        "properties": {
          "id": {
            "type": "string",
            "format": "uuid"
          },
          "run_key": {
            "type": "string",
            "minLength": 1
          },
          "run_kind": {
            "enum": [
              "forecast",
              "analysis"
            ]
          },
          "provider_run_id": {
            "type": [
              "string",
              "null"
            ]
          },
          "model_initialized_at": {
            "anyOf": [
              {
                "type": "string",
                "format": "date-time",
                "pattern": "Z$"
              },
              {
                "type": "null"
              }
            ]
          },
          "issued_at": {
            "anyOf": [
              {
                "type": "string",
                "format": "date-time",
                "pattern": "Z$"
              },
              {
                "type": "null"
              }
            ]
          },
          "issue_time_basis": {
            "enum": [
              "provider_verified",
              "unknown"
            ]
          },
          "ingested_at": {
            "type": "string",
            "format": "date-time",
            "pattern": "Z$"
          }
        },
        "required": [
          "id",
          "run_key",
          "run_kind",
          "provider_run_id",
          "model_initialized_at",
          "issued_at",
          "issue_time_basis",
          "ingested_at"
        ]
      },
      {
```

```
        "type": "null"
      }
    ]
  },
  "source_grid": {
    "anyOf": [
      {
        "type": "object",
        "additionalProperties": false,
        "properties": {
          "id": {
            "type": "string",
            "format": "uuid"
          },
          "grid_key": {
            "type": "string",
            "minLength": 1
          },
          "grid_version": {
            "type": "string",
            "minLength": 1
          },
          "identity_basis": {
            "enum": [
              "provider_cell",
              "provider_returned_coordinate",
              "synthetic"
            ]
          },
          "latitude": {
            "type": "number",
            "minimum": -90,
            "maximum": 90
          },
          "longitude": {
            "type": "number",
            "minimum": -180,
            "maximum": 180
          },
          "depth_m": {
            "type": "number",
            "minimum": 0
          }
        },
        "required": [
          "id",
          "grid_key",
          "grid_version",
          "identity_basis",
          "latitude",
          "longitude",
          "depth_m"
        ]
      },
      {
        "type": "null"
      }
    ]
  },
  "measurements": {
    "type": "array",
    "minItems": 1,
    "items": {
      "oneOf": [
        {
          "type": "object",
          "additionalProperties": false,
          "properties": {
            "id": {
              "type": "string",
              "format": "uuid"
            },
            "parameter_code": {
              "enum": [
                "water_temperature",
                "salinity",
                "current_eastward",
                "current_northward",
                "nitrate",
                "phosphate",
                "precipitation",
                "wind_speed_10m",
                "wind_direction_10m",
```

```
        "cloud_cover",
        "uv_index",
        "air_temperature_2m"
      ]
    },
    "native_value": {
      "type": [
        "number",
        "null"
      ]
    },
    "native_unit": {
      "type": "string"
    },
    "value": {
      "type": [
        "number",
        "null"
      ]
    },
    "unit": {
      "type": "string",
      "minLength": 1
    },
    "conversion_code": {
      "enum": [
        "identity",
        "unit_notation"
      ]
    },
    "value_status": {
      "enum": [
        "present",
        "missing",
        "below_detection_limit",
        "not_available"
      ]
    },
    "detection_limit": {
      "type": [
        "number",
        "null"
      ]
    },
    "ingested_at": {
      "type": "string",
      "format": "date-time",
      "pattern": "Z$"
    },
    "quality_status": {
      "enum": [
        "unreviewed",
        "accepted",
        "suspect",
        "rejected"
      ]
    },
    "kind": {
      "const": "observation"
    },
    "plot_id": {
      "type": "string",
      "format": "uuid"
    },
    "sensor_id": {
      "type": "string",
      "format": "uuid"
    },
    "source_record_key": {
      "type": "string",
      "minLength": 1
    },
    "observed_at": {
      "type": "string",
      "format": "date-time",
      "pattern": "Z$"
    }
  },
  "required": [
    "id",
    "parameter_code",
    "native_value",
    "native_unit",
```

```
      "value",
      "unit",
      "conversion_code",
      "value_status",
      "detection_limit",
      "ingested_at",
      "quality_status",
      "kind",
      "plot_id",
      "sensor_id",
      "source_record_key",
      "observed_at"
    ]
  },
  {
    "type": "object",
    "additionalProperties": false,
    "properties": {
      "id": {
        "type": "string",
        "format": "uuid"
      },
      "parameter_code": {
        "enum": [
          "water_temperature",
          "salinity",
          "current_eastward",
          "current_northward",
          "nitrate",
          "phosphate",
          "precipitation",
          "wind_speed_10m",
          "wind_direction_10m",
          "cloud_cover",
          "uv_index",
          "air_temperature_2m"
        ]
      },
      "native_value": {
        "type": [
          "number",
          "null"
        ]
      },
      "native_unit": {
        "type": "string"
      },
      "value": {
        "type": [
          "number",
          "null"
        ]
      },
      "unit": {
        "type": "string",
        "minLength": 1
      },
      "conversion_code": {
        "enum": [
          "identity",
          "unit_notation"
        ]
      },
      "value_status": {
        "enum": [
          "present",
          "missing",
          "below_detection_limit",
          "not_available"
        ]
      },
      "detection_limit": {
        "type": [
          "number",
          "null"
        ]
      },
      "ingested_at": {
        "type": "string",
        "format": "date-time",
        "pattern": "Z$"
      },
      "quality_status": {
```

```
            "enum": [
              "unreviewed",
              "accepted",
              "suspect",
              "rejected"
            ]
          },
          "kind": {
            "const": "forecast_value"
          },
          "valid_at": {
            "type": "string",
            "format": "date-time",
            "pattern": "Z$"
          },
          "temporal_support": {
            "enum": [
              "instantaneous",
              "mean",
              "sum"
            ]
          },
          "interval_status": {
            "enum": [
              "not_applicable",
              "known",
              "unknown"
            ]
          },
          "period_start_at": {
            "anyOf": [
              {
                "type": "string",
                "format": "date-time",
                "pattern": "Z$"
              },
              {
                "type": "null"
              }
            ]
          },
          "period_end_at": {
            "anyOf": [
              {
                "type": "string",
                "format": "date-time",
                "pattern": "Z$"
              },
              {
                "type": "null"
              }
            ]
          }
        },
        "required": [
          "id",
          "parameter_code",
          "native_value",
          "native_unit",
          "value",
          "unit",
          "conversion_code",
          "value_status",
          "detection_limit",
          "ingested_at",
          "quality_status",
          "kind",
          "valid_at",
          "temporal_support",
          "interval_status",
          "period_start_at",
          "period_end_at"
        ]
      }
    ]
  }
 }
},
"required": [
  "schema_version",
  "dataset_key",
  "ingestion_run",
  "raw_payload",
```

```
    "forecast_run",
    "source_grid",
    "measurements"
  ],
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://seagis.local/contracts/normalized.schema.json",
  "title": "SEAGIS normalized envelope v0.1.0",
  "allOf": [
    {
      "if": {
        "properties": {
          "dataset_key": {
            "const": "sensor_temperature_v1"
          }
        }
      },
      "then": {
        "properties": {
          "forecast_run": {
            "type": "null"
          },
          "source_grid": {
            "type": "null"
          },
          "measurements": {
            "type": "array",
            "minItems": 1,
            "items": {
              "type": "object",
              "additionalProperties": false,
              "properties": {
                "id": {
                  "type": "string",
                  "format": "uuid"
                },
                "parameter_code": {
                  "enum": [
                    "water_temperature",
                    "salinity",
                    "current_eastward",
                    "current_northward",
                    "nitrate",
                    "phosphate",
                    "precipitation",
                    "wind_speed_10m",
                    "wind_direction_10m",
                    "cloud_cover",
                    "uv_index",
                    "air_temperature_2m"
                  ]
                },
                "native_value": {
                  "type": [
                    "number",
                    "null"
                  ]
                },
                "native_unit": {
                  "type": "string"
                },
                "value": {
                  "type": [
                    "number",
                    "null"
                  ]
                },
                "unit": {
                  "type": "string",
                  "minLength": 1
                },
                "conversion_code": {
                  "enum": [
                    "identity",
                    "unit_notation"
                  ]
                },
                "value_status": {
                  "enum": [
                    "present",
                    "missing",
                    "below_detection_limit",
                    "not_available"
                  ]
```

```
            },
            "detection_limit": {
              "type": [
                "number",
                "null"
              ]
            },
            "ingested_at": {
              "type": "string",
              "format": "date-time",
              "pattern": "Z$"
            },
            "quality_status": {
              "enum": [
                "unreviewed",
                "accepted",
                "suspect",
                "rejected"
              ]
            },
            "kind": {
              "const": "observation"
            },
            "plot_id": {
              "type": "string",
              "format": "uuid"
            },
            "sensor_id": {
              "type": "string",
              "format": "uuid"
            },
            "source_record_key": {
              "type": "string",
              "minLength": 1
            },
            "observed_at": {
              "type": "string",
              "format": "date-time",
              "pattern": "Z$"
            }
          },
          "required": [
            "id",
            "parameter_code",
            "native_value",
            "native_unit",
            "value",
            "unit",
            "conversion_code",
            "value_status",
            "detection_limit",
            "ingested_at",
            "quality_status",
            "kind",
            "plot_id",
            "sensor_id",
            "source_record_key",
            "observed_at"
          ]
        }
      }
    }
  },
  "else": {
    "properties": {
      "forecast_run": {
        "type": "object",
        "additionalProperties": false,
        "properties": {
          "id": {
            "type": "string",
            "format": "uuid"
          },
          "run_key": {
            "type": "string",
            "minLength": 1
          },
          "run_kind": {
            "enum": [
              "forecast",
              "analysis"
            ]
          },
```

```
      "provider_run_id": {
        "type": [
          "string",
          "null"
        ]
      },
      "model_initialized_at": {
        "anyOf": [
          {
            "type": "string",
            "format": "date-time",
            "pattern": "Z$"
          },
          {
            "type": "null"
          }
        ]
      },
      "issued_at": {
        "anyOf": [
          {
            "type": "string",
            "format": "date-time",
            "pattern": "Z$"
          },
          {
            "type": "null"
          }
        ]
      },
      "issue_time_basis": {
        "enum": [
          "provider_verified",
          "unknown"
        ]
      },
      "ingested_at": {
        "type": "string",
        "format": "date-time",
        "pattern": "Z$"
      }
    },
    "required": [
      "id",
      "run_key",
      "run_kind",
      "provider_run_id",
      "model_initialized_at",
      "issued_at",
      "issue_time_basis",
      "ingested_at"
    ]
  },
  "source_grid": {
    "type": "object",
    "additionalProperties": false,
    "properties": {
      "id": {
        "type": "string",
        "format": "uuid"
      },
      "grid_key": {
        "type": "string",
        "minLength": 1
      },
      "grid_version": {
        "type": "string",
        "minLength": 1
      },
      "identity_basis": {
        "enum": [
          "provider_cell",
          "provider_returned_coordinate",
          "synthetic"
        ]
      },
      "latitude": {
        "type": "number",
        "minimum": -90,
        "maximum": 90
      },
      "longitude": {
        "type": "number",
```

```
        "minimum": -180,
        "maximum": 180
      },
      "depth_m": {
        "type": "number",
        "minimum": 0
      }
    },
    "required": [
      "id",
      "grid_key",
      "grid_version",
      "identity_basis",
      "latitude",
      "longitude",
      "depth_m"
    ]
  },
  "measurements": {
    "type": "array",
    "minItems": 1,
    "items": {
      "type": "object",
      "additionalProperties": false,
      "properties": {
        "id": {
          "type": "string",
          "format": "uuid"
        },
        "parameter_code": {
          "enum": [
            "water_temperature",
            "salinity",
            "current_eastward",
            "current_northward",
            "nitrate",
            "phosphate",
            "precipitation",
            "wind_speed_10m",
            "wind_direction_10m",
            "cloud_cover",
            "uv_index",
            "air_temperature_2m"
          ]
        },
        "native_value": {
          "type": [
            "number",
            "null"
          ]
        },
        "native_unit": {
          "type": "string"
        },
        "value": {
          "type": [
            "number",
            "null"
          ]
        },
        "unit": {
          "type": "string",
          "minLength": 1
        },
        "conversion_code": {
          "enum": [
            "identity",
            "unit_notation"
          ]
        },
        "value_status": {
          "enum": [
            "present",
            "missing",
            "below_detection_limit",
            "not_available"
          ]
        },
        "detection_limit": {
          "type": [
            "number",
            "null"
          ]
```

```
        },
        "ingested_at": {
          "type": "string",
          "format": "date-time",
          "pattern": "Z$"
        },
        "quality_status": {
          "enum": [
            "unreviewed",
            "accepted",
            "suspect",
            "rejected"
          ]
        },
        "kind": {
          "const": "forecast_value"
        },
        "valid_at": {
          "type": "string",
          "format": "date-time",
          "pattern": "Z$"
        },
        "temporal_support": {
          "enum": [
            "instantaneous",
            "mean",
            "sum"
          ]
        },
        "interval_status": {
          "enum": [
            "not_applicable",
            "known",
            "unknown"
          ]
        },
        "period_start_at": {
          "anyOf": [
            {
              "type": "string",
              "format": "date-time",
              "pattern": "Z$"
            },
            {
              "type": "null"
            }
          ]
        },
        "period_end_at": {
          "anyOf": [
            {
              "type": "string",
              "format": "date-time",
              "pattern": "Z$"
            },
            {
              "type": "null"
            }
          ]
        }
      },
      "required": [
        "id",
        "parameter_code",
        "native_value",
        "native_unit",
        "value",
        "unit",
        "conversion_code",
        "value_status",
        "detection_limit",
        "ingested_at",
        "quality_status",
        "kind",
        "valid_at",
        "temporal_support",
        "interval_status",
        "period_start_at",
        "period_end_at"
      ]
    }
  }
}
```

```
    }
   }
  ]
}
```

C03 A1 device packet

Atribut	Definisi
Producer → consumer	device/simulator → temperature handler + seagis_api.ingest
Interface / format	JSON / HTTP POST
Classification	explicit
Database mapping	observation + raw_payload + packet_receipt + quality_event
Compatibility	Versi konstan bila ditetapkan; negosiasi umum NOT FOUND
Evidence	Seagis-Model/supabase/functions/temperature/handler.ts:1

Field	schema_version
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Versi struktur kontrak; bukan versi model biologis
Validation / allowed values	{"const": "1.0.0"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"1.0.0"
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	sensor_id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Referensi deployment sensor
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	boot_id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Identitas sesi boot sensor untuk idempotensi
Validation / allowed values	{"type": "string", "pattern": "^[A-Za-z0-9._-]{1,64}$"}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	sequence
Type / ref	"integer"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Counter uint32 dalam satu boot; key digabung dengan boot_id
Validation / allowed values	{"type": "integer", "minimum": 0, "maximum": 4294967295}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	1
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	observed_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Instan pengukuran sensor, bukan receipt
Validation / allowed values	{"type": "string", "format": "date-time", "pattern": "^\\d{4}-\\d{2}-\\d{2}T\\d{2}:\\d{2}:\\d{2}(\\.\\d{1,6})?Z$"}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	water_temperature
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Arti	Suhu air degC atau null missing; bukan suhu udara
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	unit
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Satuan nilai normalisasi menurut registri
Validation / allowed values	{"const": "degC"}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	"degC"
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Field	value_status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Ketersediaan angka, terpisah dari kualitas
Validation / allowed values	{"enum": ["present", "missing"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"present"
DB mapping (boundary)	observation + raw_payload + packet_receipt + quality_event

Schema lengkap untuk aturan kondisional dan struktur polimorfik (required di atas berlaku dalam parent masing-masing):


```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://seagis.local/contracts/iot-packet/1.0.0",
  "title": "SEAGIS temperature packet 1.0.0",
  "type": "object",
  "additionalProperties": false,
  "required": [
    "schema_version",
    "sensor_id",
    "boot_id",
    "sequence",
    "observed_at",
    "water_temperature",
    "unit",
    "value_status"
  ],
  "properties": {
    "schema_version": {
      "const": "1.0.0"
    },
    "sensor_id": {
      "type": "string",
      "format": "uuid"
    },
    "boot_id": {
      "type": "string",
      "pattern": "^[A-Za-z0-9._-]{1,64}$"
    },
    "sequence": {
      "type": "integer",
      "minimum": 0,
      "maximum": 4294967295
    },
    "observed_at": {
      "type": "string",
      "format": "date-time",
      "pattern": "^\\d{4}-\\d{2}-\\d{2}T\\d{2}:\\d{2}:\\d{2}(\\.\\d{1,6})?Z$"
    },
    "water_temperature": {
      "type": [
        "number",
        "null"
      ]
    },
    "unit": {
      "const": "degC"
    },
    "value_status": {
      "enum": [
        "present",
        "missing"
      ]
    }
  },
  "allOf": [
```

```
    {
      "if": {
        "properties": {
          "value_status": {
            "const": "present"
          }
        }
      },
      "then": {
        "properties": {
          "water_temperature": {
            "type": "number"
          }
        }
      },
      "else": {
        "properties": {
          "water_temperature": {
            "type": "null"
          }
        }
      }
    }
  ]
}
```


C04 A3 alignment document

Atribut	Definisi
Producer → consumer	scripts/quality/alignment.py::build → seagis_quality.save_snapshot
Interface / format	JSON / SQL jsonb
Classification	explicit
Database mapping	seagis_quality.alignment_snapshot + snapshot_input
Compatibility	Versi konstan bila ditetapkan; negosiasi umum NOT FOUND
Evidence	Seagis-Model/scripts/curate_environment.py:1

Field	schema_version
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Versi struktur kontrak; bukan versi model biologis
Validation / allowed values	{"const": "a3-alignment-1"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"a3-alignment-1"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	builder_version
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	policy_version
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Referensi versi policy QC
Validation / allowed values	{"type": "string", "minLength": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	policy_sha256
Type / ref	"#/$defs/hash"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Digest isi policy
Validation / allowed values	{"$ref": "#/$defs/hash"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	code_sha256
Type / ref	"#/$defs/hash"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Digest source builder alignment
Validation / allowed values	{"$ref": "#/$defs/hash"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	source_sha256
Type / ref	"#/$defs/hash"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Digest source bundle untuk deteksi sumber berubah
Validation / allowed values	{"$ref": "#/$defs/hash"}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	plot_id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Referensi plot budidaya
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	mode
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	demo_review atau operational_review sesuai asal plot
Validation / allowed values	{"enum": ["demo_review", "operational_review"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"demo_review"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	use
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Pembatas penggunaan dokumen quality_alignment_review
Validation / allowed values	{"const": "quality_alignment_review"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"quality_alignment_review"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	training_authorized
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Harus false; review/demo bukan dataset training yang disahkan
Validation / allowed values	{"const": false}
Units / timestamp	NOT APPLICABLE / schema

Example (illustrative)	false
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	source_registry
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Salinan kontrak parameter yang dibandingkan dengan bundle SQL
Validation / allowed values	{"type": "array", "minItems": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	source_registry[]
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Salinan kontrak parameter yang dibandingkan dengan bundle SQL
Validation / allowed values	{"type": "object", "required": ["dataset_key", "parameter_code", "native_unit", "unit", "conversion_code", " native_cadence_seconds", "attribution", "licence_url"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	sensor_deployments
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Salinan identitas deployment untuk pemeriksaan konsistensi
Validation / allowed values	{"type": "array"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	sensor_deployments[]
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Salinan identitas deployment untuk pemeriksaan konsistensi
Validation / allowed values	{"type": "object"}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	start_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Awal inklusif jendela alignment
Validation / allowed values	{"type": "string", "format": "date-time"}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	end_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Akhir eksklusif jendela alignment
Validation / allowed values	{"type": "string", "format": "date-time"}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	rows
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Baris konteks per jam UTC; bukan sampel independen tambahan
Validation / allowed values	{"type": "array", "minItems": 1, "maxItems": 168}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	rows[]
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Baris konteks per jam UTC; bukan sampel independen tambahan
Validation / allowed values	{"type": "object", "required": ["at", "cells"], "additionalProperties": false}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	rows[].at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Cutoff jam UTC untuk pemilihan konteks
Validation / allowed values	{"type": "string", "format": "date-time"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	rows[].cells
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Konteks tiap dataset/parameter pada cutoff jam
Validation / allowed values	{"type": "array", "minItems": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	rows[].cells[]
Type / ref	"#/$defs/cell"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Konteks tiap dataset/parameter pada cutoff jam
Validation / allowed values	{"$ref": "#/$defs/cell"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Ringkasan QC dan coverage berbasis konteks jam
Validation / allowed values	{"type": "object", "required": ["plot_id", "start_at", "end_at", "coverage_basis", "drift_assessment", "parameters"]}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.plot_id
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Referensi plot budidaya
Validation / allowed values	{"type": "string", "format": "uuid"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"00000000-0000-4000-8000-000000000001"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.start_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Awal inklusif jendela alignment
Validation / allowed values	{"type": "string", "format": "date-time"}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.end_at
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Akhir eksklusif jendela alignment
Validation / allowed values	{"type": "string", "format": "date-time"}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	"2026-10-21T12:00:00Z"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.coverage_basis
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "string"}

Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.drift_assessment
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"const": "not_assessable_without_calibrated_colocated_reference"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"not_assessable_without_calibrated_colocated_reference"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.parameters
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "array"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.parameters[]
Type / ref	"object"
Required	True
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "object", "required": ["dataset_key", "parameter_code", "expected_hourly_contexts", "usable_contexts", "unique_selected_source_records", "native_cadence_seconds", "coverage_fraction", "value_status_counts", "flag_counts", "input_qc_flag_counts_at_final_cutoff"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.parameters[].expected_hourly_contexts
Type / ref	"integer"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Jumlah slot jam pada jendela, bukan jumlah sampel asli

Validation / allowed values	{"type": "integer", "minimum": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	1
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.parameters[].usable_contexts
Type / ref	"integer"
Required	True
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "integer", "minimum": 0}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	1
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.parameters[].unique_selected_source_records
Type / ref	"integer"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Jumlah record asal unik yang dipilih
Validation / allowed values	{"type": "integer", "minimum": 0}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	1
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	quality_report.parameters[].coverage_fraction
Type / ref	"number"
Required	True
Nullable	False
Default	NOT FOUND
Arti	usable_contexts / expected_hourly_contexts
Validation / allowed values	{"type": "number", "minimum": 0, "maximum": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	0
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.hash
Type / ref	"string"
Required	False
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi

Validation / allowed values	{"type": "string", "pattern": "^[0-9a-f]{64}$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.nullableTime
Type / ref	["string", "null"]
Required	False
Nullable	True
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": ["string", "null"], "format": "date-time"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell
Type / ref	"object"
Required	False
Nullable	False
Default	NOT FOUND
Arti	UNVERIFIED: struktur/field kontrak; arti rinci mengikuti validator dan contoh sumber, belum seluruhnya diverifikasi
Validation / allowed values	{"type": "object", "required": ["dataset_key", "parameter_code", "scope", "native_cadence_seconds", "source_id", "value", "value_status", "unit", "source_time", "available_at", "age_seconds", "vintage_age_seconds", "lineage", "spatial", "flags", "blocking_flags", "usable_context"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	{}
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.dataset_key
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Kunci dataset terdaftar
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.parameter_code
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND

Arti	Identitas parameter lingkungan
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.scope
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Asal local_sensor atau regional_model
Validation / allowed values	{"enum": ["local_sensor", "regional_model"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"local_sensor"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.native_cadence_seconds
Type / ref	"integer"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Jarak waktu asli sampel, bukan interval fetch
Validation / allowed values	{"type": "integer", "minimum": 1}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	1
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.source_id
Type / ref	["string", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Arti	Typed reference record asal terpilih, atau null
Validation / allowed values	{"type": ["string", "null"], "pattern": "^(observation|forecast_value):[0-9a-f-]{36}$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.value
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND

Arti	Nilai normalisasi; null mengikuti value_status
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.native_value
Type / ref	["number", "null"]
Required	False
Nullable	True
Default	NOT FOUND
Arti	Nilai fisik dalam satuan sumber
Validation / allowed values	{"type": ["number", "null"]}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.value_status
Type / ref	"const/enum"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Ketersediaan angka, terpisah dari kualitas
Validation / allowed values	{"enum": ["present", "missing", "below_detection_limit", "not_available"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"present"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.unit
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Satuan nilai normalisasi menurut registri
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.source_time
Type / ref	"#/$defs/nullableTime"
Required	True
Nullable	True
Default	NOT FOUND

Arti	observed_at sensor atau valid_at model
Validation / allowed values	{"$ref": "#/$defs/nullableTime"}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.available_at
Type / ref	"#/$defs/nullableTime"
Required	True
Nullable	True
Default	NOT FOUND
Arti	Waktu konservatif ketersediaan input
Validation / allowed values	{"$ref": "#/$defs/nullableTime"}
Units / timestamp	Satuan/waktu mengikuti unit dan parameter registry; UTC untuk envelope normalized.
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.age_seconds
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Arti	Umur source_time terhadap cutoff dalam detik
Validation / allowed values	{"type": ["number", "null"], "minimum": 0}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.vintage_age_seconds
Type / ref	["number", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Arti	Umur run terhadap cutoff dalam detik
Validation / allowed values	{"type": ["number", "null"], "minimum": 0}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.lineage
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND

Arti	Referensi typed record terpilih dan input pendukung QC
Validation / allowed values	{"type": "array", "uniqueItems": true}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.lineage[]
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Referensi typed record terpilih dan input pendukung QC
Validation / allowed values	{"type": "string", "pattern": "^(observation|forecast_value):[0-9a-f-]{36}$"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.spatial
Type / ref	["object", "null"]
Required	True
Nullable	True
Default	NOT FOUND
Arti	Metadata pemetaan grid, resolusi, jarak dan kedalaman
Validation / allowed values	{"type": ["object", "null"]}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	null
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.flags
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Kode QC dari penerimaan paket
Validation / allowed values	{"type": "array", "uniqueItems": true}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.flags[]
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND

Arti	Kode QC dari penerimaan paket
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.blocking_flags
Type / ref	"array"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Alasan konteks tidak dapat dipakai menurut builder
Validation / allowed values	{"type": "array", "uniqueItems": true}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	[]
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.blocking_flags[]
Type / ref	"string"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Alasan konteks tidak dapat dipakai menurut builder
Validation / allowed values	{"type": "string"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	"Lihat fixture/schema lengkap; contoh field mandiri belum diverifikasi"
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Field	$defs.cell.usable_context
Type / ref	"boolean"
Required	True
Nullable	False
Default	NOT FOUND
Arti	Keputusan QC builder untuk review; bukan izin training
Validation / allowed values	{"type": "boolean"}
Units / timestamp	NOT APPLICABLE / schema
Example (illustrative)	false
DB mapping (boundary)	seagis_quality.alignment_snapshot + snapshot_input

Schema lengkap untuk aturan kondisional dan struktur polimorfik (required di atas berlaku dalam parent masing-masing):


```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://seagis.local/contracts/alignment.schema.json",
  "title": "A3 quality and hourly alignment snapshot",
  "type": "object",
  "required": [
    "schema_version",
```

```
    "builder_version",
    "policy_version",
    "policy_sha256",
    "code_sha256",
    "source_sha256",
    "plot_id",
    "mode",
    "use",
    "training_authorized",
    "source_registry",
    "sensor_deployments",
    "start_at",
    "end_at",
    "rows",
    "quality_report"
  ],
  "additionalProperties": false,
  "properties": {
    "schema_version": {
      "const": "a3-alignment-1"
    },
    "builder_version": {
      "type": "string",
      "minLength": 1
    },
    "policy_version": {
      "type": "string",
      "minLength": 1
    },
    "policy_sha256": {
      "$ref": "#/$defs/hash"
    },
    "code_sha256": {
      "$ref": "#/$defs/hash"
    },
    "source_sha256": {
      "$ref": "#/$defs/hash"
    },
    "plot_id": {
      "type": "string",
      "format": "uuid"
    },
    "mode": {
      "enum": [
        "demo_review",
        "operational_review"
      ]
    },
    "use": {
      "const": "quality_alignment_review"
    },
    "training_authorized": {
      "const": false
    },
    "source_registry": {
      "type": "array",
      "minItems": 1,
      "items": {
        "type": "object",
        "required": [
          "dataset_key",
          "parameter_code",
          "native_unit",
          "unit",
          "conversion_code",
          "native_cadence_seconds",
          "attribution",
          "licence_url"
        ]
      }
    },
    "sensor_deployments": {
      "type": "array",
      "items": {
        "type": "object"
      }
    },
    "start_at": {
      "type": "string",
      "format": "date-time"
    },
    "end_at": {
      "type": "string",
```

```
    "format": "date-time"
  },
  "rows": {
    "type": "array",
    "minItems": 1,
    "maxItems": 168,
    "items": {
      "type": "object",
      "required": [
        "at",
        "cells"
      ],
      "additionalProperties": false,
      "properties": {
        "at": {
          "type": "string",
          "format": "date-time"
        },
        "cells": {
          "type": "array",
          "minItems": 1,
          "items": {
            "$ref": "#/$defs/cell"
          }
        }
      }
    }
  },
  "quality_report": {
    "type": "object",
    "required": [
      "plot_id",
      "start_at",
      "end_at",
      "coverage_basis",
      "drift_assessment",
      "parameters"
    ],
    "properties": {
      "plot_id": {
        "type": "string",
        "format": "uuid"
      },
      "start_at": {
        "type": "string",
        "format": "date-time"
      },
      "end_at": {
        "type": "string",
        "format": "date-time"
      },
      "coverage_basis": {
        "type": "string"
      },
      "drift_assessment": {
        "const": "not_assessable_without_calibrated_colocated_reference"
      },
      "parameters": {
        "type": "array",
        "items": {
          "type": "object",
          "required": [
            "dataset_key",
            "parameter_code",
            "expected_hourly_contexts",
            "usable_contexts",
            "unique_selected_source_records",
            "native_cadence_seconds",
            "coverage_fraction",
            "value_status_counts",
            "flag_counts",
            "input_qc_flag_counts_at_final_cutoff"
          ],
          "properties": {
            "expected_hourly_contexts": {
              "type": "integer",
              "minimum": 1
            },
            "usable_contexts": {
              "type": "integer",
              "minimum": 0
            },
            "unique_selected_source_records": {
```

```
            "type": "integer",
            "minimum": 0
          },
          "coverage_fraction": {
            "type": "number",
            "minimum": 0,
            "maximum": 1
          }
        }
      }
    }
  }
 },
 "$defs": {
  "hash": {
   "type": "string",
   "pattern": "^[0-9a-f]{64}$"
  },
  "nullableTime": {
   "type": [
    "string",
    "null"
   ],
   "format": "date-time"
  },
  "cell": {
   "type": "object",
   "required": [
    "dataset_key",
    "parameter_code",
    "scope",
    "native_cadence_seconds",
    "source_id",
    "value",
    "value_status",
    "unit",
    "source_time",
    "available_at",
    "age_seconds",
    "vintage_age_seconds",
    "lineage",
    "spatial",
    "flags",
    "blocking_flags",
    "usable_context"
   ],
   "properties": {
    "dataset_key": {
     "type": "string"
    },
    "parameter_code": {
     "type": "string"
    },
    "scope": {
     "enum": [
      "local_sensor",
      "regional_model"
     ]
    },
    "native_cadence_seconds": {
     "type": "integer",
     "minimum": 1
    },
    "source_id": {
     "type": [
      "string",
      "null"
     ],
     "pattern": "^(observation|forecast_value):[0-9a-f-]{36}$"
    },
    "value": {
     "type": [
      "number",
      "null"
     ]
    },
    "native_value": {
     "type": [
      "number",
      "null"
     ]
    },
```

```
      "value_status": {
        "enum": [
          "present",
          "missing",
          "below_detection_limit",
          "not_available"
        ]
      },
      "unit": {
        "type": "string"
      },
      "source_time": {
        "$ref": "#/$defs/nullableTime"
      },
      "available_at": {
        "$ref": "#/$defs/nullableTime"
      },
      "age_seconds": {
        "type": [
          "number",
          "null"
        ],
        "minimum": 0
      },
      "vintage_age_seconds": {
        "type": [
          "number",
          "null"
        ],
        "minimum": 0
      },
      "lineage": {
        "type": "array",
        "uniqueItems": true,
        "items": {
          "type": "string",
          "pattern": "^(observation|forecast_value):[0-9a-f-]{36}$"
        }
      },
      "spatial": {
        "type": [
          "object",
          "null"
        ]
      },
      "flags": {
        "type": "array",
        "uniqueItems": true,
        "items": {
          "type": "string"
        }
      },
      "blocking_flags": {
        "type": "array",
        "uniqueItems": true,
        "items": {
          "type": "string"
        }
      },
      "usable_context": {
        "type": "boolean"
      }
    }
  }
 }
}
```


C05 A1 history/status/replies

Atribut	Definisi
Producer → consumer	seagis_api.history/status/reply → authorized reader
Interface / format	JSON / HTTP GET
Classification	implementation-derived
Database mapping	observation + sensor + packet_receipt
Compatibility	Versi konstan bila ditetapkan; negosiasi umum NOT FOUND

Atribut	Definisi
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:1

Field / group	Type / required	Arti / validasi / mapping
from, to	string / required	UTC Z; from<to; <=7 days; observation.observed_at [from,to)
limit	integer string / optional; default 200	1..500
cursor	UUID / optional	observation.id inside plot/window; ordering observed_at,id
code, request_id	string, UUID / response	SQL reply / handler error
observations[]	array / history response	id,sensor_id,value,unit,value_status,observed_at,ingested_at,quality_status,flags,ingestion_run_id,is_synthetic
coverage	object	expected_hours,hours_with_packets,hours_with_usable_values,missing_hours,unusable_hours
next_cursor	UUID|null	last returned row if another page exists

C06 A2 adapter batch/lease

Atribut	Definisi
Producer → consumer	Adapter.envelope + claim → seagis_environment.persist
Interface / format	JSON / SQL jsonb
Classification	implementation-derived
Database mapping	raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
Compatibility	Versi konstan bila ditetapkan; negosiasi umum NOT FOUND
Evidence	Seagis-Model/scripts/environmental/common.py:1

Field	Required	Validasi / pemetaan
dataset_key	True	Kunci dataset terdaftar; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
purpose	True	Tujuan noncommercial_educational_prototype; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
terms_review	True	Identitas review ketentuan provider; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
dataset_version	True	Versi dataset yang dipin; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
provider_dataset_id	True	Identitas dataset asli penyedia; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
run_key	True	Identitas capture semantik per dataset; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
raw_sha256	True	Lihat persist/envelope; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
raw_byte_size	True	Lihat persist/envelope; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
hash_basis	True	Aturan byte/serialisasi yang di-hash; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
format	True	Media type/format representasi raw; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
started_at	True	Instan dimulainya ingestion atau attempt; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
received_at	True	Instan penerimaan raw oleh worker; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping

Field	Required	Validasi / pemetaan
metadata	True	Lihat persist/envelope; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
grids	True	Lihat persist/envelope; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
mappings	True	Lihat persist/envelope; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping
values	True	Lihat persist/envelope; raw_payload + forecast_run + forecast_value + source_grid + demo_grid_mapping

Nested grids berisi key, latitude, longitude, depth_m, identity_basis; values berisi grid_key, parameter, value, native_unit, unit, conversion_code, value_status, valid_at, temporal_support, interval_status dan period bounds. mappings membawa identitas plot dan jarak contoh. SQL memeriksa kehadiran field envelope, lease, tujuan/lisensi, hash 64 hex, jumlah grid/nilai/mapping; bukan JSON Schema universal.

C07 A3 source_bundle

Atribut	Definisi
Producer → consumer	seagis_quality.source_bundle → build / CSV exporter
Interface / format	JSON / SQL jsonb
Classification	implementation-derived
Database mapping	plot + sensor + dataset_parameter + mappings + observation + forecast_value
Compatibility	Versi konstan bila ditetapkan; negosiasi umum NOT FOUND
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1

Field	Type / required	Mapping
schema_version	constant a3-source-1	internal
plot	object	seagis.plot
start_at,end_at	timestamptz / required	query bounds
sensors	array	seagis.sensor
parameters	array	dataset_parameter + dataset + environment.policy
mappings	array	plot_grid_mapping + demo_grid_mapping + source_grid
records	array	observation + forecast_value + quality_event
source_sha256	64-hex string	seagis.jsonb_sha256(bundle)

C08 CSV experiment-csv-2

Atribut	Definisi
Producer → consumer	export_experiment_csv.py::export → offline experiments / developer
Interface / format	CSV + JSON manifest
Classification	implementation-derived
Database mapping	derived files; no database writes
Compatibility	Versi konstan bila ditetapkan; negosiasi umum NOT FOUND
Evidence	Seagis-Model/scripts/export_experiment_csv.py:1

Manifest export_version=experiment-csv-2; capture/export time, rentang, unit, jumlah record unik, jumlah CSV dan checksum per file. Suhu CSV membawa timestamp_utc, timestamp_wib, plot_id, value, raw_value, gap/synthetic/QC, observation_id,

available_at_utc, training_authorized=false, parameter_code, unit, source_kind dan value_policy. CSV regional membawa field record sumber datar, cadence, attribution, licence_url, selection_basis dan capture_at_utc. Field sumber dinamis berarti schema CSV mengikuti record bundle, bukan kontrak kolom tetap universal.

C09 iOS Assessment domain

Atribut	Definisi
Producer → consumer	MockAssessmentRepository → LoadDemoAssessment / ViewModel
Interface / format	Swift structs
Classification	implementation-derived / mock
Database mapping	NOT APPLICABLE: no backend mapping
Compatibility	Versi konstan bila ditetapkan; negosiasi umum NOT FOUND
Evidence	Seagis/Seagis/Models/Shared/Assessment.swift:1

Swift field	id
Type	String
Required	True
Meaning / mapping	Identitas baris menurut PK tabel

Swift field	plotID
Type	String
Required	True
Meaning / mapping	Field domain demo; bukan DTO backend

Swift field	title
Type	String
Required	True
Meaning / mapping	Field domain demo; bukan DTO backend

Swift field	summary
Type	String
Required	True
Meaning / mapping	Field domain demo; bukan DTO backend

Swift field	provenance
Type	AssessmentProvenance.syntheticFixture
Required	True
Meaning / mapping	Field domain demo; bukan DTO backend

Swift field	evidenceIDs
Type	[String]
Required	True
Meaning / mapping	Field domain demo; bukan DTO backend

C10 Open-Meteo response

Atribut	Definisi
Producer → consumer	Open-Meteo forecast API → OpenMeteo.parse/normalize
Interface / format	JSON hourly arrays
Classification	implementation-derived
Database mapping	forecast_value + grid + raw manifest
Compatibility	Versi konstan bila ditetapkan; negosiasi umum NOT FOUND
Evidence	Seagis-Model/scripts/environmental/open_meteo.py:1

Respons harus list sepanjang dua plot, setiap item memiliki latitude/longitude/elevation, timezone UTC/GMT, utc_offset_seconds=0, hourly_units.time=iso8601 dan array hourly sejajar. Field variabel dan unit ditetapkan config a2_sources.json. generationtime_ms tidak dipakai sebagai issue time. Null/angka mengikuti normalisasi common.point. Tidak ada response_error provider yang diklaim stabil.

C11 Copernicus subset

Atribut	Definisi
Producer → consumer	Copernicus Toolbox/xarray + STAC → Copernicus.parse/normalize
Interface / format	xarray + canonical logical JSON subset
Classification	implementation-derived
Database mapping	forecast_value + grid + raw manifest
Compatibility	Versi konstan bila ditetapkan; negosiasi umum NOT FOUND
Evidence	Seagis-Model/scripts/environmental/copernicus.py:1

STAC memerlukan id, collection/product, cube:dimensions time/elevation/latitude/longitude dan cube:variables unit/standardName. Dataset xarray memiliki satu latitude/longitude/depth dan time unik berurutan sesuai cadence; variabel di-transpose time,depth,latitude,longitude. Logical subset berisi cell,time,variables,native_units, bukan wire format provider. Metadata asli yang tidak diketahui tidak diciptakan.

Rute HTTP dan respons

Method	GET
Path (temperature base)	/health
Auth	none
Result / errors	200 service=seagis-temperature, version=1.0.0

Method	POST
Path (temperature base)	/v1/ingest
Auth	device Bearer
Result / errors	201 stored; 200 duplicate; 409 packet_conflict; 422 invalid_packet; 403 sensor_forbidden/sensor_inactive; 503 commissioning_disabled

Method	GET
Path (temperature base)	/v1/plots/{plot_id}/temperature
Auth	reader Bearer + reader_plot
Result / errors	200 observations/coverage; 400 invalid_query/invalid_plot_id; 403 plot_forbidden

Method	GET
Path (temperature base)	/v1/status

Auth	reader Bearer
Result / errors	200 checked_at,stale_after_seconds,plots[]

Method	all
Path (temperature base)	known protected paths
Auth	Bearer sgi_ pattern
Result / errors	401 unauthorized; 429 rate_limited

Method	all
Path (temperature base)	HTTP boundary
Auth	handler
Result / errors	404 not_found; 405 method_not_allowed; 415 json_required; 413 payload_too_large; 400 invalid_json/empty_body/unexpected_query; 503 temporarily_unavailable

Base deployment /functions/v1/temperature; handler juga menerima prefix /temperature. Semua respons no-store. Retry-After 60 untuk 429 dan 5 untuk 503. plots[] status berisi plot_id,plot_code,is_synthetic,latest_usable_observed_at,last_ingested_at,age_seconds,stale,observations_last_24h,rejected_last_24h. Waktu last_ingested_at tidak harus milik record latest_usable_observed_at. Null waktu/umur berarti belum tersedia.

Payload dan error A1

Contoh sintetis valid secara bentuk; penyimpanan memerlukan credential/sensor terdaftar dan policy aktif.

```
{
  "schema_version": "1.0.0",
  "sensor_id": "00000000-0000-4000-8000-000000000101",
  "boot_id": "synthetic-boot-a",
  "sequence": 1,
  "observed_at": "2026-10-21T12:00:00Z",
  "water_temperature": 29.2,
  "unit": "degC",
  "value_status": "present"
}
```

Contoh tidak valid: ubah unit menjadi kelvin. SQL mengembalikan 422 invalid_packet setelah autentikasi lolos.

```
{
  "code": "invalid_packet",
  "request_id": "00000000-0000-4000-8000-000000000999"
}
```

Respons stored memiliki observation_id, ingestion_run_id, observed_at, quality_status, flags, is_synthetic, code dan request_id. UUID dan waktu di bawah ilustratif, bukan response live.

```
{
  "observation_id": "00000000-0000-4000-8000-000000000201",
  "ingestion_run_id": "00000000-0000-4000-8000-000000000301",
  "observed_at": "2026-10-21T12:00:00.000000Z",
  "quality_status": "accepted",
  "flags": [],
  "is_synthetic": true,
  "code": "stored",
  "request_id": "00000000-0000-4000-8000-000000000999"
}
```

Boundary	A0 vs A1 packet
Rule / discrepancy	A0 fixture memakai envelope 0.1.0 dan detection_limit; A1 tepat delapan field schema 1.0.0, tidak menerima field tambahan.
Classification	VERIFIED
Evidence	Seagis-Model/contracts/iot-packet.schema.json:1

Boundary	A0 run_kind vs A2

Rule / discrepancy	0003 memperluas run_kind dengan analysis_forecast_unspecified; jangan memakai daftar A0 lama.
Classification	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1

Boundary	A3 JSON schema vs runtime
Rule / discrepancy	Schema lengkap diuji oleh test; curate/build tidak memanggil JSON Schema umum, SQL menegakkan pemeriksaan terpilih.
Classification	VERIFIED
Evidence	Seagis-Model/scripts/curate_environment.py:1

Boundary	API time vs PostgreSQL
Rule / discrepancy	A1 UTC Z; timestamptz session display / microseconds
Classification	VERIFIED
Evidence	Seagis-Model/supabase/migrations/0002_temperature_api.sql:1

Boundary	iOS Assessment vs DB
Rule / discrepancy	Tidak ada DTO/kontrak assessment backend terverifikasi.
Classification	NOT FOUND
Evidence	Seagis/Seagis/Models/Shared/Assessment.swift:1

9 Siklus Hidup Data

Pembuatan: UUID disediakan oleh caller/SQL sesuai dictionary; natural keys registry dipin seed/config. Raw, measurement, forecast, QC dan snapshot memiliki trigger pembatas mutasi. Dataset hanya dapat mengubah gate operasional tertentu; identitas dataset, parameter dan grid harus diberi versi baru bila semantiknya berubah. Sensor deployment baru diperlukan untuk perubahan lokasi/kedalaman; active dapat dinonaktifkan.

Ingestion berubah running → succeeded/failed/partial sesuai CHECK dan guard. Attempt regional berubah running → succeeded/duplicate/failed/abandoned. expired lease ditandai abandoned pada claim berikutnya. Feature snapshot berubah draft → sealed; manifest input harus cocok dan row lock induk mencegah race saat sealing. Snapshot A3 langsung immutable setelah insert. Credential dicabut dengan revoked_at atau habis lewat expires_at; ini bukan penghapusan observasi.

A0 awalnya memperbolehkan penutupan effective_until pada mapping. Migrasi 0005 menambah recorded_at dan trigger immutable_alignment_mapping sebelum UPDATE/DELETE. Karena itu instruksi lama untuk menutup mapping dengan UPDATE tidak lagi berlaku pada migrasi lengkap. Resolusi rentang baru harus dirancang tanpa menghapus histori atau melewati trigger secara diam-diam.

Tidak ditemukan job cleanup/retensi berbasis umur, soft delete umum, backup/restore teruji, replikasi lintas DB atau sinkronisasi iOS-cloud. A2 metadata_only adalah kebijakan konten raw, bukan TTL semua tabel. FK tanpa ON DELETE/UPDATE eksplisit memakai NO ACTION; tidak ada cascade delete yang boleh diasumsikan. Prototype FarmStore menghapus farmer beserta plot di memori; ini tidak menjalankan DELETE SQL backend.

Migrasi 0001 initial, 0002 temperature API, 0003 regional ingestion, 0004 provider_updating, 0005 quality alignment. SQL merupakan migrasi maju; rollback/down script NOT FOUND. Deploy scripts memakai administrator untuk provisioning dan memasang worker role; tidak dijalankan pada audit ini. Adopsi migrasi perlu backup, uji schema kosong dan uji upgrade; hal tersebut rekomendasi, bukan hasil pengujian sesi ini.

Keamanan: migration mengaktifkan RLS tabel aplikasi; FORCE RLS=false pada snapshot lama. PUBLIC/anon/authenticated dibatasi, gateway dan worker mendapat EXECUTE fungsi tertentu dengan SECURITY DEFINER serta search_path=pg_catalog. A2/A3 memakai TLS verify-full dan CA hash dipin. Token asli tidak dicetak; credential menyimpan hash. Hak administrator, role membership aktif, bucket policy, encryption-at-rest, rotasi aktual, backup dan recovery produksi UNVERIFIED. Rate-limit log, attempt, QC event dan hash menyediakan audit parsial; tidak ditemukan dead-letter queue atau alerting eksternal terkonfigurasi pada jalur ini.

Bukti implementasi utama: Seagis-Model/supabase/migrations/0001_initial_schema.sql:1;
Seagis-Model/supabase/migrations/0002_temperature_api.sql:1;
Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1;
Seagis-Model/supabase/migrations/0004_provider_update_status.sql:1;
Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1.

Bukti implementasi utama: Seagis-Model/scripts/environmental/store.py:1; Seagis-Model/scripts/quality/store.py:1;
Seagis-Model/supabase/functions/temperature/connection.ts:1.

10 Dependensi Data pada Aplikasi

Workflow suhu: perangkat mengirim packet → handler → seagis_api.ingest → receipt/observation/QC → reader terotorisasi memanggil history dengan from/to dan cursor → JSON data dan coverage. History mengurutkan observed_at,id; from inklusif, to eksklusif, maksimum tujuh hari, limit default 200/maksimal 500. Coverage memakai bucket jam UTC dan dapat berbeda dari jumlah baris pagination. Status memberi ringkasan deployment/last packet; bukan assessment penyakit.

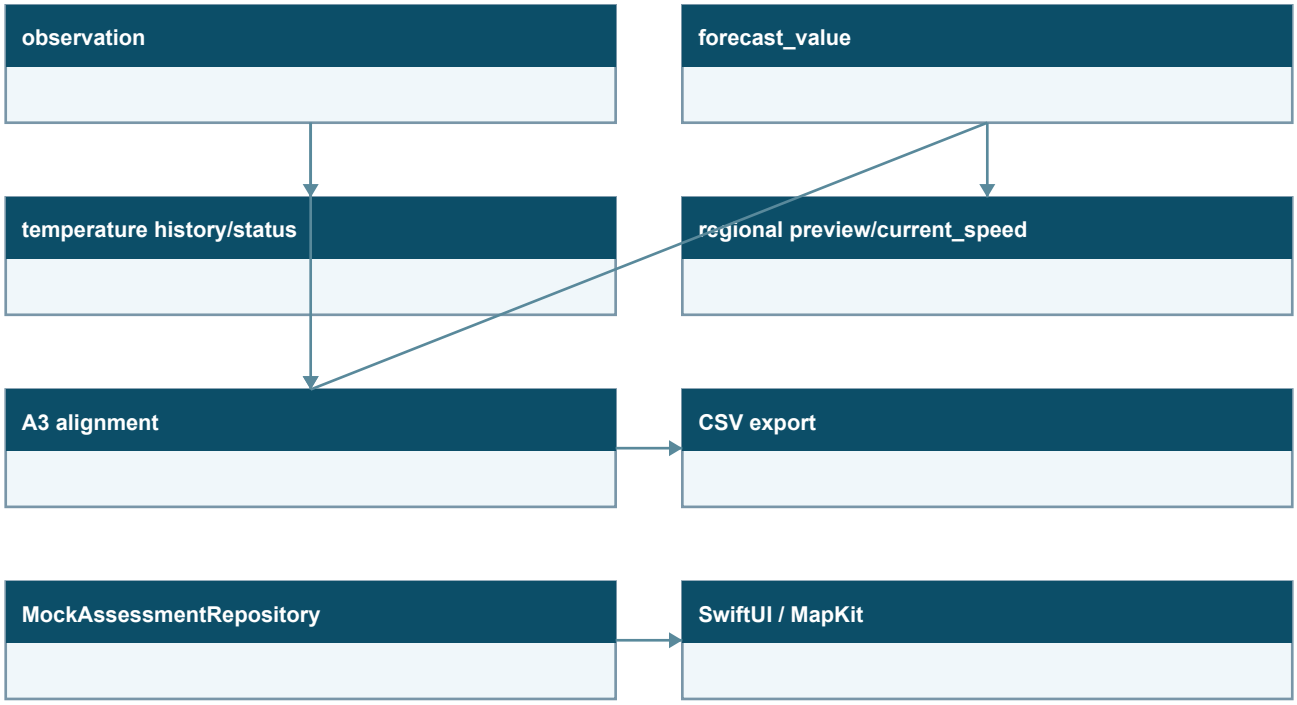
Workflow regional: dispatcher → claim/fetch/persist → preview memilih run terbaru yang tersedia pada cutoff dan tidak stale menurut policy, membatasi valid_at dua hari ke depan dan maksimum 1500 baris. current_speed adalah join inner komponen; pasangan tidak lengkap tidak menghasilkan baris view, sementara pasangan lengkap dengan missing menghasilkan speed null. Preview demo mempertahankan identitas grid dan batas representativitas.

Workflow iOS: AppContainer menyuntikkan MockAssessmentRepository → LoadDemoAssessment → BootstrapViewModel → SwiftUI. Assessment domain hanya memiliki provenance synthetic_fixture, tidak memiliki DTO backend terverifikasi. FarmStore AppPOC menyimpan Farmer/CultivationPlot dalam array, membuat UUID, men-trim nama, dan menghapus relasi secara in-memory. Tidak ada mapping ID farmer/plot prototype ke seagis.plot yang terverifikasi. Cache/persistensi lokal produksi dan konflik sinkronisasi belum ditemukan implementasinya.

Konsumen analitik yang terverifikasi berupa build QC dan CSV exporter; tabel seagis_features belum memiliki produksi feature builder/training consumer yang ditemukan. Angka coverage hourly tidak dihitung sebagai jumlah pengamatan independen. Tidak ditemukan model ML terlatih yang menghasilkan probabilitas ice-ice pada jalur backend ini.

Bukti implementasi utama: Seagis/Seagis/Repositories/Shared/MockAssessmentRepository.swift:1; AppPOC/POCApp/Shared/Data/FarmStore.swift:1; Seagis-Model/supabase/migrations/0002_temperature_api.sql:1; Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1.

Diagram reproduktif: assets/diagrams/dependencies.mmd.



Data entity	seagis.dataset
Producer / writers	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Readers	A1 ingest/history; A2 persist/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Identitas dataset, versi, cadence, lisensi dan gate ingestion.

Data entity	seagis.dataset_parameter
Producer / writers	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Readers	A1 ingest/history; A2 persist/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Kontrak variabel sumber ke parameter dan aturan konversi.

Data entity	seagis.forecast_run
Producer / writers	seagis_environment.persist; A0 fixtures
Readers	seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
API / feature	private SQL / demo / offline
Downstream	Capture/vintage model regional; waktu publikasi yang tidak diketahui tetap null.

Data entity	seagis.forecast_value
Producer / writers	seagis_environment.persist; A0 fixtures
Readers	seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
API / feature	private SQL / demo / offline
Downstream	Nilai parameter per run, grid dan valid time; tidak diduplikasi per plot.

Data entity	seagis.ingestion_run
Producer / writers	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Readers	A3 source_bundle; provenance guards
API / feature	private SQL / demo / offline
Downstream	Audit satu ingestion logis beserta status penyelesaian.

Data entity	seagis.observation
Producer / writers	seagis_api.ingest; A0 fixtures
Readers	seagis_api.history/status; A3 source_bundle; export_experiment_csv.py
API / feature	temperature history/status
Downstream	Pembacaan suhu air sensor lokal, dengan nilai native, QC dan lineage raw.

Data entity	seagis.parameter
Producer / writers	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Readers	A1 ingest/history; A2 persist/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Registri parameter dan satuan normalisasi.

Data entity	seagis.plot
Producer / writers	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Readers	A1 ingest/history; A2 persist/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Identitas plot dan status survei; koordinat nullable bukan bukti lokasi lapangan.

Data entity	seagis.plot_grid_mapping
Producer / writers	A0 fixture/admin SQL
Readers	A3 source_bundle; feature_input guards
API / feature	private SQL / demo / offline

Downstream	Pemetaan operasional plot ke grid dengan waktu efektif dan waktu tercatat.

Data entity	seagis.quality_event
Producer / writers	seagis_api.ingest; A0 fixtures
Readers	A3 source_bundle/alignment
API / feature	private SQL / demo / offline
Downstream	Log QC append-only untuk ingestion atau satu measurement.

Data entity	seagis.sensor
Producer / writers	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Readers	A1 ingest/history; A2 persist/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Identitas deployment sensor suhu, plot, kedalaman dan status aktif.

Data entity	seagis.source
Producer / writers	supabase/seed.sql; scripts/provision_a1.py; scripts/deploy_a2.py
Readers	A1 ingest/history; A2 persist/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Registri penyedia asal dataset.

Data entity	seagis.source_grid
Producer / writers	seagis_environment.persist; A0 fixtures
Readers	seagis_environment.preview/current_speed; A3 source_bundle; CSV exporter
API / feature	private SQL / demo / offline
Downstream	Identitas sel regional dan layer kedalaman; bukan plot budidaya.

Data entity	seagis_api.credential
Producer / writers	scripts/provision_a1.py; migration 0002
Readers	seagis_api.actor/rate_ok/ingest/history/status
API / feature	private SQL / demo / offline
Downstream	Identitas aktor API berupa hash token, masa berlaku dan pencabutan.

Data entity	seagis_api.packet_receipt
Producer / writers	seagis_api.ingest/reply
Readers	seagis_api.actor/rate_ok/ingest/history/status
API / feature	private SQL / demo / offline
Downstream	Kunci idempotensi paket sensor dan digest semantik.

Data entity	seagis_api.reader_plot
Producer / writers	scripts/provision_a1.py; migration 0002
Readers	seagis_api.actor/rate_ok/ingest/history/status
API / feature	private SQL / demo / offline
Downstream	Daftar plot yang boleh dibaca satu credential reader.

Data entity	seagis_api.request_event
Producer / writers	seagis_api.ingest/reply

Readers	seagis_api.actor/rate_ok/ingest/history/status
API / feature	private SQL / demo / offline
Downstream	Audit respons API terautentikasi dan dasar rate limit.

Data entity	seagis_api.temperature_policy
Producer / writers	scripts/provision_a1.py; migration 0002
Readers	seagis_api.actor/rate_ok/ingest/history/status
API / feature	private SQL / demo / offline
Downstream	Konfigurasi QC engineering pilot dan batas request; bukan ambang penyakit.

Data entity	seagis_environment.attempt
Producer / writers	seagis_environment.claim/persist/finish_failure; migration 0003
Readers	A2 status/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Audit satu upaya fetch termasuk duplicate dan abandoned.

Data entity	seagis_environment.current_speed
Producer / writers	NOT APPLICABLE: view
Readers	A2 status/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	View magnitude arus dari pasangan komponen dengan run/grid/waktu yang sama.

Data entity	seagis_environment.demo_grid_mapping
Producer / writers	seagis_environment.claim/persist/finish_failure; migration 0003
Readers	A2 status/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Bridge khusus dua plot fiktif ke grid nyata untuk preview, bukan otorisasi training.

Data entity	seagis_environment.policy
Producer / writers	seagis_environment.claim/persist/finish_failure; migration 0003
Readers	A2 status/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	Kebijakan fetch, lisensi, cadence scheduler, staleness dan retensi regional.

Data entity	seagis_environment.schedule_state
Producer / writers	seagis_environment.claim/persist/finish_failure; migration 0003
Readers	A2 status/preview; A3 source_bundle
API / feature	private SQL / demo / offline
Downstream	State mutable lease, due time, backoff dan keberhasilan terakhir.

Data entity	seagis_features.feature_input
Producer / writers	A0 fixture/test inserts; production builder NOT FOUND
Readers	snapshot/input guards; production ML consumer NOT FOUND
API / feature	private SQL / demo / offline
Downstream	Lineage FK input feature snapshot dan guard cutoff.

Data entity	seagis_features.feature_snapshot
Producer / writers	A0 fixture/test inserts; production builder NOT FOUND
Readers	snapshot/input guards; production ML consumer NOT FOUND
API / feature	private SQL / demo / offline
Downstream	Container fitur draft/sealed dengan cutoff dan hash; builder produksi belum ditemukan.

Data entity	seagis_quality.alignment_snapshot
Producer / writers	seagis_quality.save_snapshot
Readers	seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
API / feature	private SQL / demo / offline
Downstream	Dokumen QC/alignment hourly immutable; selalu training_authorized=false.

Data entity	seagis_quality.policy
Producer / writers	scripts/quality/store.py::register_policy
Readers	seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
API / feature	private SQL / demo / offline
Downstream	Registri versi kebijakan QC dan hash kode builder yang immutable.

Data entity	seagis_quality.snapshot_input
Producer / writers	seagis_quality.save_snapshot
Readers	seagis_quality.source_bundle/save_snapshot/snapshot; curate_environment.py
API / feature	private SQL / demo / offline
Downstream	Lineage measurement per jam alignment snapshot.

Data entity	seagis_raw.raw_payload
Producer / writers	seagis_api.ingest; seagis_environment.persist; A0 fixtures
Readers	A3 source_bundle; provenance guards
API / feature	private SQL / demo / offline
Downstream	Manifest raw immutable: digest, basis hash, lisensi dan mode retensi.

11 Audit Kualitas Data dan Arsitektur

Audit read-only terhadap kode dan snapshot lama; severity menyatakan dampak potensial sesuai lingkup prototype, bukan insiden produksi terukur. Tidak ada perubahan schema/aplikasi yang diterapkan.

F01 Assessment iOS belum terhubung ke backend

Atribut	Temuan
Severity / category	High / Contract / integration
Status	VERIFIED
Deskripsi / dampak	Mock mengembalikan synthetic_fixture; tidak ada adapter API assessment yang ditemukan. Konsumen baru tidak boleh menganggap model domain sebagai wire contract.
Affected	Seagis/Seagis/Repositories/Shared/MockAssessmentRepository.swift, Seagis/Seagis/Models/Shared/Assessment.swift
Evidence	Seagis/Seagis/Repositories/Shared/MockAssessmentRepository.swift:1; Seagis/Seagis/Models/Shared/Assessment.swift:1
Rekomendasi	Sepakati kontrak executable dan mapping provenance/error sebelum integrasi.

F02 Mapping immutable setelah 0005 mengubah prosedur penutupan lama

Atribut	Temuan
Severity / category	Medium / Lifecycle
Status	VERIFIED
Deskripsi / dampak	guard_mapping A0 mengizinkan penutupan interval; trigger 0005 menolak seluruh UPDATE/DELETE. Petunjuk arsitektur A0 untuk menutup mapping dapat gagal pada schema lengkap.
Affected	supabase/migrations/0001_initial_schema.sql, supabase/migrations/0005_quality_alignment.sql, docs/architecture.md
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:1; Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1; Seagis-Model/docs/architecture.md:1
Rekomendasi	Definisikan prosedur append-only untuk supersession dan uji overlap/cutoff.

F03 Riwayat tumbuh tanpa job retensi yang ditemukan

Atribut	Temuan
Severity / category	Medium / Governance / retention
Status	INFERRED
Deskripsi / dampak	request_event, observation, forecast dan snapshot append-only; TTL/cleanup terjadwal dan recovery test tidak ditemukan di checkout. Risiko kapasitas adalah inferensi, bukan hasil benchmark.
Affected	supabase/migrations/0001_initial_schema.sql, supabase/migrations/0002_temperature_api.sql, supabase/migrations/0005_quality_alignment.sql
Evidence	Seagis-Model/supabase/migrations/0001_initial_schema.sql:1; Seagis-Model/supabase/migrations/0002_temperature_api.sql:1; Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1
Rekomendasi	Tetapkan retention per domain, archival, backup dan restore drill sebelum skala produksi.

F04 SQL A3 memeriksa lineage tetapi tidak menghitung ulang seluruh QC

Atribut	Temuan
Severity / category	Medium / Validation boundary
Status	VERIFIED
Deskripsi / dampak	save_snapshot memeriksa hash policy/kode dan nilai sumber, tetapi keputusan usable_context/flags/blocking_flags tidak direkalkulasi melalui Python di DB. Role worker tetap trust boundary.
Affected	supabase/migrations/0005_quality_alignment.sql, scripts/quality/alignment.py
Evidence	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:1; Seagis-Model/scripts/quality/alignment.py:1
Rekomendasi	Dokumentasikan trust worker; bila diperlukan, verifikasi deterministik QC tambahan dan test payload tampering.

F05 Metadata-only membatasi reproduksi raw

Atribut	Temuan
Severity / category	Medium / Data provenance
Status	VERIFIED
Deskripsi / dampak	A2 tidak menyimpan raw content. Hash subset Toolbox bukan hash byte NetCDF/Zarr asli; memperoleh ulang aggregate provider tidak menjamin vintage yang sama.
Affected	scripts/environmental/copernicus.py, supabase/migrations/0003_environment_ingestion.sql
Evidence	Seagis-Model/scripts/environmental/copernicus.py:1; Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:1
Rekomendasi	Pertahankan hash_basis; bila lisensi dan biaya disetujui, usulkan penyimpanan subset/objek versi immutable.

F06 CSV capture bukan backtest historis

Atribut	Temuan
Severity / category	High / Analytics semantics
Status	VERIFIED
Deskripsi / dampak	Exporter memilih latest_available_at_capture; A3 memakai cutoff tiap jam. Memakai CSV tersebut sebagai histori availability berisiko leakage; training_authorized=false sudah dinyatakan kode.
Affected	scripts/export_experiment_csv.py, scripts/quality/alignment.py
Evidence	Seagis-Model/scripts/export_experiment_csv.py:1; Seagis-Model/scripts/quality/alignment.py:1
Rekomendasi	Gunakan source bundles seluruh vintage dan cutoff eksplisit untuk rancangan backtest; jangan melepas gate training tanpa validasi.

F07 Dokumen A0 mendahului implementasi A1–A3

Atribut	Temuan
Severity / category	Low / Documentation drift
Status	VERIFIED
Deskripsi / dampak	docs/architecture.md mengatakan ingestion/scheduler/QC belum diimplementasikan, sedangkan migrasi dan worker berikutnya ada. Pernyataan harus dibaca dengan fase dan tanggal.
Affected	docs/architecture.md, scripts/ingest_environment.py, scripts/curate_environment.py
Evidence	Seagis-Model/docs/architecture.md:1; Seagis-Model/scripts/ingest_environment.py:1; Seagis-Model/scripts/curate_environment.py:1
Rekomendasi	Gunakan indeks status fase serta source-manifest untuk pembaruan dokumentasi.

F08 Snapshot catalog berumur lima hari

Atribut	Temuan
Severity / category	Medium / Runtime verification
Status	UNVERIFIED
Deskripsi / dampak	Snapshot 22 Oktober bukan audit runtime 27 Oktober. Grants, row count, cron aktif dan migrasi cloud terkini tidak diverifikasi ulang.
Affected	runtime catalog
Evidence	schema-metadata-20261022.json inspected_at
Rekomendasi	Lakukan diff catalog read-only dan review grants sebelum keputusan produksi.

Prioritas: P1 sepakati integrasi/kontrak dan cegah leakage (F01/F06); P2 tutup trust, retensi, mapping lifecycle dan raw reproducibility (F02–F05); P3 perbarui status dokumentasi dan catalog (F07/F08). Evaluasi indeks/performa membutuhkan EXPLAIN dan beban nyata: tidak ada klaim missing index atau query lambat yang dibuktikan hanya dari nama kolom.

12 Glosarium

Term	Definisi
Origin	Asal data: sensor, model regional atau nilai turunan
Ingestion	Penerimaan dan persistence satu capture
Vintage	Versi capture model yang dipertahankan
Native cadence	Interval sampel asli, berbeda dari tick scheduler
Availability cutoff	Batas informasi yang sudah tersedia saat keputusan
Value status	Ketersediaan angka, bukan status penyakit
QC	Pemeriksaan kualitas data; bukan diagnosis
Idempotency	Retry identik tidak menambah measurement
Lineage	Jejak referensi dari hasil ke sumber
Manifest	Daftar input/berkas dengan identitas dan digest
RLS	Row Level Security; efektivitas tergantung role dan grants
FK	Foreign key: integritas referensi antarbaris
NULL	Nilai tidak tersedia; tidak sama dengan angka nol
Metadata-only	Menyimpan provenance/digest tanpa konten raw
Sealed snapshot	Snapshot yang tidak boleh diubah setelah sealing
Ice-ice	Konteks penyakit budidaya; dokumentasi ini tidak memvalidasi prediksinya

13 Lampiran

13.1 Basis bukti dan batas audit

Repository	Commit	Path
Seagis-Model	291b6c5fdc7c990660ea98b1dcf676f6c0b58148	/Users/cindyram/Documents/Apple/Seagis-Model
Seagis	109619ba0cb43035ce102e187158fbce9a7a42c1	/Users/cindyram/Documents/Apple/Seagis
AppPOC	c68d72ae90457a9e8def40b57dfae9e86607aff8	/Users/cindyram/Documents/Apple/AppPOC

File implementasi dibaca melalui pembangkit dokumentasi dan pencarian terarah; manifest mencatat hash per file. Tidak ada source eksternal yang digunakan untuk menggantikan implementasi. Tidak ada migrasi, worker ingestion, test yang menulis database atau deploy dijalankan. Test kontrak murni/offline dilaporkan terpisah pada validation.json. File .env, credential dan isi tabel tidak disalin.

13.2 Cakupan terukur

Component	Application databases
Discovered	1
ID documented	1
EN documented	1
Evidence / gaps	Kode + catalog lama; runtime terkini belum

Component	Application schemas
Discovered	6
ID documented	6
EN documented	6
Evidence / gaps	historical catalog + migrations

Component	Physical tables
Discovered	28
ID documented	28
EN documented	28
Evidence / gaps	row counts UNVERIFIED

Component	Views
Discovered	1
ID documented	1
EN documented	1
Evidence / gaps	current_speed

Component	Columns (tables + view)
Discovered	268
ID documented	268
EN documented	268
Evidence / gaps	259 tabel + 9 view; arti field tertentu masih terbatas

Component	Physical FK
Discovered	47
ID documented	47
EN documented	47
Evidence / gaps	historical catalog + SQL

Component	Column-origin mappings
Discovered	268
ID documented	268
EN documented	268
Evidence / gaps	268 VERIFIED; 0 UNVERIFIED

Component	Pipeline paths
Discovered	4
ID documented	4
EN documented	4
Evidence / gaps	A1, A2, A3, CSV export

Component	Boundary contracts
Discovered	11
ID documented	11
EN documented	11
Evidence / gaps	4 explicit; 7 implementation-derived

Component	JSON-schema field/structure entries
Discovered	147
ID documented	147
EN documented	147
Evidence / gaps	Nested semantics belum semua ditutup; schema lengkap disertakan

Component	Object dependency rows
Discovered	29
ID documented	29
EN documented	29
Evidence / gaps	writers/readers from code; deployment unverified

Component	Runtime rows/grants/backups
Discovered	UNVERIFIED
ID documented	UNVERIFIED
EN documented	UNVERIFIED
Evidence / gaps	Tidak ada akses live pada audit ini

Cakupan inventaris snapshot 29/29 objek dan 268/268 kolom bukan klaim 100% arsitektur produksi. Schema internal, sumber di luar tiga checkout, deployment runtime dan nested field ambigu tidak menjadi denominator yang diketahui.

13.2.1 Dataset file lokal teramati

Metadata CSV berikut diperiksa tanpa menyalin nilai baris. Jumlah ini milik file capture lokal, bukan row count database. CSV regional mengikuti native cadence dan latest_available_at_capture; suhu plot adalah sintetis. source_bundles/alignment/manifest JSON merupakan keluaran exporter atau curator dengan kontrak C04/C07/C08; worker.log hanya log operasional dan CA certificate hanya trust material, bukan dataset lingkungan.

File	Rows observed	Columns / contract
Seagis-Model/data/exports/forecast-20261023T015005Z/features/air_temperature_2m.csv	26	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/cloud_cover.csv	26	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/current_eastward.csv	5	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/current_northward.csv	5	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/nitrate.csv	2	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/phosphate.csv	2	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/precipitation.csv	26	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
File	Rows observed	Columns / contract
Seagis-Model/data/exports/forecast-20261023T015005Z/features/salinity.csv	5	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at,

File	Rows observed	Columns / contract
		interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url,model_initialized_at, native_cadence_seconds, native_unit, native_value,parameter_code, period_end_at, period_start_at, run_ingested_at,run_kind, selection_basis, source_grid_id, source_kind, source_time,temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/uv_index.csv	26	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/water_temperature_plot-a.csv	72	timestamp_utc, timestamp_wib, plot_id, is_gap, is_synthetic, value_status, quality_status, qc_flags, strict_source_qc_pass, observation_id, available_at_utc, training_authorized, value, raw_value, parameter_code, unit, source_kind, value_policy
Seagis-Model/data/exports/forecast-20261023T015005Z/features/water_temperature_plot-b.csv	72	timestamp_utc, timestamp_wib, plot_id, is_gap, is_synthetic, value_status, quality_status, qc_flags, strict_source_qc_pass, observation_id, available_at_utc, training_authorized, value, raw_value, parameter_code, unit, source_kind, value_policy
Seagis-Model/data/exports/forecast-20261023T015005Z/features/wind_direction_10m.csv	26	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at
Seagis-Model/data/exports/forecast-20261023T015005Z/features/wind_speed_10m.csv	26	timestamp_utc, timestamp_wib, value, unit, value_status, quality_status, qc_flags, attribution, available_at, capture_at_utc, conversion_code, dataset_key, detection_limit, forecast_run_id, id, ingested_at, interval_status, is_synthetic, issue_time_basis, issued_at, kind, licence_url, model_initialized_at, native_cadence_seconds, native_unit, native_value, parameter_code, period_end_at, period_start_at, run_ingested_at, run_kind, selection_basis, source_grid_id, source_kind, source_time, temporal_support, training_authorized, valid_at

13.3 Pertanyaan terbuka

- Apakah schema cloud dan grants berubah sejak 22 Oktober? - Bagaimana supersession pemetaan immutable dan penyelesaian overlap ditetapkan? - Siapa pemilik operasi backup/retensi dan bagaimana restore dibuktikan? - Kontrak assessment apa yang akan disetujui lintas backend/iOS? - Kapan plot/lokasi/kedalaman sensor nyata tervalidasi dan training gate dapat dievaluasi?

13.4 SQL read-only ilustratif

Query berikut belum dijalankan terhadap cloud dalam audit ini.

```
BEGIN READ ONLY;
SELECT schemaname, tablename FROM pg_catalog.pg_tables
WHERE schemaname IN ('seagis','seagis_raw','seagis_api',
 'seagis_environment','seagis_quality','seagis_features')
ORDER BY schemaname, tablename;
ROLLBACK;
```

13.5 Registry fungsi SQL

Function	Arguments	Evidence
seagis.valid_value	v double precision, native double precision, status text, lim double precision	Seagis-Model/supabase/migrations/0001_initial_schema.sql:168

Function	Arguments	Evidence
seagis.valid_support	t timestamptz, support text, status text, a timestamptz, b timestamptz	Seagis-Model/supabase/migrations/0001_initial_schema.sql:176
seagis.jsonb_sha256	value jsonb	Seagis-Model/supabase/migrations/0001_initial_schema.sql:296
seagis.reject_mutation		Seagis-Model/supabase/migrations/0001_initial_schema.sql:330
seagis.guard_ingestion		Seagis-Model/supabase/migrations/0001_initial_schema.sql:334
seagis.guard_dataset_identity		Seagis-Model/supabase/migrations/0001_initial_schema.sql:347
seagis.guard_sensor_deployment		Seagis-Model/supabase/migrations/0001_initial_schema.sql:357
seagis.guard_raw		Seagis-Model/supabase/migrations/0001_initial_schema.sql:366
seagis.guard_measurement		Seagis-Model/supabase/migrations/0001_initial_schema.sql:379
seagis.guard_run		Seagis-Model/supabase/migrations/0001_initial_schema.sql:406
seagis.guard_mapping		Seagis-Model/supabase/migrations/0001_initial_schema.sql:415
seagis.guard_quality_event		Seagis-Model/supabase/migrations/0001_initial_schema.sql:424
seagis_features.guard_input		Seagis-Model/supabase/migrations/0001_initial_schema.sql:434
seagis_features.manifest_hash	snapshot uuid	Seagis-Model/supabase/migrations/0001_initial_schema.sql:458
seagis_features.guard_snapshot		Seagis-Model/supabase/migrations/0001_initial_schema.sql:465
seagis_api.utc	t timestamptz	Seagis-Model/supabase/migrations/0002_temperature_api.sql:88
seagis_api.actor	token text, actor_kind text	Seagis-Model/supabase/migrations/0002_temperature_api.sql:90
seagis_api.reply	actor_id uuid, status integer, code text, body jsonb DEFAULT '{}', sensor uuid DEFAULT NULL	Seagis-Model/supabase/migrations/0002_temperature_api.sql:96
seagis_api.rate_ok	actor_id uuid	Seagis-Model/supabase/migrations/0002_temperature_api.sql:105
seagis_api.ingest	token text, packet jsonb, wire_sha256 text, wire_bytes integer	Seagis-Model/supabase/migrations/0002_temperature_api.sql:113
seagis_api.history	token text, target_plot uuid, spec jsonb	Seagis-Model/supabase/migrations/0002_temperature_api.sql:222
seagis_api.status	token text	Seagis-Model/supabase/migrations/0002_temperature_api.sql:274
seagis_environment.claim	k text, force_fetch boolean DEFAULT false	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:99
seagis_environment.finish_failure	lease uuid, code text	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:117
seagis_environment.persist	lease uuid, batch jsonb	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:131
seagis_environment.status		Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:212
seagis_environment.preview	plot uuid, as_of timestamptz DEFAULT now(	Seagis-Model/supabase/migrations/0003_environment_ingestion.sql:227
seagis_environment.finish_failure	lease uuid, code text	Seagis-Model/supabase/migrations/0004_provider_update_status.sql:2
seagis_quality.source_bundle	p_plot uuid,p_start timestamptz,p_end timestamptz,p_demo boolean DEFAULT false	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:47
seagis_quality.save_snapshot	document jsonb	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:121
seagis_quality.snapshot	identity uuid	Seagis-Model/supabase/migrations/0005_quality_alignment.sql:232

13.6 Register sumber

Path	Lines	SHA-256
Seagis-Model/README.md	71	4db1d01136db09e662c2bb005264cd7ff657c5fc6b2a2851b2ef9ccb5010354a
Seagis-Model/config/a2_sources.json	15	a53addfbf7a0b2733bdcedaabf6c85dad2c428d3afd05eeffbdf79af459393b2
Seagis-Model/config/a3_quality.json	33	0f4e803b19b129edd3d49be6466bcaf0e22eafb293001290c5893e23614bae5e
Seagis-Model/contracts/alignment.schema.json	81	7cf0f75151aaf3b1e73da9091ad225abaeee979aba5a1a4aa48d3bd5fff91f66
Seagis-Model/contracts/iot-packet.schema.json	19	1a03ae7fad5b1d3a4f328b879cd68e432544ebeed819f77fa97437aa627581c4
Seagis-Model/contracts/normalized.schema.json	918	e3d1605ce38afb2c77cee8ca3a669f3dfbb32ebce927155c65429b74d1b13309
Seagis-Model/contracts/source.schema.json	527	818d451814b97aba7d40d75576d60e5c0a72aea4e110cee900cff795fc7aab08
Seagis-Model/docs/a0_review.md	73	3de9ffcf90fe3abfece563347e5447dfb7bacb56dbc03c265c0d10120025966e
Seagis-Model/docs/a1/iot_handover.md	131	7fbd035f3e1b843ca05b69958f7a442515c51e4df5ca33b36bb4c1a6f0914333
Seagis-Model/docs/a1/operations.md	111	01d9de4516a22f3683b8065a5fd0b3a02941e027bba38e81307b87e821c0c36e
Seagis-Model/docs/a1/review.md	53	b2d95bdf10bff99cf560b93d0a86c2227525f882412d04b1b031d7e3cbdf705c
Seagis-Model/docs/a1/validation_20261022.json	112	1ad7b37395480199d2b6b2fa55cab7455862212901b8474c0e4bbc05577dae99
Seagis-Model/docs/a2/architecture.md	74	530a0c4d72f8d9b05b53b0c09dee3d1c30064a6446a8dba7220fc9bd019f5655
Seagis-Model/docs/a2/ingestion_report_20261022.json	432	894bb8e96fa756341075c760c2841db21ff3d95e098aa44540e7557a29c34ddd
Seagis-Model/docs/a2/licenses.md	27	c1551d495c52a5054f7974d2d7397d1e9ea3f841619b77ed4d504b4570d1743a
Seagis-Model/docs/a2/operations.md	69	d0a4e9894254025cc7514a9fe808dd25bd929c0053c389f8ea05ff4208c2acfa
Seagis-Model/docs/a2/review.md	52	ea22ff34a20cbf1691ecfcbe1b3269aeeb46d5d4d6bb5f450b8f9571667442a2
Seagis-Model/docs/a2/source_evidence_20261022.json	55	a80fd90bae4da2694c049feed7b34bde43ff0ace51d5ddbb66389536b797b0e5
Seagis-Model/docs/a3/architecture.md	59	9e209b5b3840be8dbdeb0948f2faa9c1c479aaa22bcdec70e231c6c0c94dbb24
Seagis-Model/docs/a3/operations.md	99	a5f17669b2400757e76bd6a3fd27caf29a6c6e6966a2bb3bb21d5fc5a7db676e
Seagis-Model/docs/a3/quality_report_20261022.json	1002	2e67a3b630917bda4d20fe4ef031d393d3ba61292ea90de15f4ce4a8a8500223
Seagis-Model/docs/a3/review.md	63	a0d533104fbfbedad18ab435a0a2702364fecd8392e3b7b9983410afbbd5eb33
Seagis-Model/docs/architecture.md	85	b5a7ff5659e4aad20a726218f1333ae814b60963ae5ae272071621b7c9d7f95f
Seagis-Model/docs/data_contracts.md	103	836f6ed8879a23efbd67c24fe199246d065c4aae9b73c662b9a8a47ce15a293d
Seagis-Model/docs/data_dictionary.md	115	175a21dc9d633b89ad0a605c1747296cefa287a9da134b78cac1f0a2d0f4d8c0
Seagis-Model/docs/decisions/ADR-001-data-storage.md	50	bc2fcffc9170b6034ae9707d56b4382fa54ea7a1dfda6a83acb273e4b55d49d3
Seagis-Model/docs/experiment_csv.md	75	67c912601b6c228f0140eea4726cfed34481cf3e6eaa9fe850dc33f4ad8f4b7a

Path	Lines	SHA-256
Seagis-Model/docs/metadata/source_catalogue_20261021.json	79	8ef86e2ca4db05315efb5b4cececc23951c55fa63295ffa31460cef4b96f0eee
Seagis-Model/docs/source_registry.md	54	6310bce3d80e1e7fdea1d2855fd9446febf133902a007af56304254c1a950569
Seagis-Model/fixtures/a1/hourly_synthetic.json	1970	8d833be53dbfe0c2fd5c590830821374f027a5090f8804d2085f686855add22c
Seagis-Model/fixtures/a3/synthetic_bundle.json	670	522bd51062e91a1419bf85f0b62e8b206e5e5208df17f3cf5bd81f9606c39d89
Seagis-Model/fixtures/normalized/copernicus_currents.json	79	c5fb39ed7cccb2dbba94135d3dda19464bf8287b6e082d224f006ab0461c4913
Seagis-Model/fixtures/normalized/copernicus_nutrients.json	79	a30d3dae6c700faa491662773bcda03efe08d0c93d8d136f68c84264f115fa85
Seagis-Model/fixtures/normalized/copernicus_salinity.json	60	9c262f0b2922d3f100a8d35ad326b53fc850e688e2d0d9742c4aeb0aa4e4104a
Seagis-Model/fixtures/normalized/open_meteo.json	155	8b4d8cdd2732ca0b88bcd6db6f920e12b4b6746187bb080aa135dbaadf4e94d3
Seagis-Model/fixtures/normalized/sensor.json	42	7ab0f72245e724d6ad36c5c65abcbdd24775f7708b3846d4c31a4b3b3c224d8f
Seagis-Model/fixtures/source/copernicus_currents.json	48	ee2a87156030ab83902eee430ac9ef0cbf9dc508cf2b48a5981a9780ceded27d
Seagis-Model/fixtures/source/copernicus_nutrients.json	48	5f1424796bf98e90ce129f477d063cbd286a4302f7f53e79869157ba4e342675
Seagis-Model/fixtures/source/copernicus_salinity.json	41	ba795d27a545d2830c2b05d2ffe3a8e9c4fd7c8d72874edc3493195f94436046
Seagis-Model/fixtures/source/open_meteo.json	62	fe3c4e57479ee18e2e295b23a6d4240136f89f6a3dc011950056f4f5fa5ea2f2
Seagis-Model/fixtures/source/sensor.json	20	8c18a6aff2a6b72f77bcce58e8d1a8a4a7cd1d378e2d2d70abea0150a3a9b844
Seagis-Model/pyproject.toml	19	bc2c936d59d74591a7c3714c70201d9583e5723c7a386b0e33d11954b91b4c91
Seagis-Model/scripts/check_a0.py	140	27a2ccc2ca016c93bf0436da5884f6ec32598847856fdcc36f5533a9499f728d
Seagis-Model/scripts/check_a1.py	10	9223b11860bc099ef330c604d51fb6e53afdeec7651d110ab1dcba32ff27b792
Seagis-Model/scripts/check_a1_http.py	161	19e104cc1cdcd406ef60b1c83a95ef5f9d68460c68dda5b7acc73772bcd773c2
Seagis-Model/scripts/check_a1_live.py	109	6c5992eb844030de9dd96f255c273b46d1aed6bbb3e9fc43d13f0c8c066dbe6d
Seagis-Model/scripts/curate_environment.py	63	a16b2cce80ef1bc84d514e616c06609a4ac6ecc3f1a3d40bad1ab5f137018c33
Seagis-Model/scripts/deploy_a1.py	130	38bb8c515829ddfb81d2698d27d2a197a4ec49f5d82ef670ccd8327fe8f09c0f
Seagis-Model/scripts/deploy_a2.py	36	70b9e672dc5ab270984998320c329fe7ba24b0f6700989e8033c8ac461bc0ab2
Seagis-Model/scripts/deploy_a3.py	56	24bd90b43ea4ec355745f335a51520cc693ce249f9c8289e016a335eb876c9fe
Seagis-Model/scripts/environmental/__init__.py	1	1f689d46a84d43a484eae691eb87eb1c5a18fcde06b197d7a02f80441e5126d1
Seagis-Model/scripts/environmental/common.py	158	83237385df19e57171552ab6c42551fcd531d156500780f84e838a964e0adae8
Seagis-Model/scripts/environmental/copernicus.py	133	a55dc9b11c14f9b094e72ffb832844b6cc4f530963270f0f6d98b23801e1ef04

Path	Lines	SHA-256
Seagis-Model/scripts/environmental/open_meteo.py	65	684ab244e25e6eafac81a2f910462fa326444bfdab26a49c06ed450085eae361
Seagis-Model/scripts/environmental/store.py	68	6e81e722210f5d39f15595e0039f16185ef31d18521c8a206f5de5424ea76708
Seagis-Model/scripts/export_experiment_csv.py	245	7a3eea3739e9aa8938a32b403599d911515b3d4fefe86fdb065ef12c7072c4fa
Seagis-Model/scripts/ingest_environment.py	76	10a83e77f4f42ed58095385aedb95aa92e620dd458bc673341de074086952ebb
Seagis-Model/scripts/install_a2_scheduler.py	52	1d3bfd452f5b1a6adf850b7d5169d698967d8691785bec1aa37006995ef36fb0
Seagis-Model/scripts/provision_a1.py	111	224bc7a7ce5dcce83e290fd6261c8ddd6349a86d07e4041ac51d0741405503bc
Seagis-Model/scripts/quality/__init__.py	1	c5b07be1b5223a81fc5ea6a9cf99f794c7a96360a552d23eb3c4300783592ebf
Seagis-Model/scripts/quality/alignment.py	387	dde4eb5b79edfc3fff1e0ed62befdeea58b162541d9e53962f5dc0f508384ca6
Seagis-Model/scripts/quality/store.py	66	204f411108496fe63d242a8fdc6c338f7e17f52def9cc909affef4e010115802
Seagis-Model/scripts/rotate_worker_credentials.py	105	2a8f5cd0606861abb48ffa145bd4a4f2629f244272cfd931f1d1da31edd6aa73
Seagis-Model/scripts/run_a2_worker.py	31	00fb29d0cb12ef882e729d965bbf788aefdb2f47a2597c5d446cd02c1016e3cb
Seagis-Model/scripts/simulate_temperature.py	216	1c452139a18e64d91d4292e0d1370fb010988a58a310a2632c253d312240f4fe
Seagis-Model/scripts/supabase_admin.py	81	ff4f5e9b5d2849a435fb3b8508b11c4e20b255840c4a89c67e10a04b382fa21d
Seagis-Model/scripts/validate_contracts.py	263	bd0c9589beb03ae16c5c74141c739d73c980599eda46fb078578df07cfad2a0b
Seagis-Model/supabase/config.toml	420	5e2f8fab3c2bcd9dd70e290b6a739257df5d8e2b5729084bb4314174fe9fcceb
Seagis-Model/supabase/functions/temperature/connection.ts	27	9eb48afaba8117353276eabfdfc76a37b24c3d1eaac576049c233fc564b27f1a
Seagis-Model/supabase/functions/temperature/deno.json	15	a16b21ff8a72b6d280187e532a40636be001322c6c86fda1df87485160b8feec
Seagis-Model/supabase/functions/temperature/handler.test.ts	88	7f231639b4160931ce21e2f32bdf8f420ba83a6bb5ab1c219c46f865bf7067e7
Seagis-Model/supabase/functions/temperature/handler.ts	113	e8391e427cab3979b1076a4aba695da63072a770923d6f7716ef6d4776d08bd7
Seagis-Model/supabase/functions/temperature/index.ts	31	1e74cd3a0eadd83a8145b352603fd9ef3c116057c7663d7183327c5c09fa1b44
Seagis-Model/supabase/migrations/0001_initial_schema.sql	544	f83e16be7d676cd1552266f2648444edbe12c01aac6b6ac94d4c6d32161c9b42
Seagis-Model/supabase/migrations/0002_temperature_api.sql	297	c1547f70b3bf29ea5a54171ff547177713bbd0f4d9bac8e33d47d1c940fd992f
Seagis-Model/supabase/migrations/0003_environment_ingestion.sql	261	d49a5897bca967b33e3ab79745bb4f1bb9e76babc318544bfc7c024c8993d1f5
Seagis-Model/supabase/migrations/0004_provider_update_status.sql	15	bf92788ed13dd0ffd60a8f21b25f84627976f5e9b46e27f98b00f4062a4cbe0a
Seagis-Model/supabase/migrations/0005_quality_alignment.sql	249	afc5d564ed54adf9cf353865f0859f414469016b02823707f3d872453608da65
Seagis-Model/supabase/seed.sql	10	4809f15911388f46d0b94fcc09d413e341f7aadf4eb67fb88dadf6e6f0748f79

Path	Lines	SHA-256
Seagis-Model/tests/bootstrap_roles.sql	8	4fe6ea493501b15e67aa693fce29a08f24096f1c95d5f6f52af79e3943338f08
Seagis-Model/tests/test_a1.py	268	49ffb98dbf091d5343be35d7f0ef532e27beccdef5af9b2a343bc30b1e8e758b
Seagis-Model/tests/test_a1_tools.py	85	e9a79d56845040e695774e8205a2aa5ef9ea85bc3dd6448f07479ffc5e0a0bb9
Seagis-Model/tests/test_a3.py	418	607f4ec2cf2bb737a93d4412294da99b49d5bfea04ff1ae886d677f23a2f1df1
Seagis-Model/tests/test_contracts.py	108	af73d14cd7b5d49e91555b1040a9520d2ca2eb2ccc7def15f40e59ab31296ea9
Seagis-Model/tests/test_database.py	489	19be7574738537be1358ed6b36ca1cf586be5affd9e9b846f92032e755d56e44
Seagis/AGENTS.md	28	9aeaecda1a0926efa09176c2adbb543b06c4c7041d7fed671cb7a43177fee1e2
Seagis/README.md	124	c40fef74d33539a20f283c742175e2f1fd4a747e4d909a9aa6cc152ef29682ba
Seagis/SPEC_REFERENCE.md	25	dc29fa4fcf7879059b83c255fd604ab0a6d8eb7376deb2a0114ca544db329c8a
Seagis/Seagis/App/AppContainer.swift	20	30f78023407d3e3f85dd4f1f3de7be9e8c0db823843ac2435f3134ebc0f85421
Seagis/Seagis/App/SeagisApp.swift	10	d6f0f8373869116965c49bbf404a64a1138cc939fd2a6703cdf4ee71ab99797a
Seagis/Seagis/Models/FarmMap/FarmPlotDisplayStatus.swift	25	30863de6a7400f137297a0668d175b3c5814f7db7720ad3699325b56ee07a43e
Seagis/Seagis/Models/PlotCapture/CaptureMapArea.swift	20	c58772b18a16045b75b00f60e86a56a2f034e3da4ba0cb80d4b3c47b030f169f
Seagis/Seagis/Models/PlotDetail/ConditionDay.swift	16	10924ce38ee6f243f009d260ca0e099bc7cdd05f12b07c02ffce73009f78fd9e
Seagis/Seagis/Models/PlotDetail/ConditionPoint.swift	23	916c2b770f0bf4311257d690d03b3fe2091fa8e9e54f7b41428bae9ade438249
Seagis/Seagis/Models/PlotDetail/ParameterColorScale.swift	64	182ab73e68f99e7b1af35d363bffd616a8b810401e54a4df75c43f45148ddcb8
Seagis/Seagis/Models/PlotDetail/PlotDetailStatus.swift	30	1701f9fca3172013ffdc396ee32f5b55cc10db9c12a1c7e8976b0602bffbf50e
Seagis/Seagis/Models/PlotDetail/WaterParameter.swift	55	1b1c583b79a385e9b1327ba5466dc0c1fb7eea2e2ae3d00e3f58d7f6a2256137
Seagis/Seagis/Models/PlotDetail/WaterParameterInfo.swift	32	10782cf6ba9b090cc5e8acd5c0614f2b76813b650dd44462a1f49207e7b744f8
Seagis/Seagis/Models/Shared/Assessment.swift	25	349be21ee4ddaf9dec672875837b2876893658b53c5343ac9a67dc4dda07a70e
Seagis/Seagis/Models/Shared/ConditionWindow.swift	37	22019f46b70b9ef012f97407e7df3dba338baa6dcd4f79419a2ecba3e2478a54
Seagis/Seagis/Models/Shared/MetricSummaryItem.swift	56	9705a185e0d121695da2018450df2f49662e68834a7c9d752b5968e5dee46d2d
Seagis/Seagis/Repositories/Shared/AssessmentRepository.swift	17	37c4c063a83327523bb7d3ab186cddf6c780d8c651e498f8fe6165425e52df2f
Seagis/Seagis/Repositories/Shared/MockAssessmentRepository.swift	21	41618019f73ba0774b04dd43fb2b1643f9b11e17ba7caaff41811907bee4effc
Seagis/Seagis/Services/Intelligence/InsightGenerator.swift	6	5cde5f99c69be2fd73e14f8984928dbb930a732e308429542cb22d673bdafcdc
Seagis/Seagis/Services/Intelligence/TemplateInsightGenerator.swift	13	945cfd6d3ef1e12ed8d3eef12d58e8a7157618a969d2c7485a0ad9eeae2eae17

Path	Lines	SHA-256
Seagis/Seagis/UseCases/Bootstrap/LoadDemoAssessment.swift	13	657ae092c4e13b79772fd05be04606b860cc3897574068535738ac3f6652b9f9
Seagis/Seagis/Utilities/PlotDetail/ConditionDayFormatter.swift	26	7ad2a51263fee6b90eaf933a6ebf0457a776214011e47f93d76c57ab64e44f1f
Seagis/Seagis/Utilities/PlotDetail/DailyConditionSummaryFormatter.swift	29	98775dad31d7afa55b4bb62b076b0ddcd5e11a167daeaad56fb15f254c37bd52
Seagis/Seagis/ViewModels/Bootstrap/BootstrapViewModel.swift	33	bc8f0d8ee67ae582bddfefe68043892be053d75e8083226d19423f871998d142
Seagis/Seagis/Views/Components/BootstrapView.swift	75	55d25cf1d0f7a6a8786a918c46ed9b989b11b649c19eee9425fbb757ba747b75
Seagis/Seagis/Views/Components/ComponentGalleryPreview.swift	92	c3a459bc889713e0201bbe669012e5244f5c6c41a719a15419bd56e9ce566a07
Seagis/Seagis/Views/Components/FarmMap/FarmMapSheetHeader.swift	165	4ed2fca1dbaaf83fcf9a41ff4e2c41fe5c72b38b487e54774a145f61a0c7ff7d
Seagis/Seagis/Views/Components/FarmMap/FarmerRow.swift	95	4e21e4e2b7620a06a04132d07af57f3c55e523b1892c02dd995b667526cdbb92
Seagis/Seagis/Views/Components/FarmMap/PlotMapLabels.swift	102	5c87d86f283db0fc7e590dbc69da2f0d346b2923d36b86c52ec09efc90c77fbb
Seagis/Seagis/Views/Components/FarmMap/PlotRow.swift	60	6c1dec4c8355c3f7df0787275d46d54723b93115fc7d70c955b6b9d3ae47388d
Seagis/Seagis/Views/Components/PlotCapture/ARSceneView.swift	56	34785ca13eecd86ab33b063f8e4dcf6c2035409b4b09c0e898270dacf9532a8e
Seagis/Seagis/Views/Components/PlotCapture/CaptureChrome.swift	223	bb7a4f5b14b98aef0fdb7a02985255be9d494dae2cea391c5840d1c291a5e60f
Seagis/Seagis/Views/Components/PlotCapture/CaptureControls.swift	141	d7cb9b0d0c169567c1e21788cd04bea0da420f5e24a4e533c58e8a1a68c44761
Seagis/Seagis/Views/Components/PlotCapture/CaptureMapView.swift	364	de0facdb408e2cda3cbe8701783863accd8417558486de75780324cdfa304d54
Seagis/Seagis/Views/Components/PlotCapture/CaptureStepsOverlay.swift	80	e643bea1cbd3373a5a22cac83cf7b93846f4a1332a001c0007093d643824fb84
Seagis/Seagis/Views/Components/PlotCapture/HatchedOutline.swift	61	4fdacc0fd4b5ff87139787b8020605536ea141107e294923bca595a71ac67bc0
Seagis/Seagis/Views/Components/PlotCapture/OutlinePreview.swift	110	98724e172d88169bcf13fb11453d5cdbea88c1a67f8ed2e64faee8da72184195
Seagis/Seagis/Views/Components/PlotDetail/ConditionWidget.swift	91	732c4cb26477995528a12f41f4728ffa77d18a7ab3206abc2e28c561d8bfe31d
Seagis/Seagis/Views/Components/PlotDetail/CurrentWidget.swift	85	48d87c6ff150c26d8e27bafad217c6114a1d616fe9bfac6d035c1450bd7c4122
Seagis/Seagis/Views/Components/PlotDetail/GaugeWidget.swift	48	7655e21e12000acbc45586fd5e7a349c4f132be1fe51663e743be0e14ec5e302
Seagis/Seagis/Views/Components/PlotDetail/HourlyConditionChart.swift	96	c0ebef731591ac7c84c634950e0ca0352ce52a1454b2cecbb8d45f3fa49b1cf0
Seagis/Seagis/Views/Components/PlotDetail/ParameterDayCarousel.swift	73	525e536ecda2a4dfddc2d7940a27bef59f18f6f798b97dad78ad35a507ab5c21
Seagis/Seagis/Views/Components/PlotDetail/PlotDetailPreviewData.swift	29	d2d0b94eee2d321b75d4f0ebc338032172a423e54bce95272f1783920e4b441e

Path	Lines	SHA-256
Seagis/Seagis/Views/Components/PlotDetail/PlotOutlineShape.swift	30	d9f42846399d4c218fa4efabf55943c4bd5e46e31ff520dcf910d9daaa3b9f52
Seagis/Seagis/Views/Components/PlotDetail/PlotStatusCard.swift	88	4b9e3437150e8013e5154079f647c97e24d3d6344f3fe6b60cb259f2fc51b4ff
Seagis/Seagis/Views/Components/PlotDetail/TemperatureWidget.swift	95	401025ba3aa721025753419da186b8fcbda7ee3bcd23e91c45d12e4a218b6935
Seagis/Seagis/Views/Components/Shared/AdaptiveGlassModifier.swift	43	070be23bb434650845804f3227749464f01283aec5e22c9083f68b2d9b0122c2
Seagis/Seagis/Views/Components/Shared/CircleIconButton.swift	125	9f8458edcff549fd16c0df7d9fccdca5f28990e7de4fe71cb2a252cd2069d276
Seagis/Seagis/Views/Components/Shared/StatusBadge.swift	115	0f7de149618bfe6c82e7ed946f70de7a8d09c18269da97307f917d8cb60a9348
Seagis/Seagis/Views/Components/Shared/StatusBanner.swift	93	7eab25a99e8c8c006c060ebd8e6ac33ea262cc6ae67f1e5bc135691dff7360f1
Seagis/SeagisTests/Services/Intelligence/TemplateInsightGeneratorTests.swift	15	1a23b61f6a886ec61810f3e60725321fec4b4f29dba2ed93355f4bd839488e99
Seagis/SeagisTests/UseCases/Bootstrap/LoadDemoAssessmentTests.swift	17	f12da530e126fbd6e0eabcb87b1706b5de3b9bd1b26f252d769f680ef5259ae2
Seagis/SeagisTests/ViewModels/Bootstrap/BootstrapViewModelTests.swift	47	29c0ee996c8aba71f50e15936b01affa343ed2c4abc989851b7c1dee389c4368
Seagis/docs/architecture.md	67	9a3a73e6763f42c40d2609e8ff35f9e1680894340a9fc560ab5855662bd1b1a2
Seagis/docs/decisions/ADR-001-engine-placement.md	27	0292e6f7fbc53ea13e987c619e862f4a58d3429a4bdf4c4ba27621d3480b7b0f
Seagis/docs/development-workflow.md	27	b6202666a9d68357b8d07f8aef9762410319be76d3eecb1d8fe8addbd28e4e73
Seagis/docs/integration-boundaries.md	29	a214c3e3b6432fcbc27afed03009e95d854cbf3b03dee4c013f8dddab5242a61
Seagis/docs/ios2-scaffold.md	85	377a42d94241f06d4b5adb8588c746a6992c4cdc1e6024026c28941c4c47e34b
Seagis/docs/ios3-reusable-components.md	1347	c866cb9afc1d1c7a30501c015a90f4410a4f6e2f5262cbf0f48c63d0384dbab4
Seagis/docs/iot-dataml-architecture-mapping.md	149	200129d604a399f5da6e0f15928c2bc5f7583484b48f2adcb0e992c24d0ab088
Seagis/docs/open-decisions.md	17	3a1c1673c8337da046d48935f57b97c0347c7fc01501eab9b02b1cf85196fcd1
AppPOC/AGENTS.md	9	39ac4b865dfefe8101538df5c7703e9dc039e7da7ca2c0b7ca873ec0470c8919
AppPOC/Documentation/MODEL_CARD.md	29	86ce30190d9d267bb6757d57e857f6120411178ab0ef96ca063a40427aa07e5d
AppPOC/Documentation/SOURCE_REQUIREMENTS.md	150	3c130872b1e857653e8ff47af3f9d82e571e7ad0e1655860f69d7d263d3896ed
AppPOC/Documentation/UI_REDESIGN.md	34	52013450c9b0e04a263a452cfba15f8553272dc07a353356a57734c991342b4d
AppPOC/Documentation/VERIFICATION.md	25	5237b4ff2b43b40fc98cf68647c485e79787d75dbc8e9c884f43e3c75494f494
AppPOC/POCApp/App/POCApp.swift	9	8e14c4b8dde5c3b11de73ed22dddf1861cf2d69bd382e8dc9b51e065ad388ab2

Path	Lines	SHA-256
AppPOC/POCApp/Features/FarmMap/ViewModels/FarmMapViewModel.swift	151	e6eb210fc07dfec6f38df729129707343905498d8c30e23485eefc0636e11881
AppPOC/POCApp/Features/FarmMap/Views/FarmMapSheetHeader.swift	53	296dc9226891ad521b2df0c18e8d1190d57782be1d800d442a91f5ee9a6c60be
AppPOC/POCApp/Features/FarmMap/Views/FarmMapView.swift	127	06f85b64f137a63b64cd839fc39fead1b1f5824e1258cb9582f00cba08e9b134
AppPOC/POCApp/Features/FarmMap/Views/FarmerListView.swift	22	ee5b06f9d4aa1d1e3ed1eab2aecd0925f5c01480abdf0f692a49e0e38e25e77d
AppPOC/POCApp/Features/FarmMap/Views/FarmerRow.swift	29	2bb17c91bd9ee33de4929f1070a7d544415068fe75035d9c919598f044537285
AppPOC/POCApp/Features/FarmMap/Views/PlotListView.swift	47	47f3db3934b6aded271d696420487ae58b58a081512e2fceee9a746f698ec578
AppPOC/POCApp/Features/FarmMap/Views/PlotMapLabels.swift	46	9c00ca470f4921b0f440573f9f54b26b3b518f0654adca72db7f5b991aef7389
AppPOC/POCApp/Features/PlotCapture/Models/ARSite.swift	110	c68a4dcff8c56b8094e70deb68ee43821040e8b8383caab967d97a74c340a57b
AppPOC/POCApp/Features/PlotCapture/Models/CaptureArea.swift	44	f6709a92b1898e1fddb15e31c73224830af42342444bb95fe81f60c16b7514e0
AppPOC/POCApp/Features/PlotCapture/ViewModels/ARCaptureViewModel.swift	445	673b9744f7e416d9bea503f09caf54d848d49468a59d24a8000ec94ac301362b
AppPOC/POCApp/Features/PlotCapture/ViewModels/CaptureNudge.swift	20	b3987dab6afbc02080a2450f0297ec5c028f9c3cf2b6c35688b1a3d09f3f8625
AppPOC/POCApp/Features/PlotCapture/ViewModels/GPSCaptureViewModel.swift	122	885bef577472ce779e1e296f98659ce30b9be3d086d13d29bb8686608e47a2fa
AppPOC/POCApp/Features/PlotCapture/Views/ARCaptureView.swift	71	3dd05c13eea600040c88d516c1224af4533fdf6bbff8b4ec003315f3ab19eaf9
AppPOC/POCApp/Features/PlotCapture/Views/ARSceneView.swift	19	16c02fd37f704d0840992a1a34f3beb27b03c37a5a77d732f2558a45edf2c667
AppPOC/POCApp/Features/PlotCapture/Views/CaptureMapView.swift	201	c60881ad2bd06fbd2e603a07bfdc1e783d5a13baf86e74a02a88b38d2b190ecb
AppPOC/POCApp/Features/PlotCapture/Views/Components/CaptureChrome.swift	56	5d311360e4bcbd2d5750b02d513fe5e9c3129398c3e9fb705f7a4f6f80cccb77
AppPOC/POCApp/Features/PlotCapture/Views/Components/CaptureStepsOverlay.swift	45	ed7dfe27275a27e141a7783eb8cf4130922c673f1de10a78e7dbda9000528438
AppPOC/POCApp/Features/PlotCapture/Views/Components/GlassControls.swift	82	59cb3693cf0192152574e8baba54d830909890dfb3a5cf44ea04499c8c483af7
AppPOC/POCApp/Features/PlotCapture/Views/GPSCaptureView.swift	125	94cc2cff9e9c98e7ef305be4002e2260e97fe9fa784c38e7d3e4cfb07f089da8
AppPOC/POCApp/Features/PlotCapture/Views/PlotCaptureFlow.swift	49	2e36716eba568cb10ce43f290824045c4da81731835581c1c030cb124c91aa91
AppPOC/POCApp/Features/PlotCapture/Views/SavePlotSheet.swift	191	f3c02cba58ab38c054d254de7bdd2a10d972a1941489cebe5f3511f168eecdbc

Path	Lines	SHA-256
AppPOC/POCApp/Features/PlotDetail/Models/DailySummary.swift	44	893581a30491604589955e6902042ff015b465c235cdb73c984ffd9f87af441b
AppPOC/POCApp/Features/PlotDetail/Models/ParameterColorScale.swift	60	8ef76246dead359265e5c94ce1105a757c271c8183b4632ae785c1baed4e1848
AppPOC/POCApp/Features/PlotDetail/Models/PlotConditions.swift	53	680e31bb9aab5efad8d867b0e02255adad2247979b225837377fa6b34d26df1f
AppPOC/POCApp/Features/PlotDetail/Models/PlotStatus.swift	40	8f70f34713993f4392eb319f854bbe7817c6872ead25547bf4e6f5f1d06174ec
AppPOC/POCApp/Features/PlotDetail/Models/WaterParameter.swift	61	78a6dca94962d9e271c77fc2ddf3ef6692eb9987e796b46e8bd43a4207735564
AppPOC/POCApp/Features/PlotDetail/Models/WaterParameterInfo.swift	34	3b4582d97e9394fc28c73491127d5cbbb8eea446e03b8aa7c2ce145e4a452e5b
AppPOC/POCApp/Features/PlotDetail/Services/WaterConditionProvider.swift	52	e4db4bc86a9737e1ae041b1e8a89eb0f5b02d91d7ae7e3602cdf3028d4beb125
AppPOC/POCApp/Features/PlotDetail/ViewModels/PlotDetailViewModel.swift	41	cd36ed23c519d7e076cc32e77061febcd37396a624928dc38373cff1ebdc7ba1
AppPOC/POCApp/Features/PlotDetail/Views/ConditionDayFormat.swift	31	2557c452929fb3da14a8b4a044b526245b3af87bbc1ab81b11cb69456c176ef5
AppPOC/POCApp/Features/PlotDetail/Views/Detail/HourlyConditionChart.swift	115	4258f8d2bc53cf3fb7b9cc2b2ba4f7df73390f8ea920e16a9822cd91afb0a3d3
AppPOC/POCApp/Features/PlotDetail/Views/Detail/ParameterDayCarousel.swift	59	ff39a72b4e7b4a287c311f243836cbac1a0be3742eb860da13e753f6b0cdfd51
AppPOC/POCApp/Features/PlotDetail/Views/Detail/ParameterDetailView.swift	126	b3cbd2b5bd967678b6c863a89d9275d7553ddc80f864e93fd0cedd944e5e3fc2
AppPOC/POCApp/Features/PlotDetail/Views/Detail/PlotOutlineShape.swift	19	e59eb2b7041c030193659355a9f8f65604b54fec564ff6097cd85296fb8a4dbe
AppPOC/POCApp/Features/PlotDetail/Views/PlotDetailView.swift	57	8d076b7b181e40108086fe24a57d9ff04cb84dce3fdba2d07b167600065980c7
AppPOC/POCApp/Features/PlotDetail/Views/StatusCard.swift	83	4a6fe7b331dcb0accd4ccd1cb784e47ae1243a78e45a0b7a62d0a6409d845b52
AppPOC/POCApp/Features/PlotDetail/Views/Widgets/ConditionWidget.swift	76	7b8865207bb5d336feb7dd3ac4b93e596723863833d563647aeb01a79ebeae88
AppPOC/POCApp/Features/PlotDetail/Views/Widgets/CurrentWidget.swift	74	3334408a2ecc5d412aaf2cb6f297fc0f940c3e58428f5f5b0b38c87e582c65c9
AppPOC/POCApp/Features/PlotDetail/Views/Widgets/GaugeWidget.swift	30	498490bb259c057497df3f47bd7a108447b14111efef2a932f71c8e340d9cf7d
AppPOC/POCApp/Features/PlotDetail/Views/Widgets/TemperatureWidget.swift	100	59c03a89512341b8f2fd2c7c9852a7943fcce45de6c142c5d674d9cfd126633d
AppPOC/POCApp/Shared/Data/FarmStore.swift	56	35e892e5fa3bc5c8e7a6bcc6b50d03e2fa02712366099d48b0416597b1d3b78f
AppPOC/POCApp/Shared/DesignSystem/CircleButton.swift	25	b5032491bff7586d8b555e4e3e1a741592e357304a699b650026cc6b7bff13b2

Path	Lines	SHA-256
AppPOC/POCApp/Shared/DesignSystem/Glass.swift	13	e53326f3fdd43b3043bedfe64dde78791469d0060e8407b99f022646fb4e43d9
AppPOC/POCApp/Shared/DesignSystem/Marine.swift	17	09771fbfbbb29f7c24fb72c2f85c496d43ea3523c6729c3cefbeee8b6f171061
AppPOC/POCApp/Shared/Geo/Geo.swift	102	5a98921b657536f8ad0e25f24e9deda3655415d0fcba80dc185538145f364928
AppPOC/POCApp/Shared/Location/LocationManager.swift	77	e907be2920f002826f63db9988e95b67521888a9f75e304c088137226fe116b1
AppPOC/POCApp/Shared/Models/CultivationPlot.swift	134	beb25cbb8ab28def3e2529dcae63974825b4767a31a51633358f3723cde23df6
AppPOC/POCApp/Shared/Models/Farmer.swift	27	4497a6d1e7caf1af6344e1700d4b52b6dfe05a16432b038054509b1fad589a5d
AppPOC/Packages/TidungCore/Package.swift	3	44dfbfd9d7b80fb3c3995154378836659ea002a57425bb7006c2cb79405a484b
AppPOC/Packages/TidungCore/Sources/TidungCore/Models.swift	112	163620afbf41029cc36b880fdde51076991c69adeeecf8899347bb0c7700f8a5
AppPOC/Packages/TidungCore/Sources/TidungCore/NASAPowerClient.swift	85	166aaf4d7e9a79f0ce49b077b3c93961210c1bab2af08c4535737e19a8aaef48
AppPOC/Packages/TidungCore/Sources/TidungCore/SeriesAnalysis.swift	25	577b6b80d785724744f6c26c4cec939e36363067613b464c8228a4cbbf779342
AppPOC/Packages/TidungCore/Sources/TidungCore/SolarModel.swift	44	2b5c297043fd37d50c141b8edcafaf2a381596c6f452b80ccd87de824b15e41a
AppPOC/Packages/TidungCore/Tests/TidungCoreTests/TidungCoreTests.swift	92	7cd85a87fd93510f3e9e8dc73547a762313fe40eed8e7e12411dc478bddfc72d
AppPOC/README.md	114	7b87e44385e144ae3332357b22ceadb26f89295bcdccaca1444b5e1b4396afc8
AppPOC/TidungMonitor/App/Farms.swift	63	41f88166cc3201e029873c9aa866c685231e70ae2aea25eb1dd4e5cf9b3d5c75
AppPOC/TidungMonitor/App/MonitorViewModel.swift	58	8527556638d36d5ebc6d1710aa0df3192e24c212a76bec4fe6761fecb9d32341
AppPOC/TidungMonitor/App/TidungMonitorApp.swift	89	21d94c9480a177df12654b298ae605a2a69ec8d483af263c052a2d6eb68a9aea
AppPOC/TidungMonitor/App/WaterConditions.swift	114	6cf2c87e96e463862a1558fa813bb747332531a669257e6f5a5407a3f59f697f
AppPOC/TidungMonitor/Views/ConditionCards.swift	250	0bd541537b7b04aa4023ee6da43eb87572060737fe0e266e7aa51e99450fe17b
AppPOC/TidungMonitor/Views/DataSourcesView.swift	35	8673864e9e44548b37d752ca53b468164bc78ae12adae8688a90484392103942
AppPOC/TidungMonitor/Views/HistoryView.swift	127	9ef084f459060ab27f7b8764e144dfbcc0e6e25047d63f7e9ac92f5a40f0aa2f
AppPOC/TidungMonitor/Views/ModelView.swift	32	1ba7dcf9e3286a29b40f5cc82e0808587dcaa634ab06b3015f7ea24297f4abd2
AppPOC/TidungMonitor/Views/OverviewView.swift	266	402c6ffb87943bed0d7775ab10b62cd9c062c44ed7de857a5dcc987cd8b79787
AppPOC/TidungMonitor/Views/PlotDetailView.swift	76	5c9a4650ca71eccd92be843424225b448d5d6731757124d058d82ae309b982ac
AppPOC/UITests/TidungMonitorUITests.swift	73	70b60d71083ec127a51ddff7482f006d12985c66efbbd5ff4e7487e1574c3f7e

Salinan migrasi dan empat kontrak tersedia di assets/evidence. source-manifest.json, coverage.json dan validation.json berada di assets/generated. File pembangkit dokumentasi dan PDF disertakan agar hasil dapat direproduksi.